Business Analytics Toolkit Portfolio Final

Course: BUSI 6532 - Business Analytics & Data Visualization

Submitted to: Dr. Kyung Lee

Submitted by: Muhammad Hassan Shahid

B00767748

Dalhousie University • 2026

Table of Contents

Chapter 1 Microsoft Excel Pivot Tables ---------- 5
1.1 Introduction of Microsoft Excel Pivot Tables ---------- 5
1.2 Dataset and Research Questions ---------- 5
1.3 Deriving Business Insights and Findings ---------- 5
1.3.1 How did Global Bike Inc.'s sales change from year to year, and which year had the strongest overall performance? 5
1.3.2 Which product category has the highest total cost of goods sold (COGS)? ---------- *6*
1.3.3 How does the regional net sales per unit sold in a country look like? ---------- *7*
1.3.4 Which are the top 5 profitable products (as per the Gross Margin)? ---------- *8*
1.3.5 What is the monthly sales revenue trend, yearly? Which months show spike and which shows dip? ---------- *8*
1.3.6 How is the total discount distributed across different products? ---------- *9*
1.4 Critical Analysis of Using Excel Pivot Tables ---------- 10
1.5 Conclusion ---------- 10

Chapter 2 FAO SDG Data using Microsoft Excel for Cleansing and SAC for Visualization ---------- 11
2.1 Microsoft Excel vs SAP Analytics Cloud ---------- 11
2.2 Datasets and Suggested Questions ---------- 11
2.3 Application of Data Cleansing Techniques and Visualization Tool ---------- 11
2.3.1 How were irrelevant and unwanted columns identified and removed to align the dataset with the analytical objectives? ---------- *11*
2.3.2 How were three-year average columns restructured into a consistent annual format, and how was the RIGHT() function in Microsoft Excel used? ---------- *14*
2.3.3 How were missing and blank data points identified maintain analytical accuracy and comparability across countries and years? ---------- *14*
2.3.4 How was the final dataset further cleaned and validated to ensure it was fully analysis-ready? ---------- *16*
2.3.5 How does the contribution of CH4 emissions compare to CO2 and N2O emissions across countries? ---------- *16*
2.3.6 What is the emissions trend, and which specific pollutant shows the most significant steady increase over the ten-year period? ---------- *17*
2.3.7 For CO2 emissions from 2014 to 2023, which countries demonstrate the highest contribution to global emissions? ---------- *18*
2.3.8 What is the emissions trend for N2O emissions by area post-covid? ---------- *18*
2.4 Critical Analysis: Excel vs. SAP Analytics Cloud ---------- 19
2.5 Conclusion ---------- 19

Chapter 3 SAP Analytics Cloud ---------- 19
3.1 Introduction of SAP Analytics Cloud ---------- 19
3.2 Dataset and Research Questions ---------- 19
3.3 Deriving Business Insights and Findings ---------- 20

3.3.1 Which product category (E-Bike, Offroad, Road, Touring, or Trend) has seen the most significant growth in revenue share between 2011 and 2023? ---- 20
3.3.2 Within the United States, which specific locations (e.g., Silicon Valley Bikes vs. Peachtree Bikes) are the top three contributors to sales volume? ---- 21
3.3.3 Which customers maintain the highest Gross Margin Ratio, and is there a negative correlation between high sales volume and high margin percentage? ---- 21
3.3.4 Based on the revenue trends from 2011 to 2023, what is the projected revenue forecast for the first half of 2024, and what is the degree of uncertainty (confidence bound) for those months? ---- 22
3.3.5 How do high-volume, low-profit products impact the Sales Organisation efficiency? ---- 23
3.3.6 How is the total discount distributed across different products? ---- 24
3.4 Critical Analysis of Using SAP Analytics Cloud ---- 25
3.5 Conclusion ---- 25

Chapter 4 Business Data Analysis Using Tableau: Insights from the Sample Superstore Dataset ---- 25
4.1 Tableau vs SAP Analytics Cloud ---- 25
4.2 Datasets and Suggested Questions ---- 26
4.3 Application of Data Cleansing Techniques and Visualization Tool ---- 26
4.3.1 Which Geographic Regions Generate the Highest Profit Margin and which state had highest profit margin in the east market? ---- 26
4.3.2 How Do Sales and Profit Vary Across Different Customer Segments over years? ---- 29
4.3.3 Which state contributes the most to sales? ---- 31
4.3.4 Does faster shipping lead to higher sales or profitability? ---- 31
4.3.5 What is the Relationship Between Discount and Profit? ---- 32
4.3.6 Which product sub-categories act as the primary profit drivers for the business?---- 33
4.4 Critical Analysis: Tableau vs. SAP Analytics Cloud ---- 34
4.5 Conclusion ---- 34

Chapter 5 Reporting with SAP BW using SAP Crystal Reports ---- 34
5.1 Introduction of SAP Crystal Reports ---- 34
5.2 Dataset and Questions ---- 35
5.3 Application of Report Designing and Formatting in SAP Crystal Reports ---- 35
5.3.1 How can record selection be applied in SAP Crystal Reports to filter the dataset for a specific division?---- 35
5.3.2 How can the sales data be organized and grouped by Calendar Year and Country in the report? ---- 37
5.3.3 How can a clear and professional report header be added to improve report readability? ---- 38
5.3.4 How can a page footer with page numbers and copyright information be included in the report? ---- 39
5.4 Critical Analysis: SAP Crystal Reports ---- 40
5.5 Conclusion ---- 40

Chapter 6 Tableau Desktop ---------- 40
6.1 Introduction of Tableau for Textual and Sentiment Analysis ---------- 41
6.2 Dataset and Data Enrichment Workflow ---------- 41
6.3 Deriving Business Insights and Findings ---------- 41
6.3.1 How do average wine prices and rating scores compare across different continents and countries?---------- 41
6.3.2 What is the relationship between wine price and rating points across countries and provinces of origin?-------- 43
6.3.3 How do average wine prices vary geographically across countries based on a choropleth map visualization? - 45
6.3.4 How does the distribution of sentiment scores differ across continents?---------- 47
6.3.5 Do higher-priced wines tend to receive more positive sentiment in reviews across different countries?---------- 49
6.3.6 Which country has the lowest average sentiment score, indicating the most negative reviews?---------- 51
6.4 Critical Analysis of Using Tableau for Unstructured Data ---------- 52
6.5 Conclusion ---------- 52

Chapter 7 Association Analysis & Time Series Forecasting using SAP Analytics Cloud (SAC) ---------- 52
7.1 Introduction to SAP Analytics Cloud ---------- 52
7.2 Datasets and Analytical Objectives ---------- 53
7.3 Association Analysis Using Titanic Dataset ---------- 54
7.3.1 What does the analysis reveal about passenger survivability patterns on the Titanic? ---------- 54
7.3.2 How does increasing the support threshold (from 0.01 to 0.04) affect the association rules generated? ---------- 55
7.3.3 How does decreasing the confidence threshold (from 0.8 to 0.5) impact the strength and number of rules? ------- 57
7.4 Time Series Forecasting Using Global Bike Sales Data ---------- 59
7.4.1 How does the predictive model use data aggregation and MAPE to validate seasonal sales forecasts for different geographic entities? ---------- 59
7.4.2 How do forecasted and actual sales compare for December 2025 in Germany (EUR)? ---------- 60
7.4.3 What are the forecasted sales for July 2025 in the United States (USD), and how do they align with historical trends? ---------- 60
7.5 Critical Analysis of SAP Analytics Cloud Capabilities ---------- 62
7.6 Conclusion ---------- 62

Chapter 8 Executive Summary ---------- 63

Chapter 1 Microsoft Excel Pivot Tables

1.1 Introduction of Microsoft Excel Pivot Tables

Microsoft Excel Pivot Tables are a useful starting tool for making sense of large datasets without needing advanced technical skills. They allow users to quickly summarize and reorganize raw data by simply dragging and dropping fields, making it easy to view information from different perspectives, such as sales by region or by time period. Since Excel is commonly used in most business environments, pivot tables are an accessible and familiar way to generate basic business insights and is still effective for turning large amounts of data into clear summaries and simple visuals that support decision-making.

1.2 Dataset and Research Questions

This section presents the analysis conducted using Microsoft Excel Pivot Tables based on the Global Bike Inc. wholesale dataset. The dataset consists of approximately 29,000 transactional records, covering multiple years of wholesale sales activity from 2007 to 2011. Key data fields used in the analysis include Calendar Year and Month, Customer Description, Country, Region, Sales Quantity, Revenue, Net Sales, Cost of Goods, and Discount. Each research question is addressed using a dedicated pivot table and supported by a corresponding visualization, allowing patterns and comparisons to be identified clearly below:

- How did Global Bike Inc.'s sales change from year to year, and which year had the strongest overall performance?
- Which product category has the highest total cost of goods sold (COGS)?
- How does the regional net sales per unit sold in a country look like?
- Which are the top 5 profitable products (as per the Gross Margin)?
- What is the monthly sales revenue trend, yearly? Which months show spike and which shows dip?
- How is the total discount distributed across different products?

1.3 Deriving Business Insights and Findings

1.3.1. How did Global Bike Inc.'s sales change from year to year, and which year had the strongest overall performance?

To analyze changes in sales performance over time, a pivot table was created (Fig. 1.1) using Calendar Year as the row variable and Revenue as the value field, summarized using the sum function (Fig. 1.2). This compared total sales across different years and helped identify which year performed the strongest overall. In some cases, Sales Quantity was also included as an additional value field to provide further context (Fig. 1.3). Including sales quantity makes it possible to determine whether changes in revenue were driven by higher sales volume or by other factors such as pricing or product mix.

Row Labels	Sum of Revenue
2007	$ 60,715,831.69
2008	$ 59,444,067.11
2011	$ 56,625,742.94
2010	$ 55,854,415.97
2009	$ 52,610,815.06
Grand Total	$ 285,250,872.77

(Figure – 1.1)

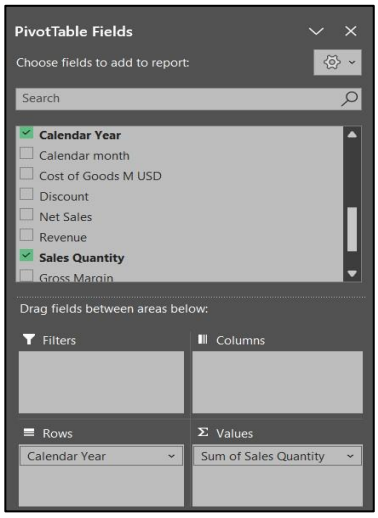


(Figure – 1.2)

The analysis shows that 2007 was the strongest overall year for Global Bike Inc.'s wholesale business. During this year, the company recorded the highest total revenue, reaching $60,715,831.69, which was higher than any other year in the dataset. In addition to revenue, 2007 also had the highest sales volume, with 37,537 units sold.

Row Labels	Sum of Sales Quantity
2007	37,537
2008	36,088
2010	32,237
2011	31,880
2009	31,112
Grand Total	**168,854**

(Figure – 1.3)



(Figure – 1.4)

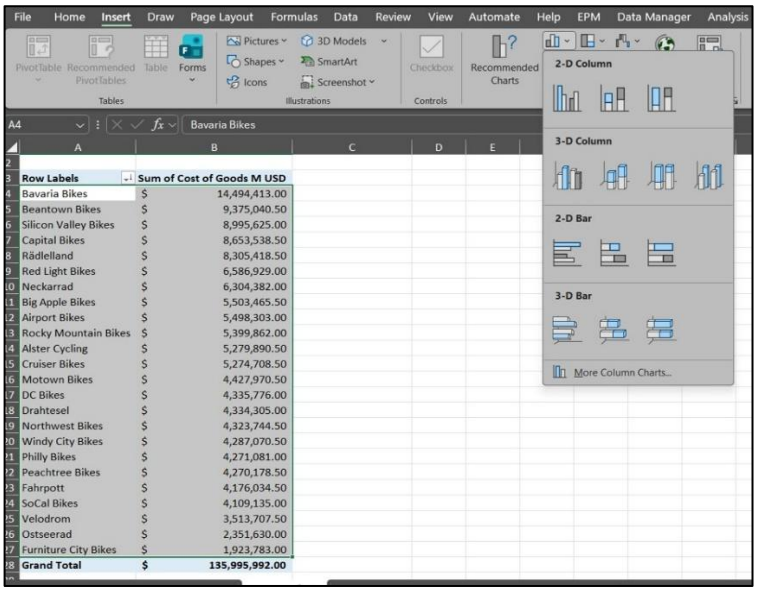
When revenue was compared with sales quantity, it suggested that Global Bike Inc. was able to maintain a higher average price per unit in 2007 compared to later years such as 2011. This indicates that strong performance in 2007 was driven not only by higher sales volume, but also by more efficient pricing.

1.3.2. Which product category has the highest total cost of goods sold (COGS)?

To analyze cost performance, a Pivot Table was built using Customer Desc for the rows and Sum of Cost of Goods M USD for the values. Sorting this data in descending order highlighted Bavaria Bikes as the top cost driver at $14,494,413.00 as shown in the (Fig. 1.5). To make these figures easier to understand, a 2-D Column chart was created via the Insert tab (Fig. 1.6). As shown in the PivotChart Fields pane, placing Customer Desc in the "Axis" area and the cost sum in "Values" generated a dynamic bar graph. (Fig. 1.7)

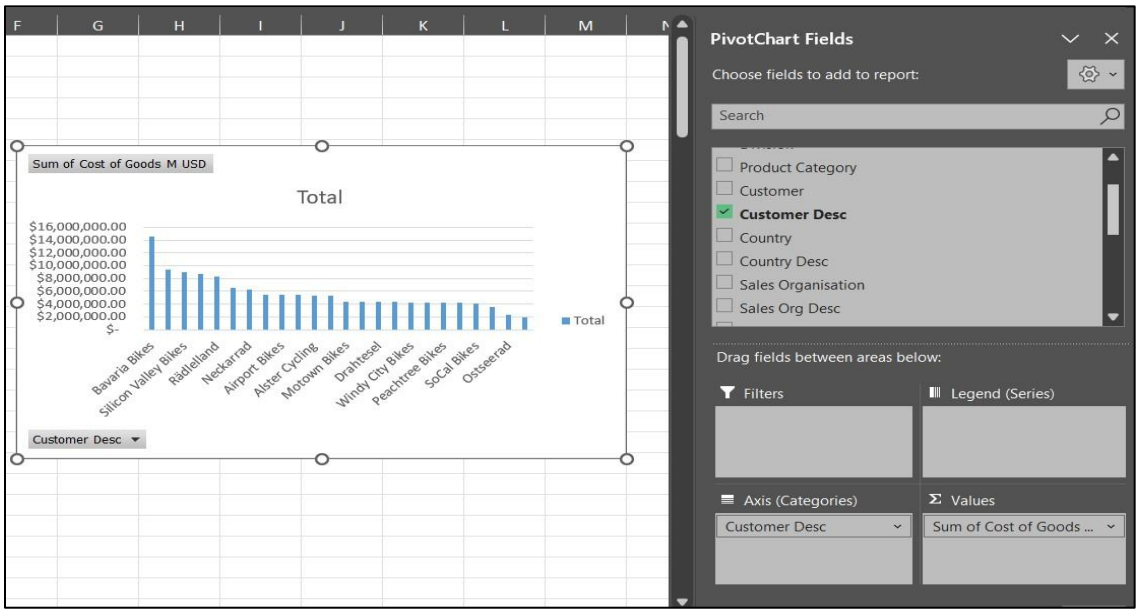
Row Labels	Sum of Cost of Goods M USD
Bavaria Bikes	$ 14,494,413.00
Beantown Bikes	$ 9,375,040.50
Silicon Valley Bikes	$ 8,995,625.00
Capital Bikes	$ 8,653,538.50
Rädlelland	$ 8,305,418.50
Red Light Bikes	$ 6,586,929.00
Neckarrad	$ 6,304,382.00
Big Apple Bikes	$ 5,503,465.50
Airport Bikes	$ 5,498,303.00
Rocky Mountain Bikes	$ 5,399,862.00
Alster Cycling	$ 5,279,890.50
Cruiser Bikes	$ 5,274,708.50
Motown Bikes	$ 4,427,970.50
DC Bikes	$ 4,335,776.00
Drahtesel	$ 4,334,305.00
Northwest Bikes	$ 4,323,744.50
Windy City Bikes	$ 4,287,070.50
Philly Bikes	$ 4,271,081.00
Peachtree Bikes	$ 4,270,178.50
Fahrpott	$ 4,176,034.50
SoCal Bikes	$ 4,109,135.00
Velodrom	$ 3,513,707.50
Ostseerad	$ 2,351,630.00
Furniture City Bikes	$ 1,923,783.00
Grand Total	**$ 135,995,992.00**

(Figure – 1.5)



(Figure – 1.6)

To analyze costs, a Pivot Table and 2-D Column chart were used to identify key cost drivers. Bavaria Bikes emerged as the top customer contributor with a Cost of Goods Sold (COGS) of ($14,494,413.00), followed by Beantown Bikes ($9,375,040.50) and Silicon Valley Bikes ($8,995,625.00). These visuals highlight a concentrated market where a few top tier partners drive the majority of wholesale volume.



(Figure – 1.7)

1.3.3. How does the regional net sales per unit sold in a country look like?

To create the visual analysis for regional net sales per unit, a combination of Excel Pivot Tables and manual formula calculations was utilized across figures 1.8 and 1.9. First, the raw data was organized into a Pivot Table using the columns Country Desc and Sales Desc (such as Germany North and US West) as the row labels, with Sum of Net Sales and Sum of Sales Quantity assigned as the value fields. To calculate the net sales per unit sold of each region, a manual formula was applied outside the Pivot Table as demonstrated in the (Fig. 1.8) where total Net Sales was divided by Sales Quantity (e.g., =B13/C13) to derive the specific value for each unit sold. These results were then consolidated into a clean, formatted table to compare the four main regions, revealing that Germany South achieved the highest net sales per unit at $1,750.46, while the US West had the lowest at $1,527.01.

Row Labels	Sum of Net Sales	Sum of Sales Quantity	
Germany	**$ 158,897,607.43**	**91,939**	
Germany North	$ 85,978,819.57	50,282	
Germany South	$ 72,918,787.86	41,657	
United States	**$ 117,483,539.23**	**76,915**	
US East	$ 65,585,038.22	42,928	
US West	$ 51,898,501.01	33,987	
Grand Total	**$ 276,381,146.66**	**168,854**	
	Net Sales	Sales Quantity	Net Sales per Unit Sold
Germany	**$ 158,897,607.43**	**91,939**	=B13/C13
Germany North	$ 85,978,819.57	50,282	
Germany South	$ 72,918,787.86	41,657	
United States	**$ 117,483,539.23**	**76,915**	
US East	$ 65,585,038.22	42,928	
US West	$ 51,898,501.01	33,987	

(Figure – 1.8)

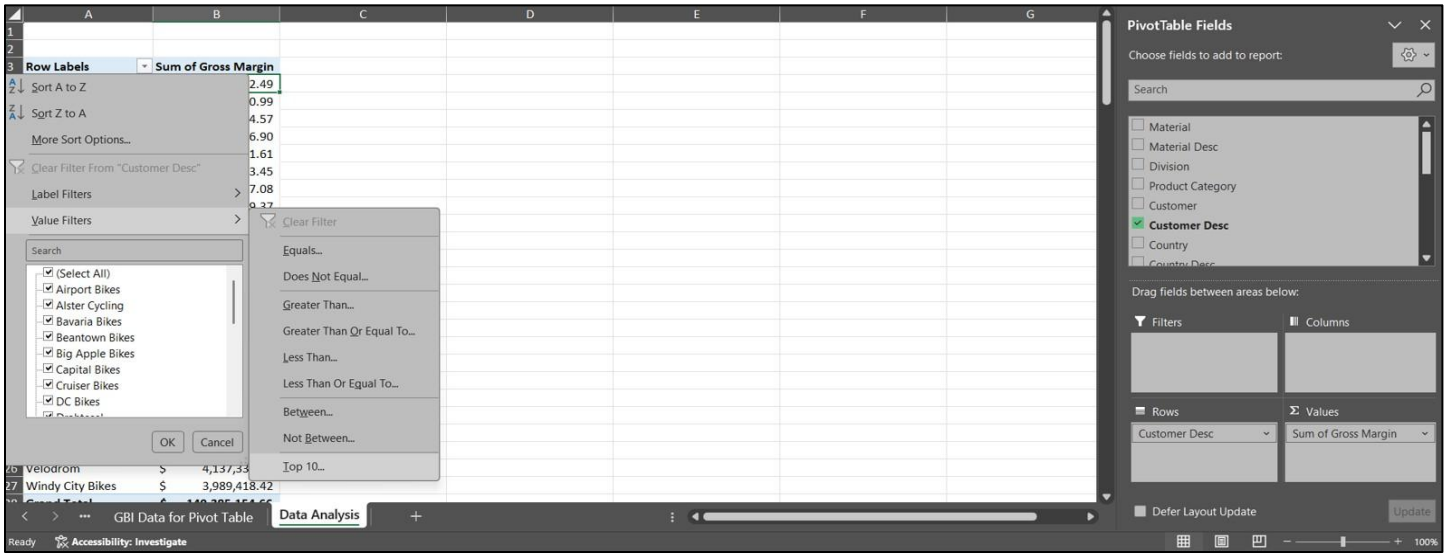
	Net Sales	Sales Quantity	Net Sales per Unit Sold
Germany South	$ 72,918,787.86	41,657	$ 1,750.46
Germany North	$ 85,978,819.57	50,282	$ 1,709.93
US East	$ 65,585,038.22	42,928	$ 1,527.79
US West	$ 51,898,501.01	33,987	$ 1,527.01

(Figure – 1.9)

Looking at the data in the above figures, the biggest takeaway is that not all sales volumes are created equal across regions. While the total revenue numbers might look impressive, the Net Sales per Unit reveals a significant gap in the average revenue earned per unit between the German and US markets. Germany South is the clear leader, pulling in $1,750.46 per unit, whereas the US West lags behind at just $1,527.01 per unit.

1.3.4. Which are the top 5 profitable products (as per the Gross Margin)?

To get the top 5 performing customers, Excel's Value Filters were utilized. After building a Pivot Table with Customer Desc in the Rows and Sum of Gross Margin in the Values area, filter dropdown on the Row Labels was accessed. From there, Top 10... option (Fig. 1.10) was selected and then modified to specifically show only the Top 5 items based on the Sum of Gross Margin.



(Figure – 1.10)

By applying the Top 5 filter to the Sum of Gross Margin, Bavaria Bikes clearly stands out as our most profitable product, contributing a massive $15,787,424.57. The rest including Capital Bikes, Rädlelland, Beantown Bikes, and Silicon Valley Bikes; all perform strongly in the $7.8M to $9.7M range as can be seen in (Fig. 1.11).

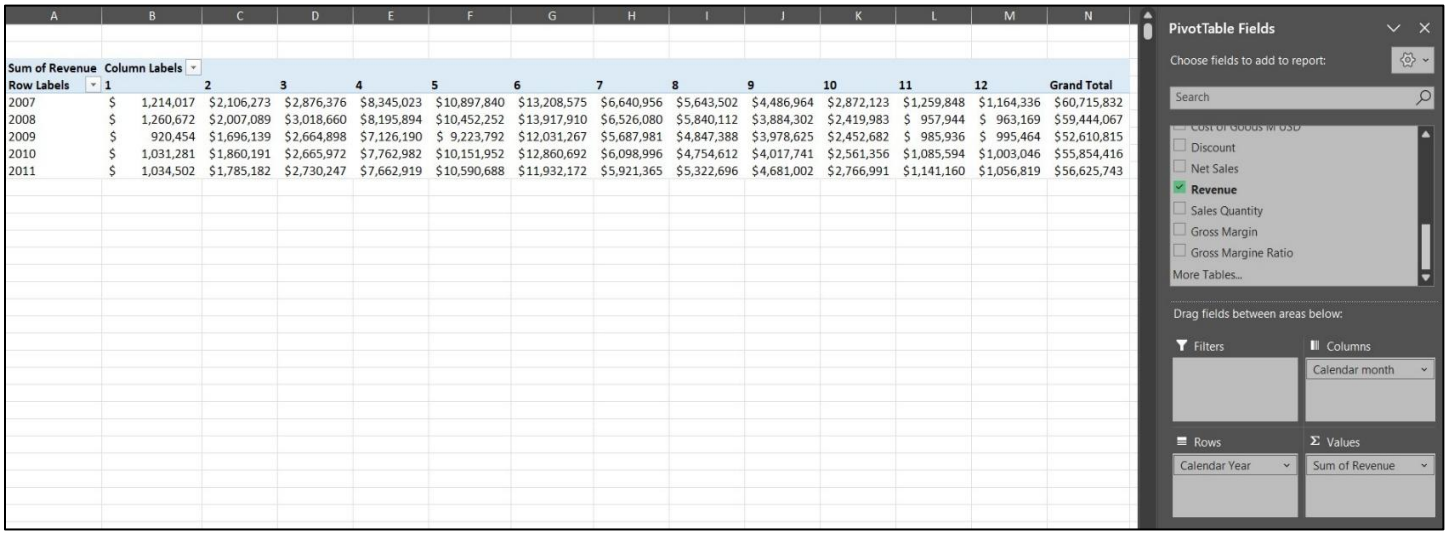
Row Labels	Sum of Gross Margin
Bavaria Bikes	$ 15,787,424.57
Capital Bikes	$ 9,715,803.45
Rädlelland	$ 9,266,373.78
Beantown Bikes	$ 8,361,276.90
Silicon Valley Bikes	$ 7,888,045.74

(Figure – 1.11)

1.3.5. What is the monthly sales revenue trend, yearly? Which months show spike and which shows dip?

To build this visual, the data was organized into a matrix-style Pivot Table to get a clear view of sales cycles. Calendar Year was placed in the Rows and Calendar month in the Columns, creating a grid that makes it easy to compare years and spot seasonal trends. By putting Revenue in the Values area, sales were totaled for every single month. The final result is a table

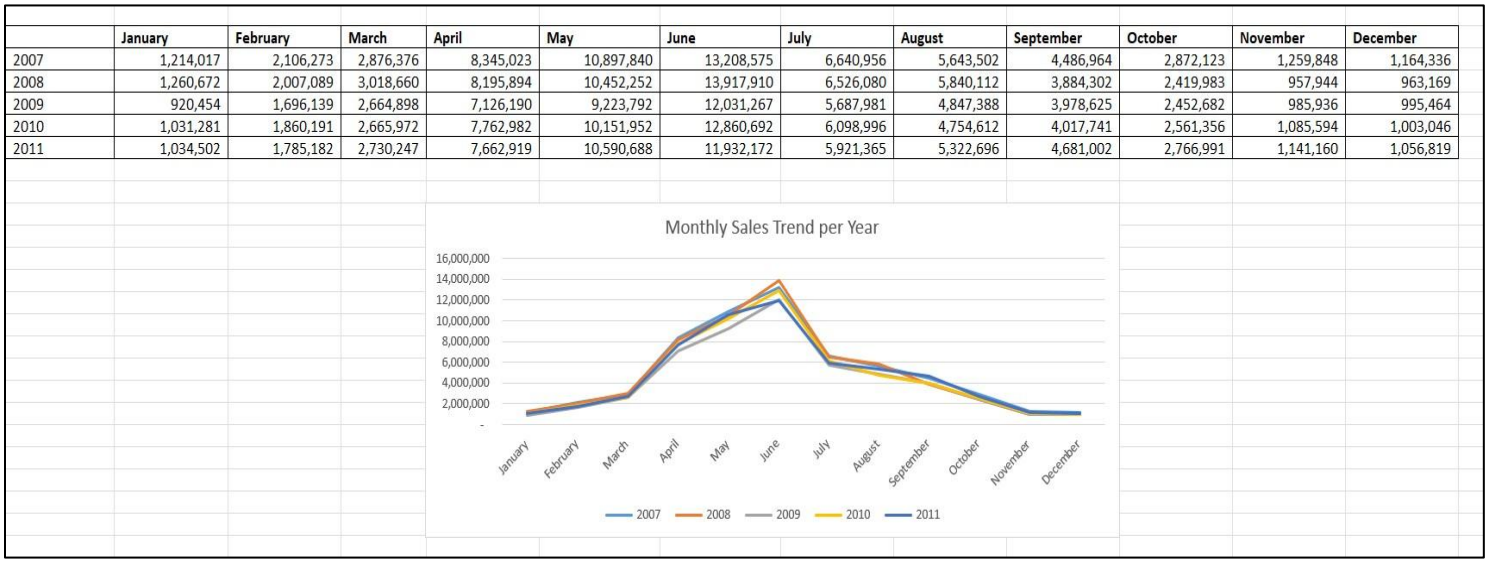
where each row shows a year from 2007 to 2011 and each column shows a month, ending with a Grand Total for each year as shown in (Fig. 1.12)



Sum of Revenue	Column Labels												
Row Labels	1	2	3	4	5	6	7	8	9	10	11	12	Grand Total
2007	$ 1,214,017	$2,106,273	$2,876,376	$8,345,023	$10,897,840	$13,208,575	$6,640,956	$5,643,502	$4,486,964	$2,872,123	$1,259,848	$1,164,336	$60,715,832
2008	$ 1,260,672	$2,007,089	$3,018,660	$8,195,894	$10,452,252	$13,917,910	$6,526,080	$5,840,112	$3,884,302	$2,419,983	$ 957,944	$ 963,169	$59,444,067
2009	$ 920,454	$1,696,139	$2,664,898	$7,126,190	$ 9,223,792	$12,031,267	$5,687,981	$4,847,388	$3,978,625	$2,452,682	$ 985,936	$ 995,464	$52,610,815
2010	$ 1,031,281	$1,860,191	$2,665,972	$7,762,982	$10,151,952	$12,860,692	$6,098,996	$4,754,612	$4,017,741	$2,561,356	$1,085,594	$1,003,046	$55,854,416
2011	$ 1,034,502	$1,785,182	$2,730,247	$7,662,919	$10,590,688	$11,932,172	$5,921,365	$5,322,696	$4,681,002	$2,766,991	$1,141,160	$1,056,819	$56,625,743

(Figure – 1.12)

The line chart shows a very consistent seasonal pattern where revenue climbs sharply from March and peaks every June. During this surge, sales hit between $11.9 million and $13.9 million. After the summer high, business declines, bottoming out in December and January as shown in (Fig. 1.13). This "revenue mountain" shape stayed the same even during the 2009 dip, proving the business is tied heavily to the cycling season.

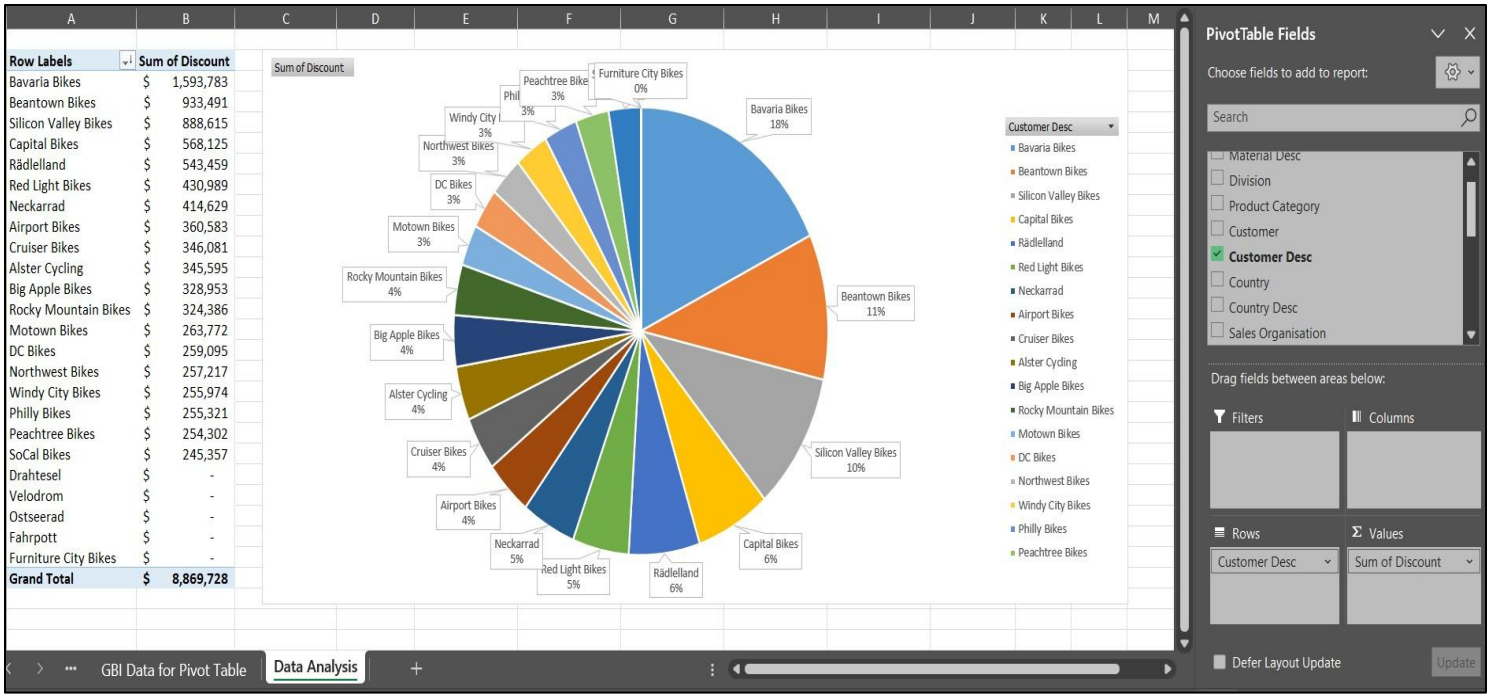


	January	February	March	April	May	June	July	August	September	October	November	December
2007	1,214,017	2,106,273	2,876,376	8,345,023	10,897,840	13,208,575	6,640,956	5,643,502	4,486,964	2,872,123	1,259,848	1,164,336
2008	1,260,672	2,007,089	3,018,660	8,195,894	10,452,252	13,917,910	6,526,080	5,840,112	3,884,302	2,419,983	957,944	963,169
2009	920,454	1,696,139	2,664,898	7,126,190	9,223,792	12,031,267	5,687,981	4,847,388	3,978,625	2,452,682	985,936	995,464
2010	1,031,281	1,860,191	2,665,972	7,762,982	10,151,952	12,860,692	6,098,996	4,754,612	4,017,741	2,561,356	1,085,594	1,003,046
2011	1,034,502	1,785,182	2,730,247	7,662,919	10,590,688	11,932,172	5,921,365	5,322,696	4,681,002	2,766,991	1,141,160	1,056,819

(Figure – 13)

1.3.6. How is the total discount distributed across different products?

To build the discount visual, the data was organized into a Pivot Table by dragging Customer Desc into the Rows area and Sum of Discount into the Values area. A Pie Chart was inserted to visualize how the total discount pool is distributed among various wholesalers. To make the chart more readable, data labels were added to show the percentage each customer receives, and a legend was included to help distinguish the different accounts at a glance.



(Figure – 1.14)

The pie chart above shows that nearly half of all discounts go to just a few big customers. Bavaria Bikes gets the biggest slice at 18%, followed by Beantown Bikes at 11% and Silicon Valley Bikes at 10%.

1.4 Critical Analysis of Using Excel Pivot Tables

From my experience working with this dataset, Microsoft Excel pivot tables are very effective for quickly exploring large amounts of transactional data. Since most business users are already familiar with Excel, it feels very accessible and easy to start working with. The drag-and-drop functionality of pivot tables made it simple to reorganize rows, columns, and values, allowing me to analyze revenue, sales quantity, discounts, and costs without needing to write complex formulas. However, I also noticed some limitations while working with pivot tables and visualizations. As the analysis became more detailed, it was easy to lose track of applied filters, which could lead to confusion or incorrect interpretation if not checked carefully. In addition, while Excel provides many charting options, turning pivot tables into clean and professional-looking visuals often requires a lot of manual effort. Overall, Excel pivot tables are extremely useful for initial basic exploration and everyday data analysis, especially for small to medium-sized datasets. They provide quick insights and flexibility with minimal setup. However, for more advanced analysis, more specialized analytical or visualization tools may be better suited.

1.5 Conclusion

Overall, Excel proved to be a practical and accessible tool for initial business data analysis. While it has some limitations when it comes to automation and advanced visual designing, pivot tables are highly effective for quickly summarizing data and supporting basic decision-making. This exercise demonstrated how even simple analytical techniques can generate meaningful insights when applied thoughtfully.

Chapter 2 Data Challenge with 'Sustainability' Data using Microsoft Excel for Cleansing and SAC for Visualization

2.1 Microsoft Excel vs SAP Analytics Cloud

I. The Power of Cleansing: Excel – Cleaning data in Excel is a deeply meaningful part of the data analysis process because it transforms a cluttered mess into a reliable dataset by allowing for a hands-on approach to cleaning out the junk, manually removing thousands of irrelevant rows and highlighting inconsistencies that would otherwise ruin the final analysis.

II. Storytelling via SAP Analytics Cloud – If Excel is the engine room, SAP Analytics Cloud (SAC) is the stage where the data finally tells its story. Instead of getting lost in endless rows of numbers, SAC uses visual cues to make the most important trends jump off the screen immediately.

2.2 Datasets and Suggested Questions

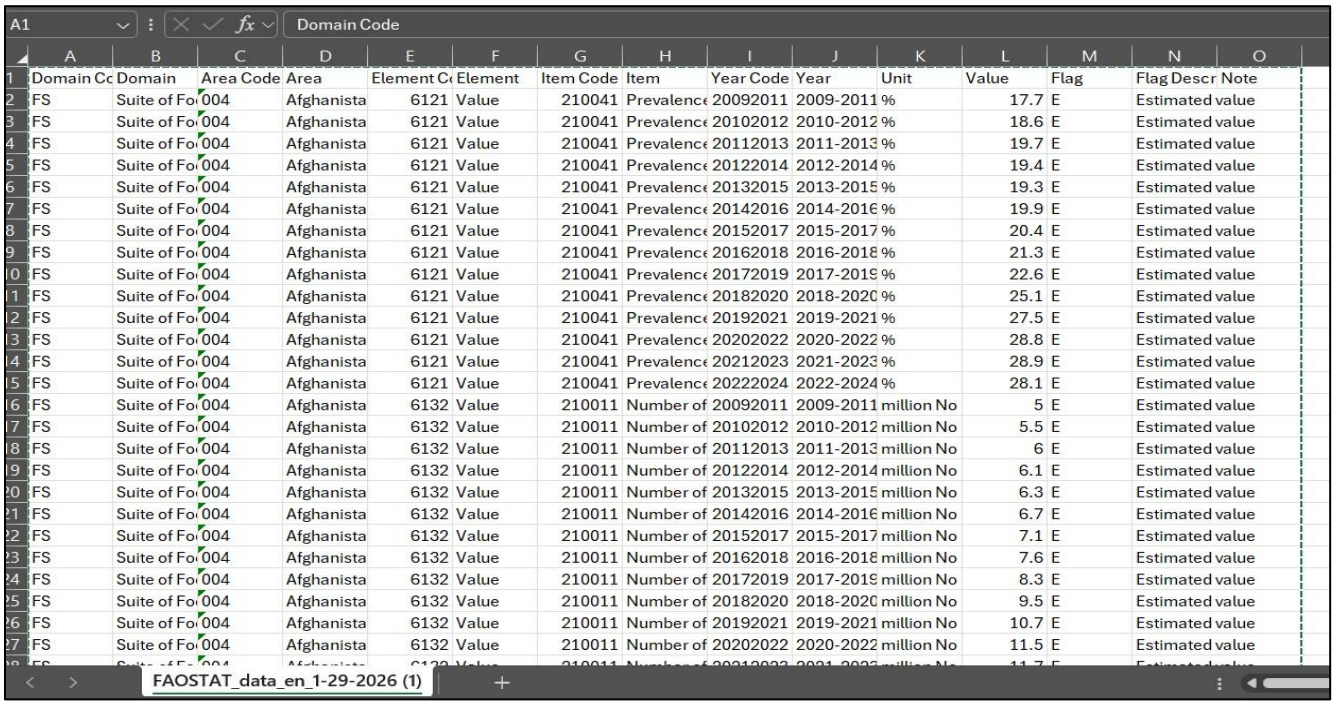
The analysis uses the FAOSTAT – Suite of Food Security Indicators dataset. Initially, the dataset included 27 indicators spanning 15 years. Data cleansing was carried out using Microsoft Excel to address missing values and inconsistencies. Following this process, 9 indicators with 7 years of reliable data were retained for analysis. On the other hand, The FAOSTAT – Emissions from Pre- and Post-Agricultural Production dataset was used to develop visualizations related to greenhouse gas emissions on SAP Analytics Cloud. The dataset contained the following variables: Area, Latitude, Longitude, Year, Emissions (CH4), Emissions (CO2), and Emissions (N2O). The data cleansing process for this dataset followed the same structured methodology applied to the Food Security Indicators dataset. This included ensuring year-wise consistency, removing irrelevant or incomplete records, identifying and handling missing values, and retaining only those variables with sufficient data coverage for accurate analysis. Below are the questions on which the cleansing was done and visualization were crafted:

- How were irrelevant and unwanted columns identified and removed to align the dataset with the analytical objectives?
- How were three-year average columns restructured into a consistent annual format, and how was the RIGHT() function in Microsoft Excel used?
- How were missing and blank data points identified maintain analytical accuracy and comparability across countries and years?
- How was the final dataset further cleaned and validated to ensure it was fully analysis-ready?
- How does the contribution of (CH4) emissions compare to CO2 and N2O emissions across countries?
- What is the emissions trend, and which specific pollutant shows the most significant steady increase over the ten-year period?
- For CO2 emissions from 2014 to 2023, which countries demonstrate the highest contribution to global emissions?
- What is the emissions trend for N2O emissions by area post-covid?

2.3 Application of Data Cleansing Techniques and Visualization Tool

2.3.1. How were irrelevant and unwanted columns identified and removed to align the dataset with the analytical objectives?

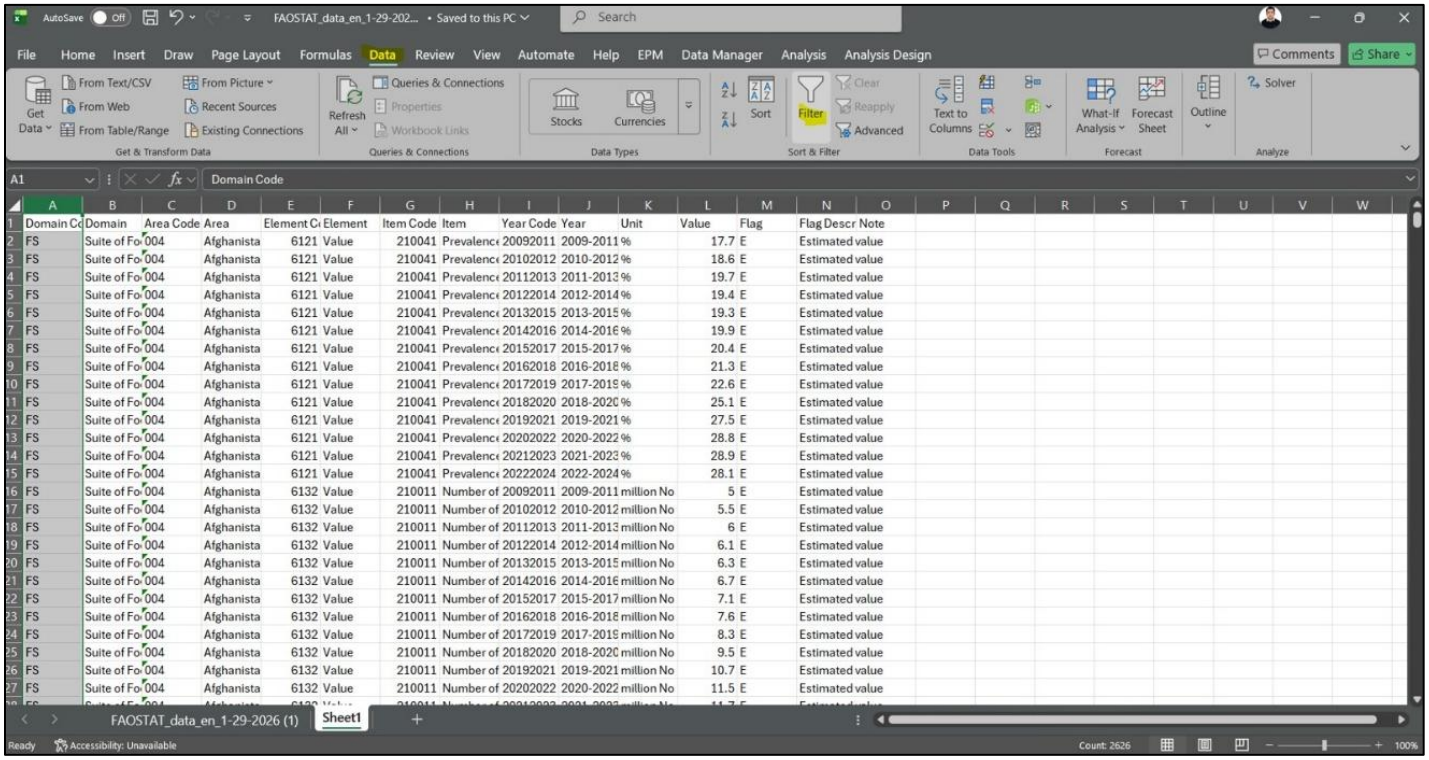
To get the FAOSTAT data ready for work, the first step was selecting the entire dataset to make sure not a single row, or entry was left behind (Fig. 2.1).



Domain Code	Domain	Area Code	Area	Element Code	Element	Item Code	Item	Year Code	Year	Unit	Value	Flag	Flag Description	Note
FS	Suite of Food	004	Afghanistan	6121	Value	210041	Prevalence	20092011	2009-2011	%	17.7	E	Estimated value	
FS	Suite of Food	004	Afghanistan	6121	Value	210041	Prevalence	20102012	2010-2012	%	18.6	E	Estimated value	
FS	Suite of Food	004	Afghanistan	6121	Value	210041	Prevalence	20112013	2011-2013	%	19.7	E	Estimated value	
FS	Suite of Food	004	Afghanistan	6121	Value	210041	Prevalence	20122014	2012-2014	%	19.4	E	Estimated value	
FS	Suite of Food	004	Afghanistan	6121	Value	210041	Prevalence	20132015	2013-2015	%	19.3	E	Estimated value	
FS	Suite of Food	004	Afghanistan	6121	Value	210041	Prevalence	20142016	2014-2016	%	19.9	E	Estimated value	
FS	Suite of Food	004	Afghanistan	6121	Value	210041	Prevalence	20152017	2015-2017	%	20.4	E	Estimated value	
FS	Suite of Food	004	Afghanistan	6121	Value	210041	Prevalence	20162018	2016-2018	%	21.3	E	Estimated value	
FS	Suite of Food	004	Afghanistan	6121	Value	210041	Prevalence	20172019	2017-2019	%	22.6	E	Estimated value	
FS	Suite of Food	004	Afghanistan	6121	Value	210041	Prevalence	20182020	2018-2020	%	25.1	E	Estimated value	
FS	Suite of Food	004	Afghanistan	6121	Value	210041	Prevalence	20192021	2019-2021	%	27.5	E	Estimated value	
FS	Suite of Food	004	Afghanistan	6121	Value	210041	Prevalence	20202022	2020-2022	%	28.8	E	Estimated value	
FS	Suite of Food	004	Afghanistan	6121	Value	210041	Prevalence	20212023	2021-2023	%	28.9	E	Estimated value	
FS	Suite of Food	004	Afghanistan	6121	Value	210041	Prevalence	20222024	2022-2024	%	28.1	E	Estimated value	
FS	Suite of Food	004	Afghanistan	6132	Value	210011	Number of	20092011	2009-2011	million No	5	E	Estimated value	
FS	Suite of Food	004	Afghanistan	6132	Value	210011	Number of	20102012	2010-2012	million No	5.5	E	Estimated value	
FS	Suite of Food	004	Afghanistan	6132	Value	210011	Number of	20112013	2011-2013	million No	6	E	Estimated value	
FS	Suite of Food	004	Afghanistan	6132	Value	210011	Number of	20122014	2012-2014	million No	6.1	E	Estimated value	
FS	Suite of Food	004	Afghanistan	6132	Value	210011	Number of	20132015	2013-2015	million No	6.3	E	Estimated value	
FS	Suite of Food	004	Afghanistan	6132	Value	210011	Number of	20142016	2014-2016	million No	6.7	E	Estimated value	
FS	Suite of Food	004	Afghanistan	6132	Value	210011	Number of	20152017	2015-2017	million No	7.1	E	Estimated value	
FS	Suite of Food	004	Afghanistan	6132	Value	210011	Number of	20162018	2016-2018	million No	7.6	E	Estimated value	
FS	Suite of Food	004	Afghanistan	6132	Value	210011	Number of	20172019	2017-2019	million No	8.3	E	Estimated value	
FS	Suite of Food	004	Afghanistan	6132	Value	210011	Number of	20182020	2018-2020	million No	9.5	E	Estimated value	
FS	Suite of Food	004	Afghanistan	6132	Value	210011	Number of	20192021	2019-2021	million No	10.7	E	Estimated value	
FS	Suite of Food	004	Afghanistan	6132	Value	210011	Number of	20202022	2020-2022	million No	11.5	E	Estimated value	

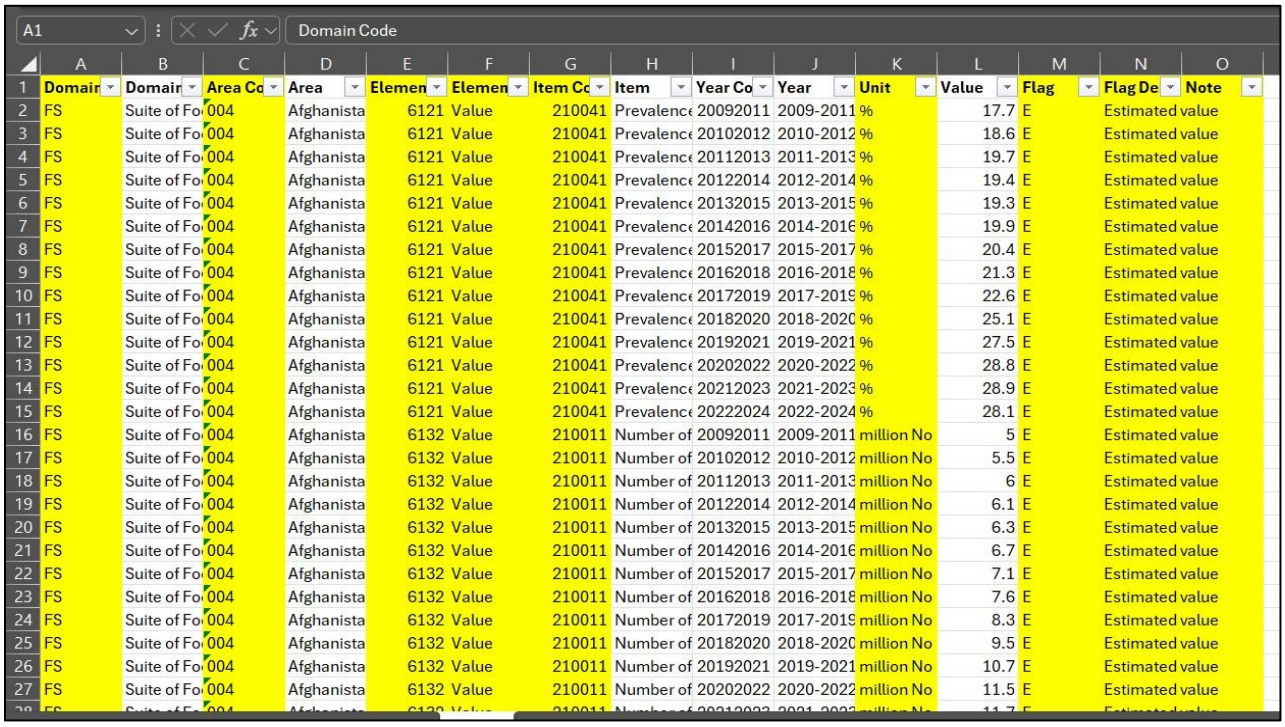
(Figure – 2.1)

Once everything was highlighted, the Filter tool was clicked for the header row, which allows quickly sorting through thousands of lines by country, year, or item type as shown in the (Fig. 2.2).



(Figure – 2.2)

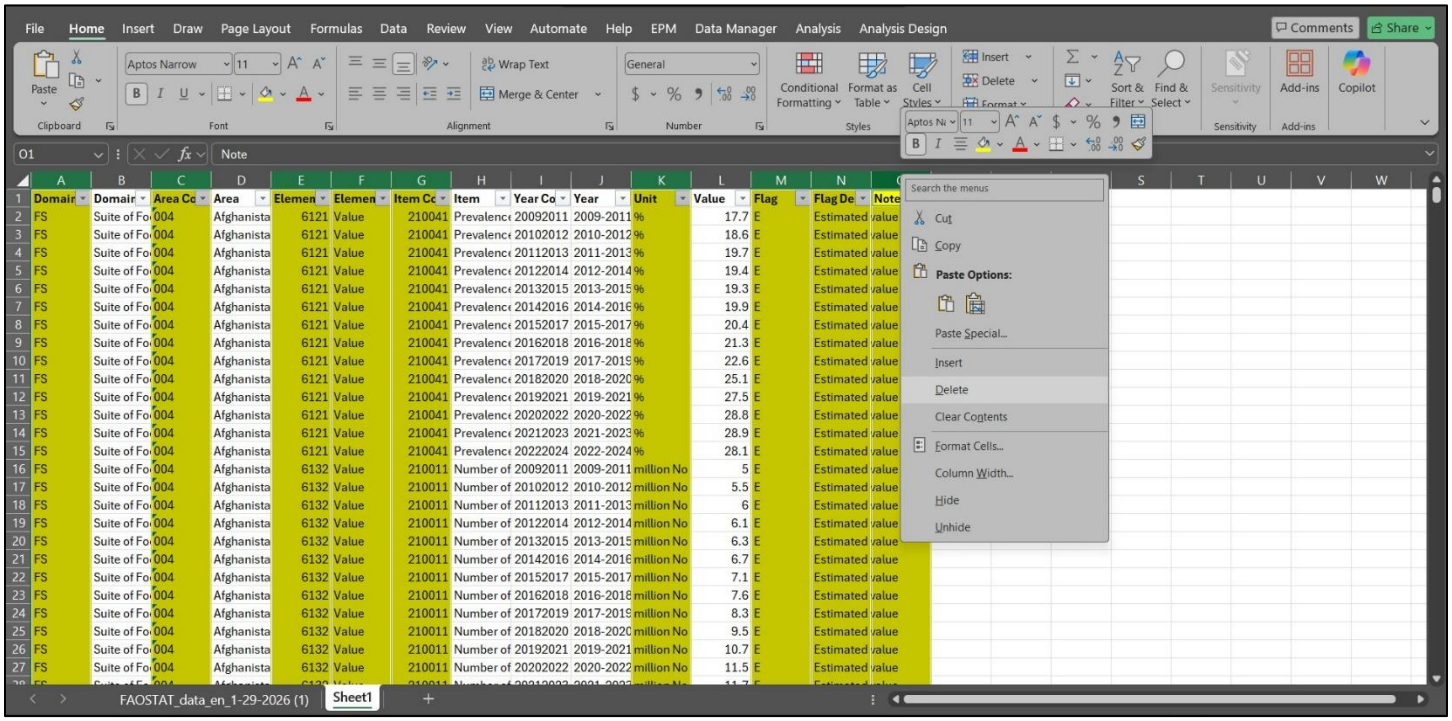
With the filters in place, the next task was to figure out what was worth keeping. There were several columns — like domain, area code, elements, and flags etc. that didn't add any real value to the analysis, so they were highlighted in bright yellow in the picture below.



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	Domain	Domain	Area Co	Area	Elemen	Elemen	Item Co	Item	Year Co	Year	Unit	Value	Flag	Flag De	Note
2	FS	Suite of Fo	004	Afghanista	6121	Value	210041	Prevalence	20092011	2009-2011	%	17.7	E	Estimated value	
3	FS	Suite of Fo	004	Afghanista	6121	Value	210041	Prevalence	20102012	2010-2012	%	18.6	E	Estimated value	
4	FS	Suite of Fo	004	Afghanista	6121	Value	210041	Prevalence	20112013	2011-2013	%	19.7	E	Estimated value	
5	FS	Suite of Fo	004	Afghanista	6121	Value	210041	Prevalence	20122014	2012-2014	%	19.4	E	Estimated value	
6	FS	Suite of Fo	004	Afghanista	6121	Value	210041	Prevalence	20132015	2013-2015	%	19.3	E	Estimated value	
7	FS	Suite of Fo	004	Afghanista	6121	Value	210041	Prevalence	20142016	2014-2016	%	19.9	E	Estimated value	
8	FS	Suite of Fo	004	Afghanista	6121	Value	210041	Prevalence	20152017	2015-2017	%	20.4	E	Estimated value	
9	FS	Suite of Fo	004	Afghanista	6121	Value	210041	Prevalence	20162018	2016-2018	%	21.3	E	Estimated value	
10	FS	Suite of Fo	004	Afghanista	6121	Value	210041	Prevalence	20172019	2017-2019	%	22.6	E	Estimated value	
11	FS	Suite of Fo	004	Afghanista	6121	Value	210041	Prevalence	20182020	2018-2020	%	25.1	E	Estimated value	
12	FS	Suite of Fo	004	Afghanista	6121	Value	210041	Prevalence	20192021	2019-2021	%	27.5	E	Estimated value	
13	FS	Suite of Fo	004	Afghanista	6121	Value	210041	Prevalence	20202022	2020-2022	%	28.8	E	Estimated value	
14	FS	Suite of Fo	004	Afghanista	6121	Value	210041	Prevalence	20212023	2021-2023	%	28.9	E	Estimated value	
15	FS	Suite of Fo	004	Afghanista	6121	Value	210041	Prevalence	20222024	2022-2024	%	28.1	E	Estimated value	
16	FS	Suite of Fo	004	Afghanista	6132	Value	210011	Number of	20092011	2009-2011	million No	5	E	Estimated value	
17	FS	Suite of Fo	004	Afghanista	6132	Value	210011	Number of	20102012	2010-2012	million No	5.5	E	Estimated value	
18	FS	Suite of Fo	004	Afghanista	6132	Value	210011	Number of	20112013	2011-2013	million No	6	E	Estimated value	
19	FS	Suite of Fo	004	Afghanista	6132	Value	210011	Number of	20122014	2012-2014	million No	6.1	E	Estimated value	
20	FS	Suite of Fo	004	Afghanista	6132	Value	210011	Number of	20132015	2013-2015	million No	6.3	E	Estimated value	
21	FS	Suite of Fo	004	Afghanista	6132	Value	210011	Number of	20142016	2014-2016	million No	6.7	E	Estimated value	
22	FS	Suite of Fo	004	Afghanista	6132	Value	210011	Number of	20152017	2015-2017	million No	7.1	E	Estimated value	
23	FS	Suite of Fo	004	Afghanista	6132	Value	210011	Number of	20162018	2016-2018	million No	7.6	E	Estimated value	
24	FS	Suite of Fo	004	Afghanista	6132	Value	210011	Number of	20172019	2017-2019	million No	8.3	E	Estimated value	
25	FS	Suite of Fo	004	Afghanista	6132	Value	210011	Number of	20182020	2018-2020	million No	9.5	E	Estimated value	
26	FS	Suite of Fo	004	Afghanista	6132	Value	210011	Number of	20192021	2019-2021	million No	10.7	E	Estimated value	
27	FS	Suite of Fo	004	Afghanista	6132	Value	210011	Number of	20202022	2020-2022	million No	11.5	E	Estimated value	

(Figure – 2.3)

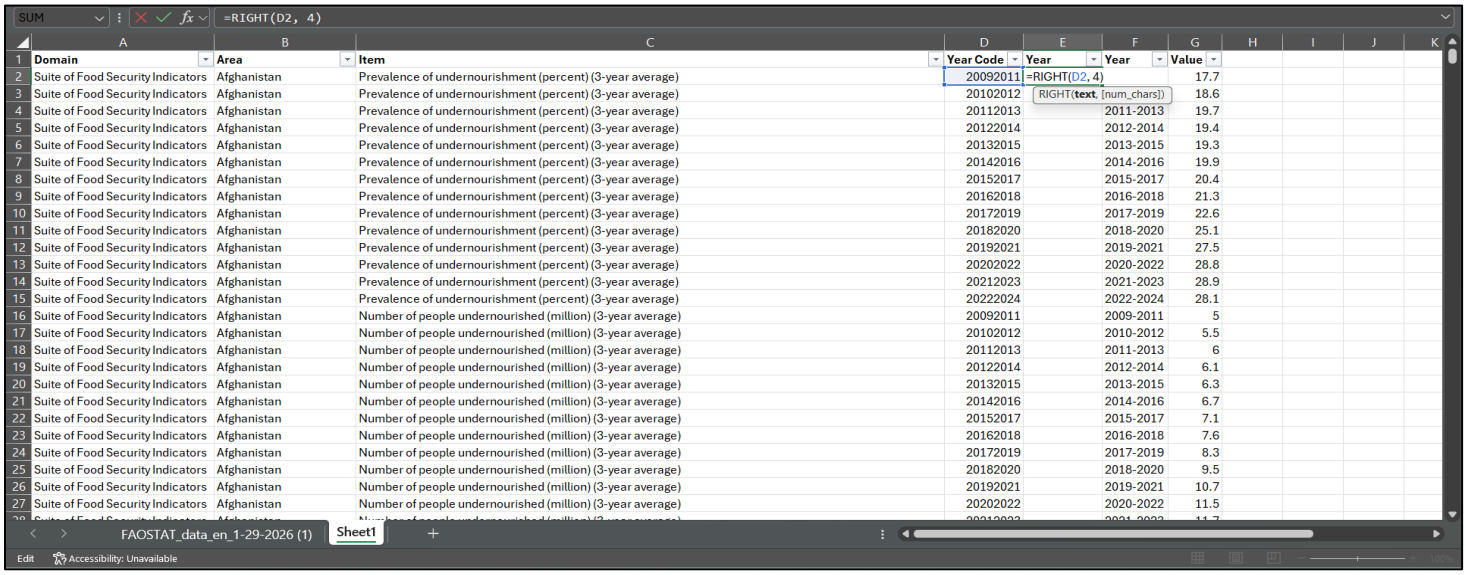
After selecting the entire dataset and applying filters to manage the information, the highlighted columns in yellow were deleted to leave behind a clean and focused spreadsheet as shown in (Fig. 2.4)



(Figure – 2.4)

2.3.2. How were three-year average columns restructured into a consistent annual format, and how was the RIGHT() function in Microsoft Excel used?

To extract year values, a new column named Year (Column E) was inserted next to Column D to extract absolute year values to ensure consistency. Within this column, the formula = RIGHT(D2, 4) was applied to extract the final four digits from the "Year Code". As shown in the visual (Fig. 19), this function converted range formats like "20092011" into a singular "2011" entry.

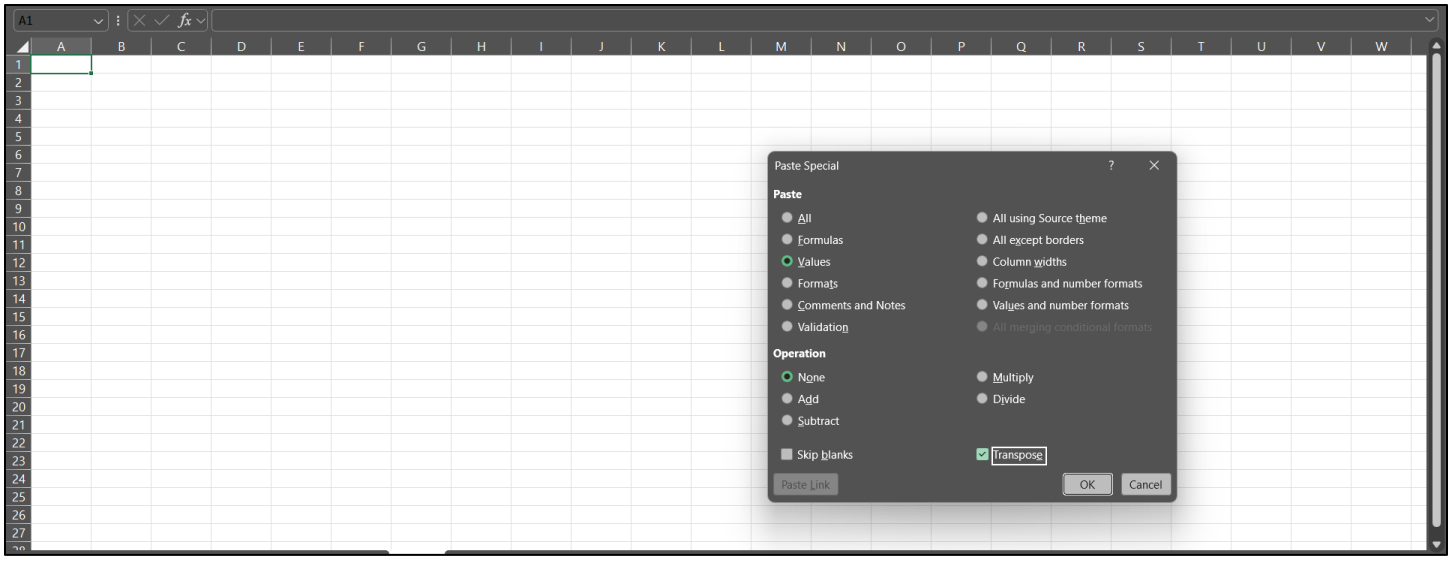


Domain	Area	Item	Year Code	Year	Value
Suite of Food Security Indicators	Afghanistan	Prevalence of undernourishment (percent) (3-year average)	20092011	2009-2011	17.7
Suite of Food Security Indicators	Afghanistan	Prevalence of undernourishment (percent) (3-year average)	20102012	2010-2012	18.6
Suite of Food Security Indicators	Afghanistan	Prevalence of undernourishment (percent) (3-year average)	20112013	2011-2013	19.7
Suite of Food Security Indicators	Afghanistan	Prevalence of undernourishment (percent) (3-year average)	20122014	2012-2014	19.4
Suite of Food Security Indicators	Afghanistan	Prevalence of undernourishment (percent) (3-year average)	20132015	2013-2015	19.3
Suite of Food Security Indicators	Afghanistan	Prevalence of undernourishment (percent) (3-year average)	20142016	2014-2016	19.9
Suite of Food Security Indicators	Afghanistan	Prevalence of undernourishment (percent) (3-year average)	20152017	2015-2017	20.4
Suite of Food Security Indicators	Afghanistan	Prevalence of undernourishment (percent) (3-year average)	20162018	2016-2018	21.3
Suite of Food Security Indicators	Afghanistan	Prevalence of undernourishment (percent) (3-year average)	20172019	2017-2019	22.6
Suite of Food Security Indicators	Afghanistan	Prevalence of undernourishment (percent) (3-year average)	20182020	2018-2020	25.1
Suite of Food Security Indicators	Afghanistan	Prevalence of undernourishment (percent) (3-year average)	20192021	2019-2021	27.5
Suite of Food Security Indicators	Afghanistan	Prevalence of undernourishment (percent) (3-year average)	20202022	2020-2022	28.8
Suite of Food Security Indicators	Afghanistan	Prevalence of undernourishment (percent) (3-year average)	20212023	2021-2023	28.9
Suite of Food Security Indicators	Afghanistan	Prevalence of undernourishment (percent) (3-year average)	20222024	2022-2024	28.1
Suite of Food Security Indicators	Afghanistan	Number of people undernourished (million) (3-year average)	20092011	2009-2011	5
Suite of Food Security Indicators	Afghanistan	Number of people undernourished (million) (3-year average)	20102012	2010-2012	5.5
Suite of Food Security Indicators	Afghanistan	Number of people undernourished (million) (3-year average)	20112013	2011-2013	6
Suite of Food Security Indicators	Afghanistan	Number of people undernourished (million) (3-year average)	20122014	2012-2014	6.1
Suite of Food Security Indicators	Afghanistan	Number of people undernourished (million) (3-year average)	20132015	2013-2015	6.3
Suite of Food Security Indicators	Afghanistan	Number of people undernourished (million) (3-year average)	20142016	2014-2016	6.7
Suite of Food Security Indicators	Afghanistan	Number of people undernourished (million) (3-year average)	20152017	2015-2017	7.1
Suite of Food Security Indicators	Afghanistan	Number of people undernourished (million) (3-year average)	20162018	2016-2018	7.6
Suite of Food Security Indicators	Afghanistan	Number of people undernourished (million) (3-year average)	20172019	2017-2019	8.3
Suite of Food Security Indicators	Afghanistan	Number of people undernourished (million) (3-year average)	20182020	2018-2020	9.5
Suite of Food Security Indicators	Afghanistan	Number of people undernourished (million) (3-year average)	20192021	2019-2021	10.7
Suite of Food Security Indicators	Afghanistan	Number of people undernourished (million) (3-year average)	20202022	2020-2022	11.5

(Figure – 2.5)

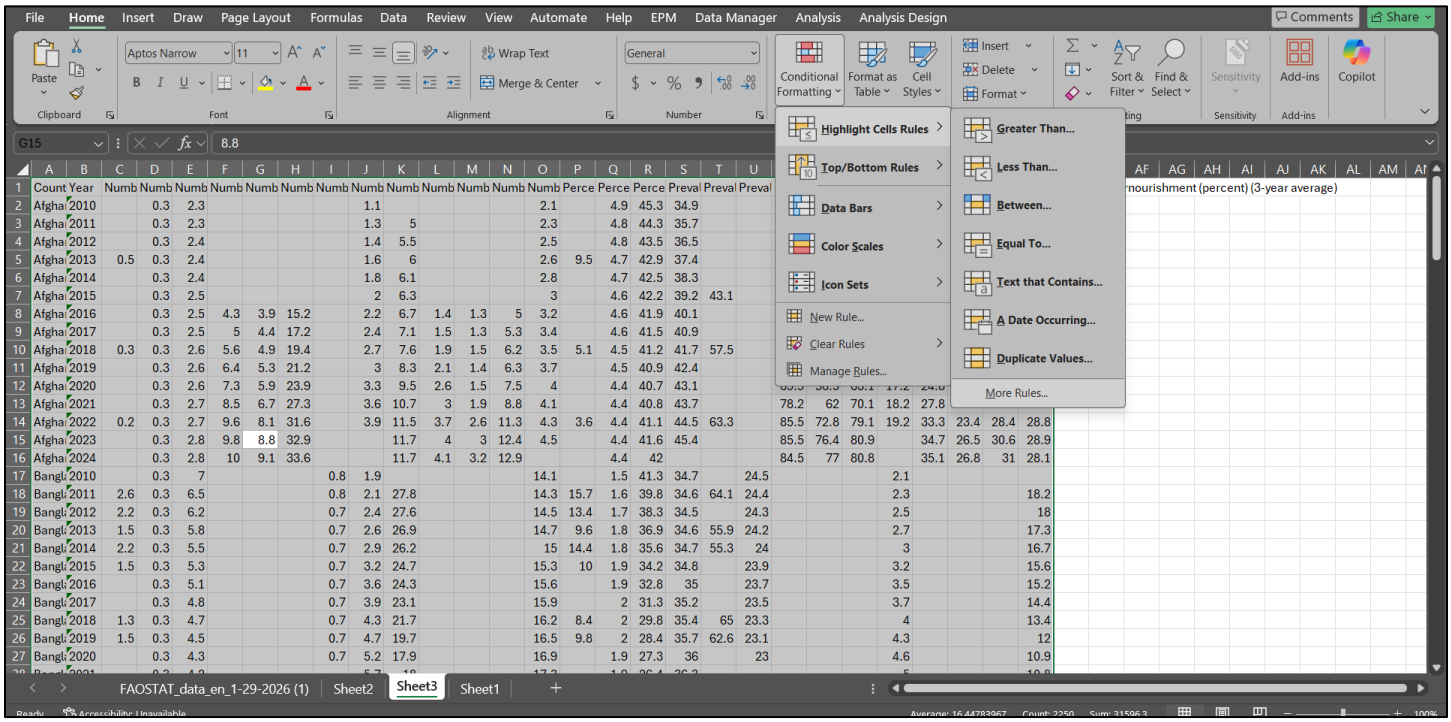
2.3.3. How were missing and blank data points identified maintain analytical accuracy and comparability across countries and years?

After ensuring consistency across all selected columns, a new worksheet was created. The cleaned data was copied and pasted as values using the TRANSPOSE function to restructure the dataset and align columns and years in a consistent, analysis-ready format, as shown in (Figure 2.6).

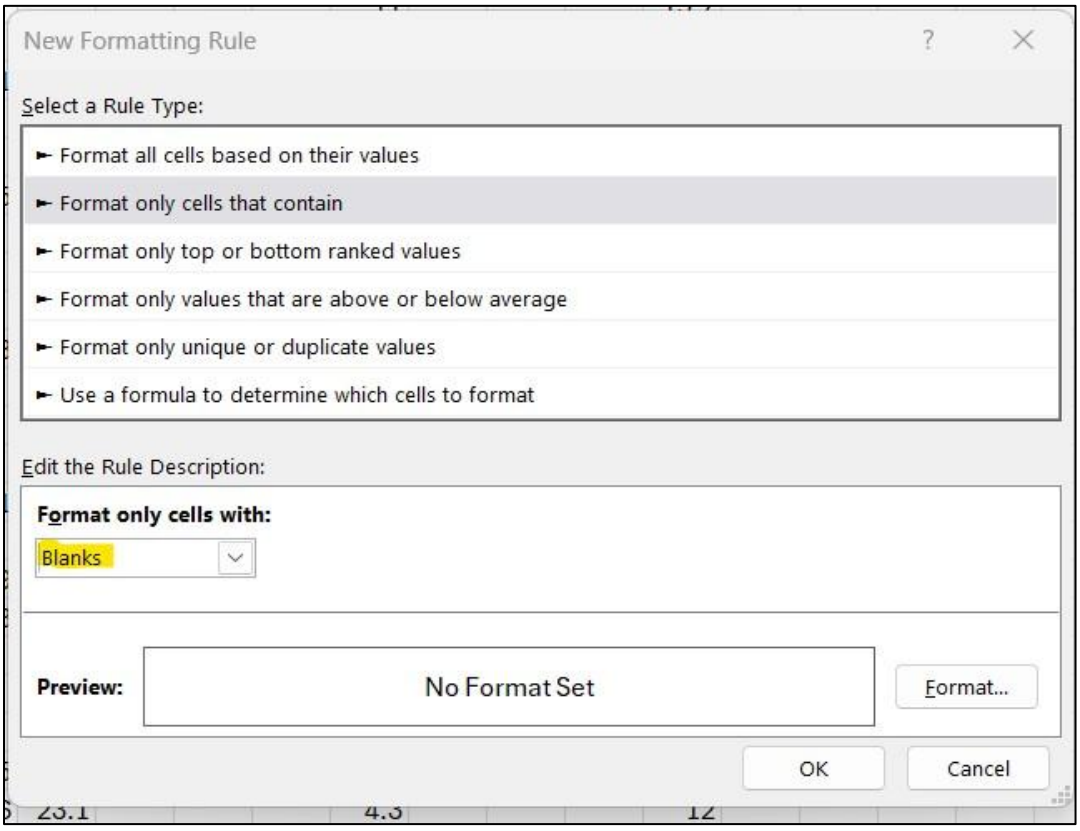


(Figure – 2.6)

Once the data was consolidated, a substantial number of null and blank values were observed across multiple columns. To identify these gaps, the dataset was copied and the Conditional Formatting feature was applied to highlight missing values. Blank cells were visually flagged in yellow, enabling clear identification and assessment of data completeness, as illustrated in Figures 2.7 and 2.8.



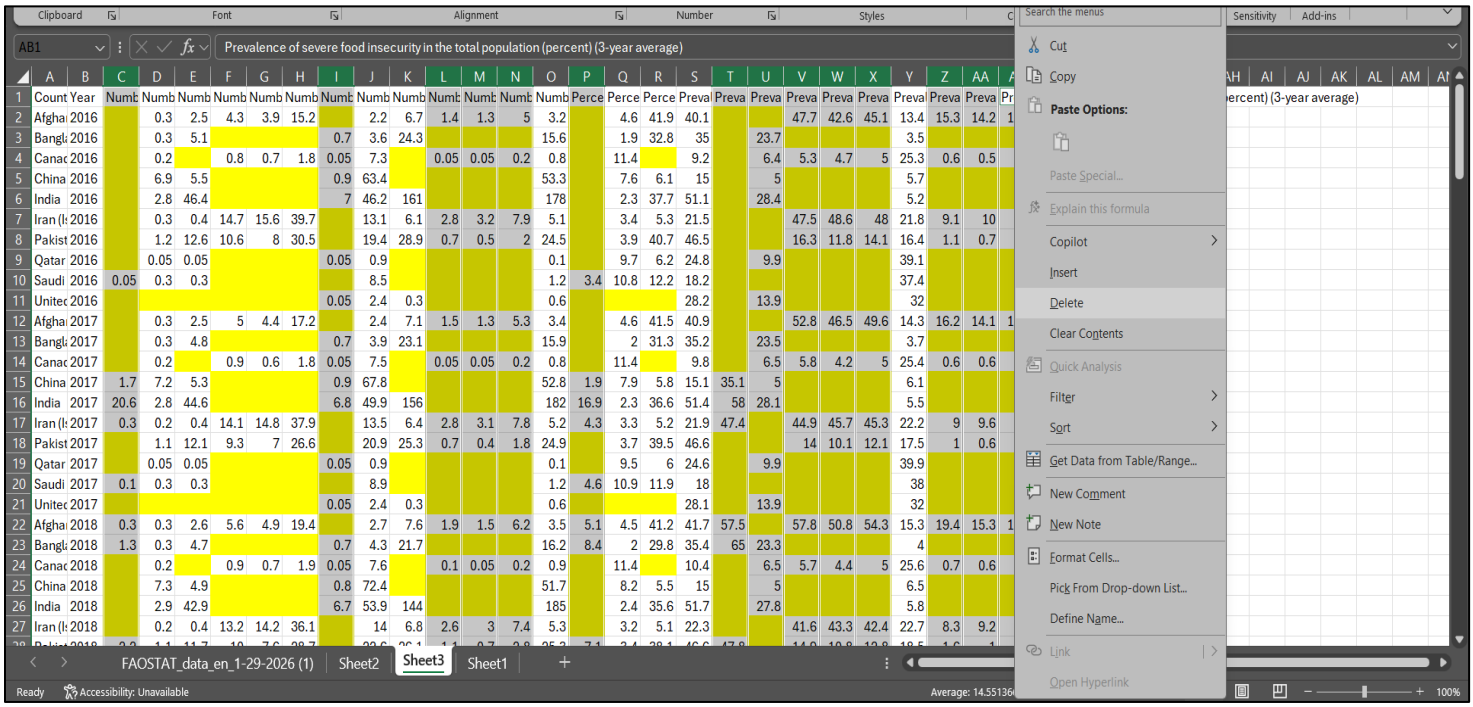
(Figure – 2.7)



(Figure – 2.8)

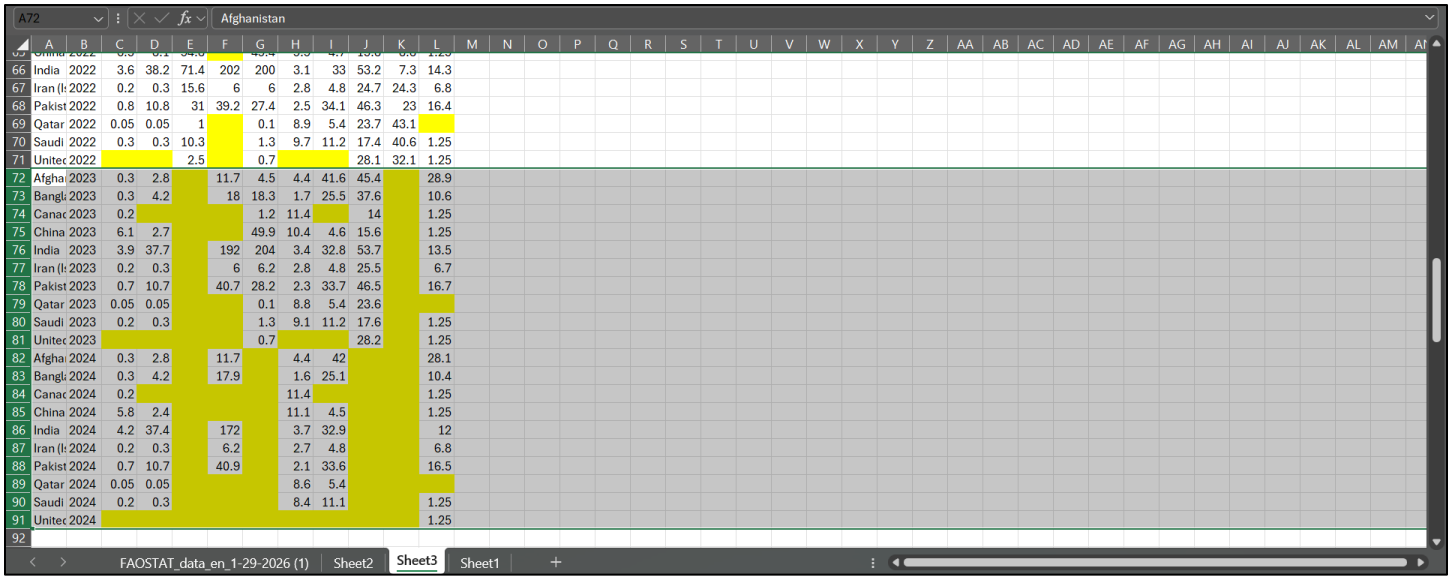
2.3.4. How was the final dataset further cleaned and validated to ensure it was fully analysis-ready?

To finalize the data cleaning process, the dataset was first sorted on a year basis to ensure proper temporal alignment across all columns. Following this, columns were reviewed to assess data completeness. Columns containing a high proportion of null or missing values were identified and selected for removal as shown in (Fig. 2.9) below.



(Figure – 2.9)

These columns were then deleted in a stepwise manner, retaining only those columns with sufficient and consistent data coverage across countries and years. This remaining iterative cleaning process is illustrated in (Figure 2.11) below

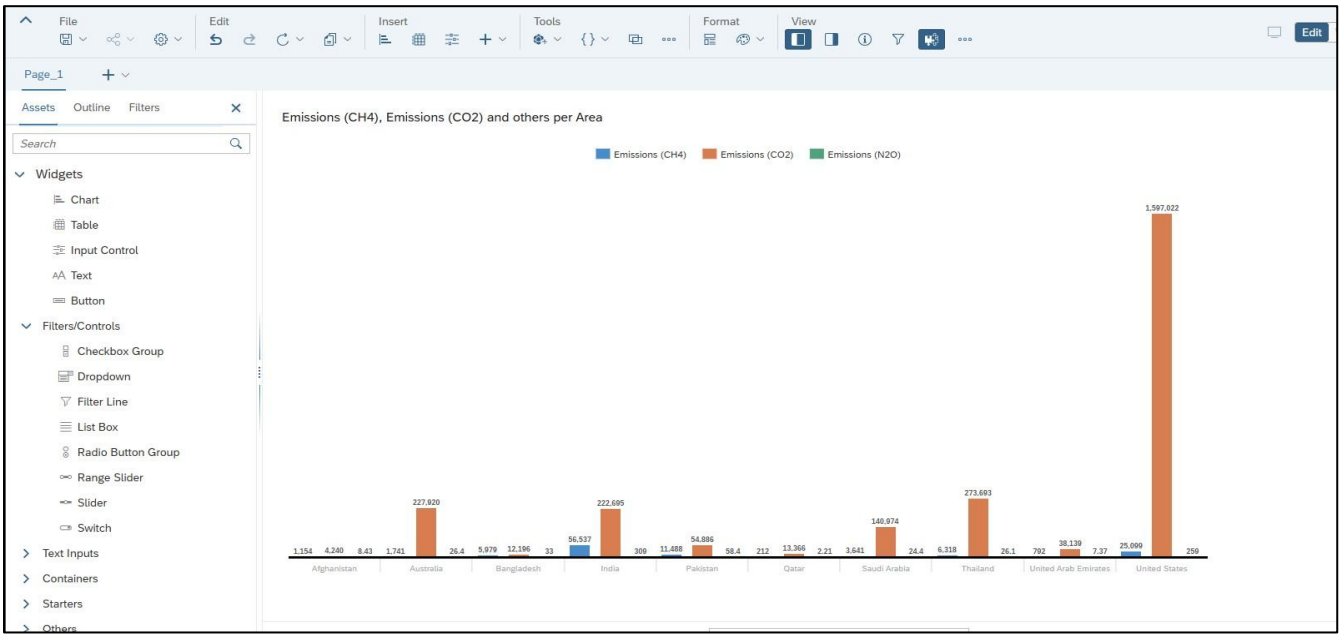


(Figure – 2.11)

2.3.5. How does the contribution of methane (CH4) emissions compare to CO2 and N2O emissions across countries?

(CH4) emissions are significantly lower than CO2 emissions across all observed areas but remain noticeably higher than (N2O) emissions. Countries with large agricultural sectors such as India, Pakistan, and the United States show relatively

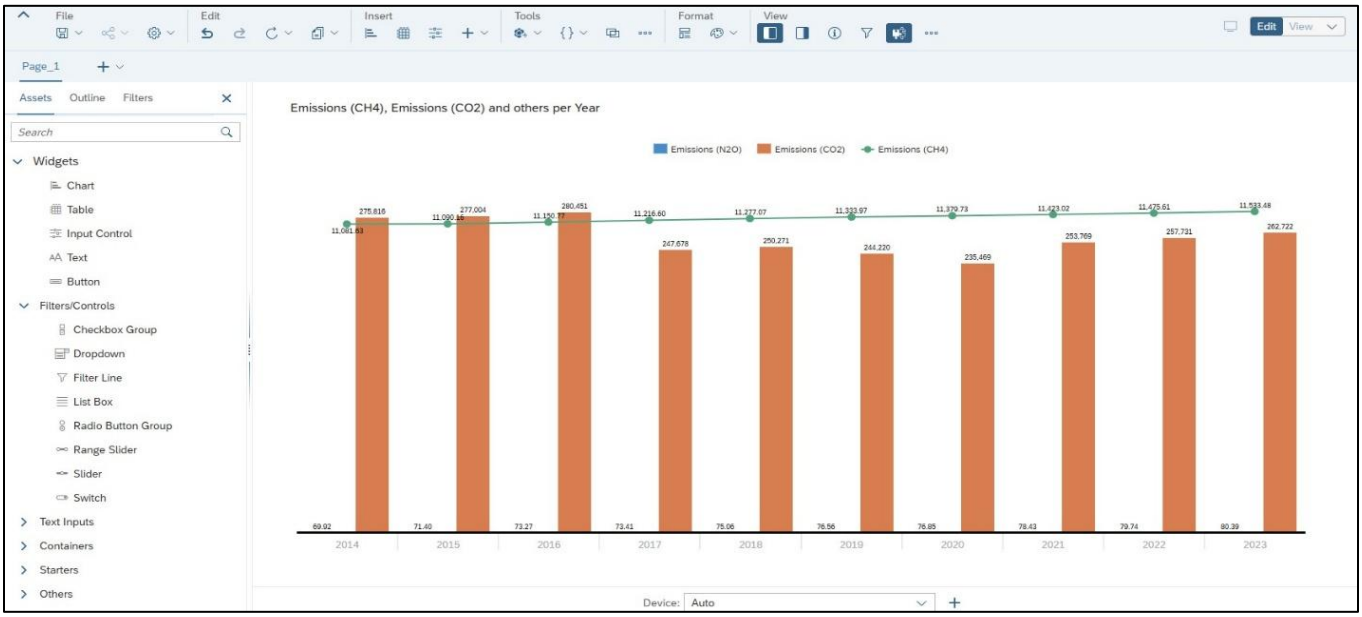
higher CH4 emissions, suggesting the influence of livestock production and rice cultivation. However, CH4 does not exceed CO2 in any country, indicating that CO2 remains the primary emission source in (Fig. 2.12) below.



(Figure – 2.12)

2.3.6. What is the emissions trend, and which specific pollutant shows the most significant steady increase over the ten-year period?

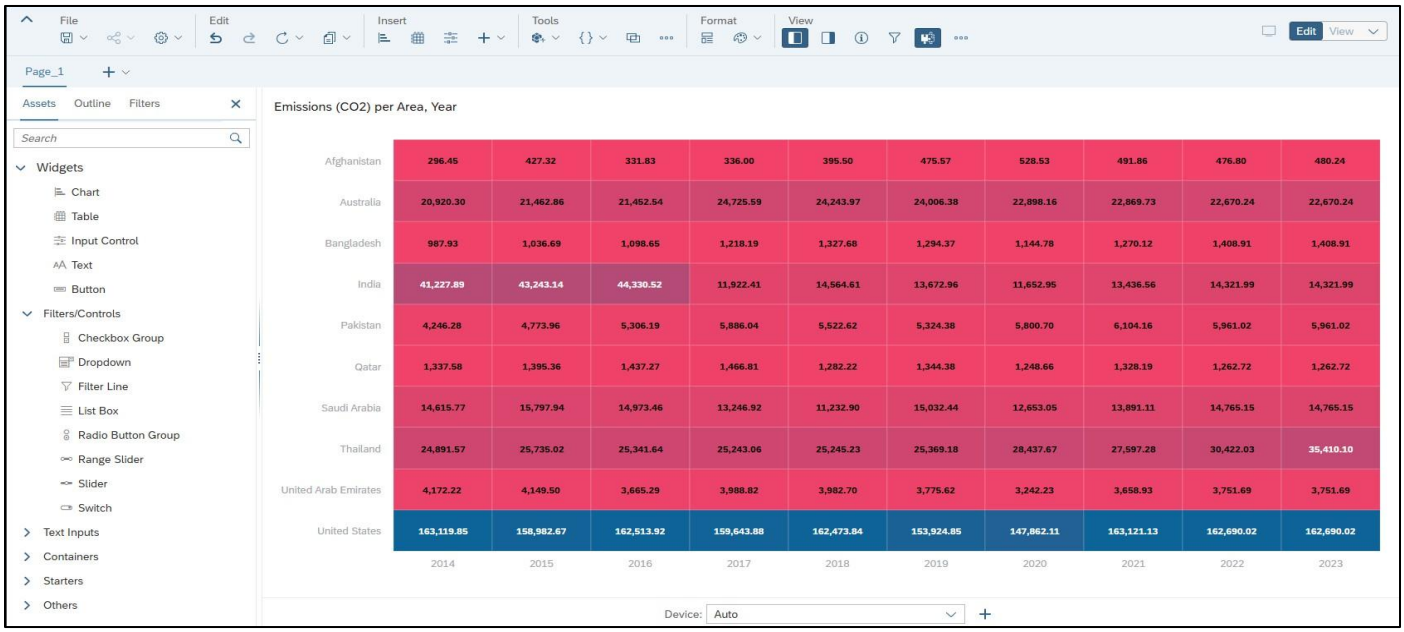
Figure 2.13 highlights a mix of steady growth and unexpected dips. CH4 is the most consistent; it has climbed every single year without fail, starting at 11,081.63 and hitting 11,533.48 by 2023. On the other hand, CO2 levels are much worse. While it hit a massive peak of 280,451 back in 2016, it actually saw a significant drop over the next few years, bottoming out around 2020 before starting to climb again recently. Finally, while N2O exists in much smaller volumes, it has quietly grown from 69.92 to over 80 in the same period, following the general global trend of rising emissions.



(Figure – 2.13)

2.3.7. For CO2 emissions from 2014 to 2023, which countries demonstrate the highest contribution to global emissions?

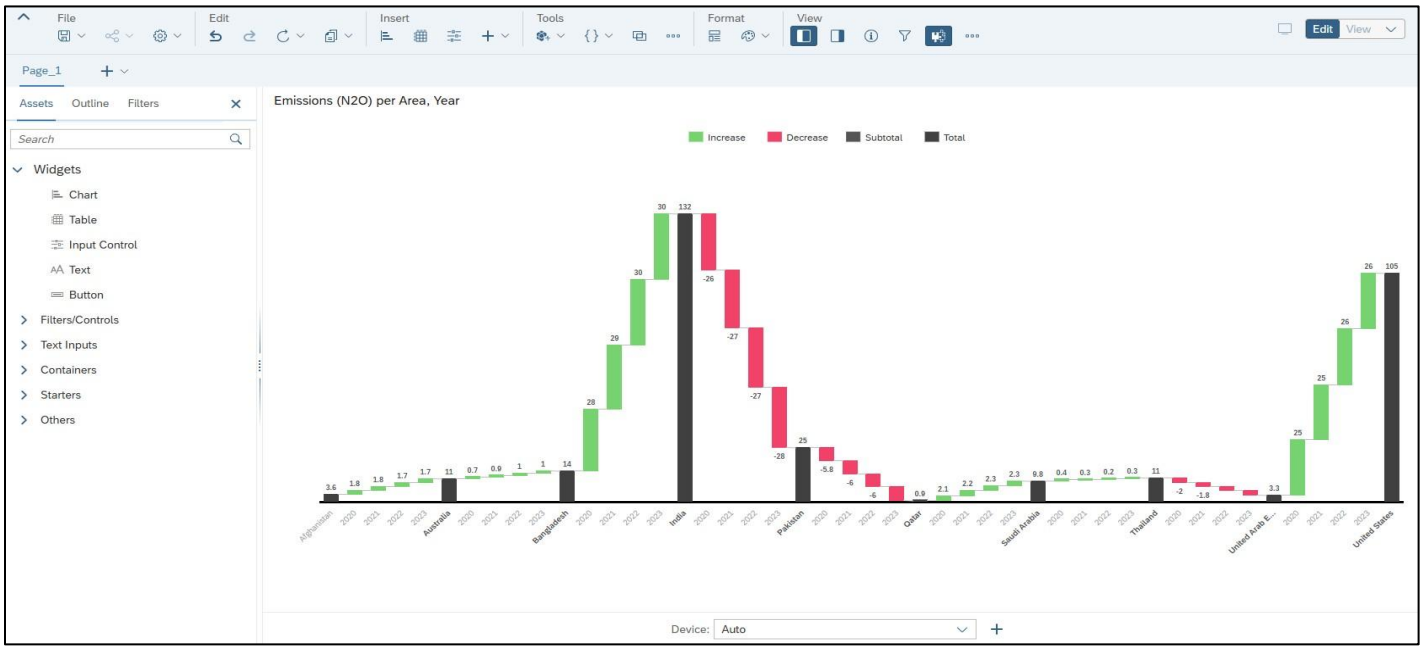
The heatmap clearly in the figure 27 shows the US as the top emitter, with values consistently high between 162,000 and 163,000. India experienced a major change, dropping from over 40,000 to the 11,000 – 14,000 range after 2016, which drastically lightened its color on the map. Meanwhile, Thailand's emissions climbed from 24,891 to over 35,410, showing steady industrial growth.



(Figure – 2.14)

2.3.8. What is the emissions trend for N2O emissions by area post-covid?

Post-COVID, the waterfall chart describing N2O trends show a sharp divide between rising and falling emitters. India and the United States saw the biggest surges, reaching totals of 132 and 105, respectively, by 2023. Bangladesh also trended upward during this period. Conversely, Pakistan and Qatar managed to reduce their emissions, marked by consistent yearly declines. Australia and Thailand stayed almost entirely flat with minimal change to their footprints as shown in (Fig. 2.15)



(Figure – 2.15)

2.4 Critical Analysis: Excel vs. SAP Analytics Cloud

Excel was highly effective for hands-on data cleaning tasks, enabling quick filtering of irrelevant rows and the use of conditional formatting to identify missing or blank values. Functions such as RIGHT() were critical for transforming inconsistent three-year average strings into clean, annual date values. However, Excel requires caution, as manual operations like deleting highlighted columns can introduce errors if not carefully managed. Once cleaned, the data was transferred to SAP Analytics Cloud (SAC), shifting the focus from preparation to analysis. While Excel supports detailed tabular work, SAC excels at pattern discovery. For instance, the Heat Map clearly exposed the emission gap between the United States and other countries, and the Waterfall Chart effectively highlighted country-level contributions to the Post-Covid N2O increase insights that are difficult to extract from spreadsheets alone.

2.5 Conclusion

To sum up, Excel is where the messy work happens, while SAP Analytics Cloud is where the results actually come to life. Excel is the perfect tool for getting hands-on with the data cleaning out the junk, highlighting what wasn't needed, and fixing broken dates so everything lined up correctly. It took a confusing pile of information and turned it into a solid, organized base. The real discoveries, though, happened in SAC. Instead of staring at endless rows of numbers, visuals like the Heat Map and Waterfall Chart made it obvious where the biggest problems were, like the sudden shifts in India's data or the rising trends in the US. In short, I believe Excel makes the data clean, but SAC makes the data make sense.

Chapter 3 SAP Analytics Cloud

3.1 Introduction of SAP Analytics Cloud

SAP Analytics Cloud (SAC) is a cloud-based analytics tool that helps organizations analyze data and present it in a clear visual format. Instead of working through large tables of numbers, users can create charts, dashboards, and interactive reports that make patterns and trends easier to understand. One of the main advantages of SAP Analytics Cloud is that it brings data visualization, analysis, and business planning together in one platform. It can connect to different data sources and transform raw datasets into visual stories that highlight important insights. Because of this, SAC is widely used in businesses for performance tracking, reporting, and supporting data-driven decisions.

3.2 Dataset and Research Questions

This section presents the analysis conducted using SAP Analytics Cloud based on the Global Bike Inc. wholesale dataset. The dataset consists of approximately 29,000 transactional records, covering multiple years of wholesale sales activity from 2011 to 2023. Key data fields used in the analysis include Calendar Year and Month, Customer Description, Country, Region, Sales Quantity, Revenue, Net Sales, Cost of Goods, and Discount. Each research question is addressed using a dedicated pivot table and supported by a corresponding visualization, allowing patterns and comparisons to be identified clearly below:

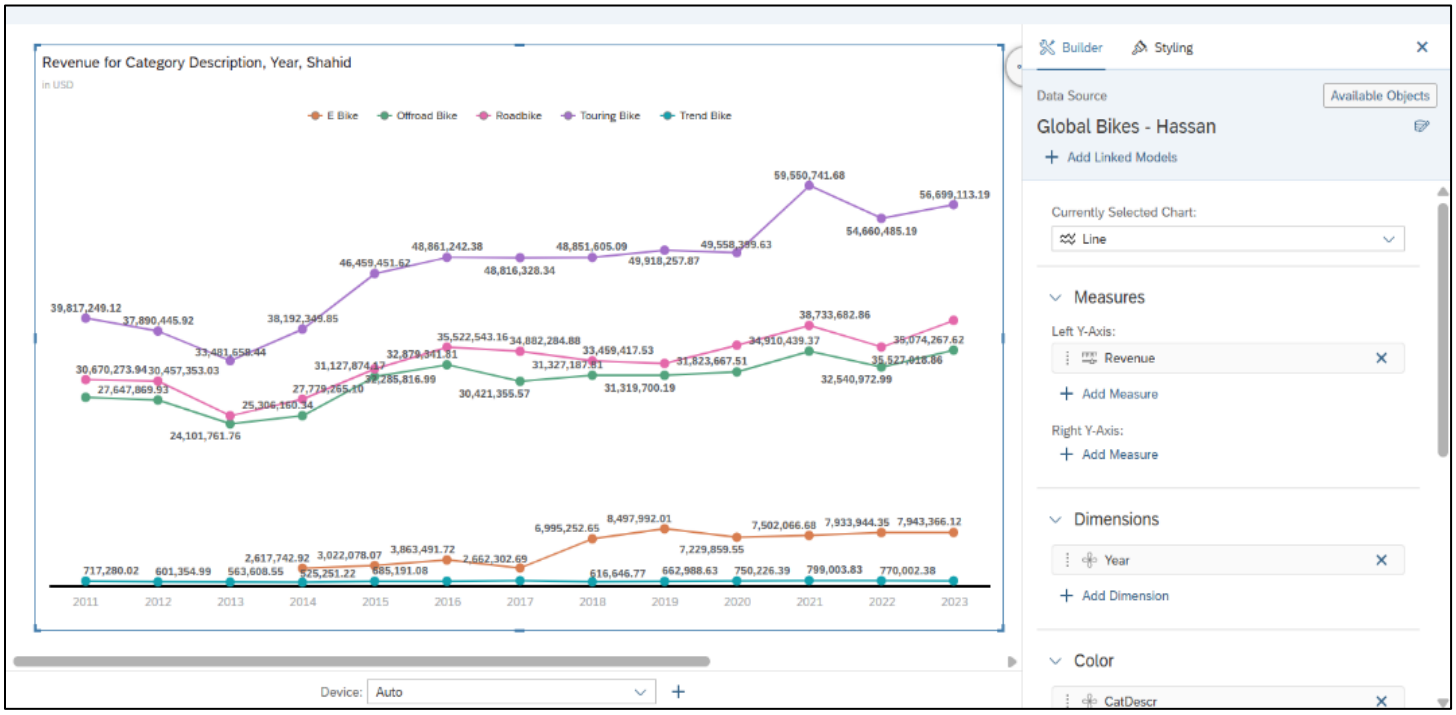
- Which product category (E-Bike, Offroad, Road, Touring, or Trend) has seen the most significant growth in revenue share between 2011 and 2023?

- Within the United States, which specific locations (e.g., Silicon Valley Bikes vs. Peachtree Bikes) are the top three contributors to sales volume?
- Which customers maintain the highest Gross Margin Ratio, and is there a negative correlation between high sales volume and high margin percentage?
- Based on the revenue trends from 2011 to 2023, what is the projected revenue forecast for the first half of 2024, and what is the degree of uncertainty (confidence bound) for those months?
- How do high-volume, low-profit products impact the Sales Organisation efficiency?
- What are the primary drivers of revenue leakage across Global Bike's operations, and how do discounting strategies impact the profit efficiency of its largest customers?

3.3 Deriving Business Insights and Findings

3.3.1. Which product category (E-Bike, Offroad, Road, Touring, or Trend) has seen the most significant growth in revenue share between 2011 and 2023?

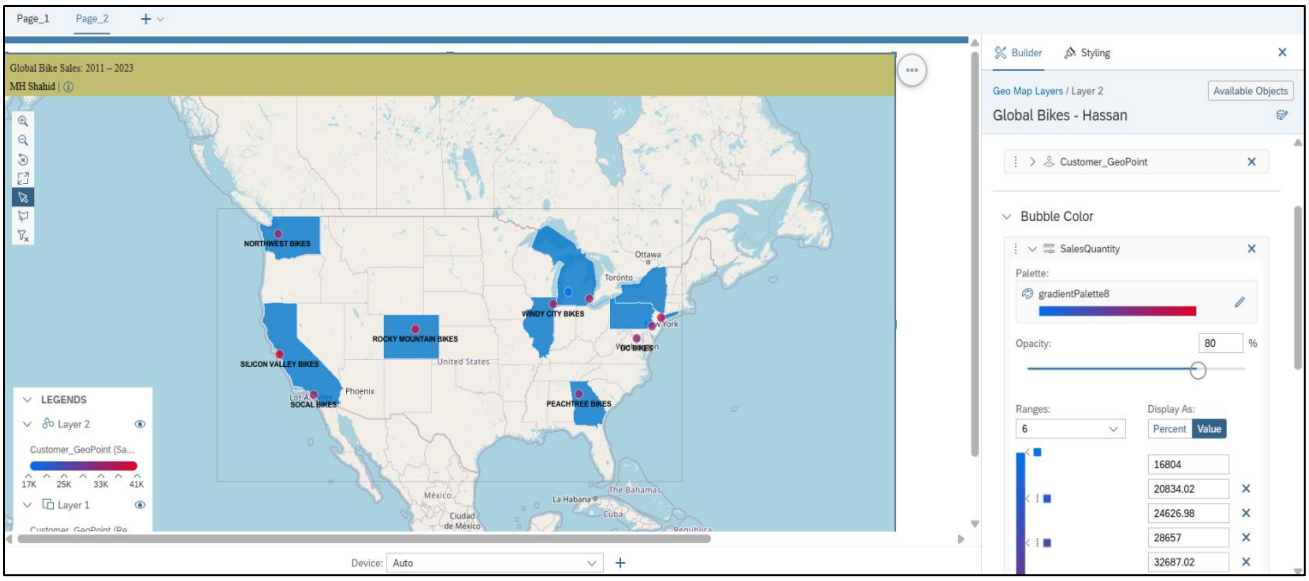
To capture the long-term trends of our product lineup, I built a line chart within the SAP Analytics Cloud. I placed "Year" on the horizontal X-axis and "Revenue" on the vertical Y-axis to track our financial performance over the 2011–2023 period. The "secret sauce" that really brought the data to life was dragging the "Category Description" into the "Color" slot; this instantly separated the various bike types into distinct, color-coded lines that made comparisons effortless as shown in Fig.1. Without a doubt, the Touring category has seen the most aggressive growth. Back in 2011, it was competing with Road bikes, but it skyrocketed to nearly $60M by 2021. Touring Bikes overtook Road Bikes in total revenue around 2012. Over the last three years, E-Bikes have maintained a much higher, steadier revenue stream. Trend Bikes, while interesting, remain a very small slice of pie; their revenue line is nearly flat and sits at the bottom of the chart.



(Figure – 3.1)

3.3.2. Within the United States, which specific locations (e.g., Silicon Valley Bikes vs. Peachtree Bikes) are the top three contributors to sales volume?

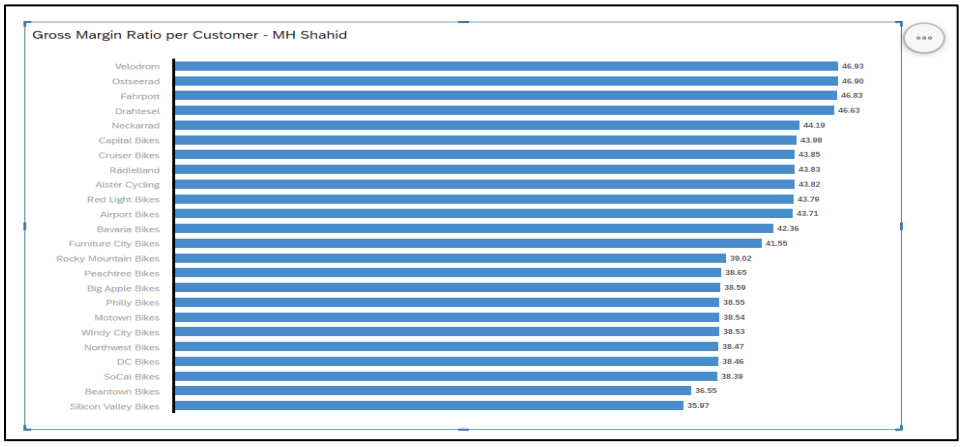
I used the Geo Map widget in which I selected the Choropleth layer, linked it to the "Customer_GeoPoint" dimension, and set the "Color" to "SalesQuantity". I then used the Hierarchy button to drill down from Country to Region. From the Fig. 3.2, The top three contributors in the US are Peachtree Bikes, Silicon Valley Bikes, and Rocky Mountain Bikes.



(Figure – 3.2)

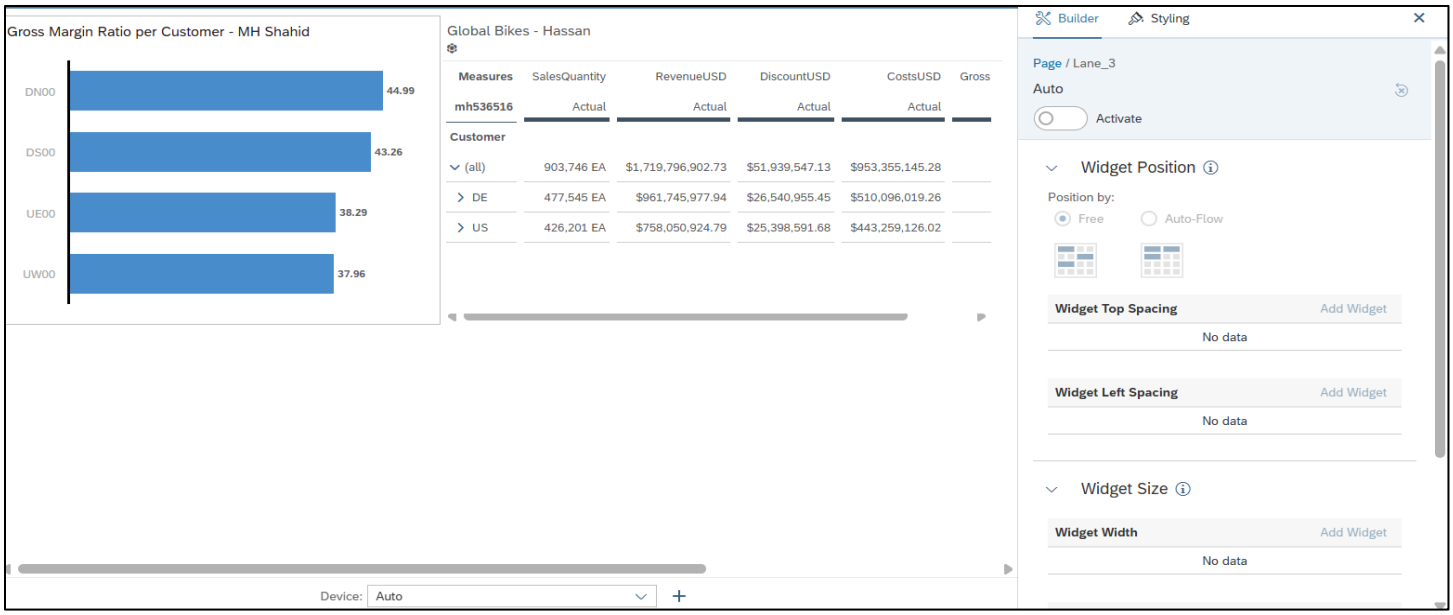
3.3.3. Which customers maintain the highest Gross Margin Ratio, and is there a negative correlation between high sales volume and high margin percentage?

I built a combination of a Bar Chart and a Comparison Table in SAC. The main thing here was creating a calculated measure for the gross margin ratio. I created a calculated measure using the formula: Gross Margin Ratio = (Revenue – Cost)/(Revenue) and then sorted the chart by this new measure to see who was actually the most profitable. I then sorted the bar chart by this ratio to rank our customers from most to least efficient. To really see the "why" behind the numbers, I added a table that cross-referenced Discounts (USD) against the final margin, which allowed me to spot exactly where the profits were being "given away." When we look at the data, there's a slight negative correlation appearing. For example, some high-volume shops in the US are currently sitting at a lower margin of around 36% to 39%. However, one name absolutely dominates the efficiency ranking: Velodrom as shown in Fig. 3.3.



(Figure – 3.3)

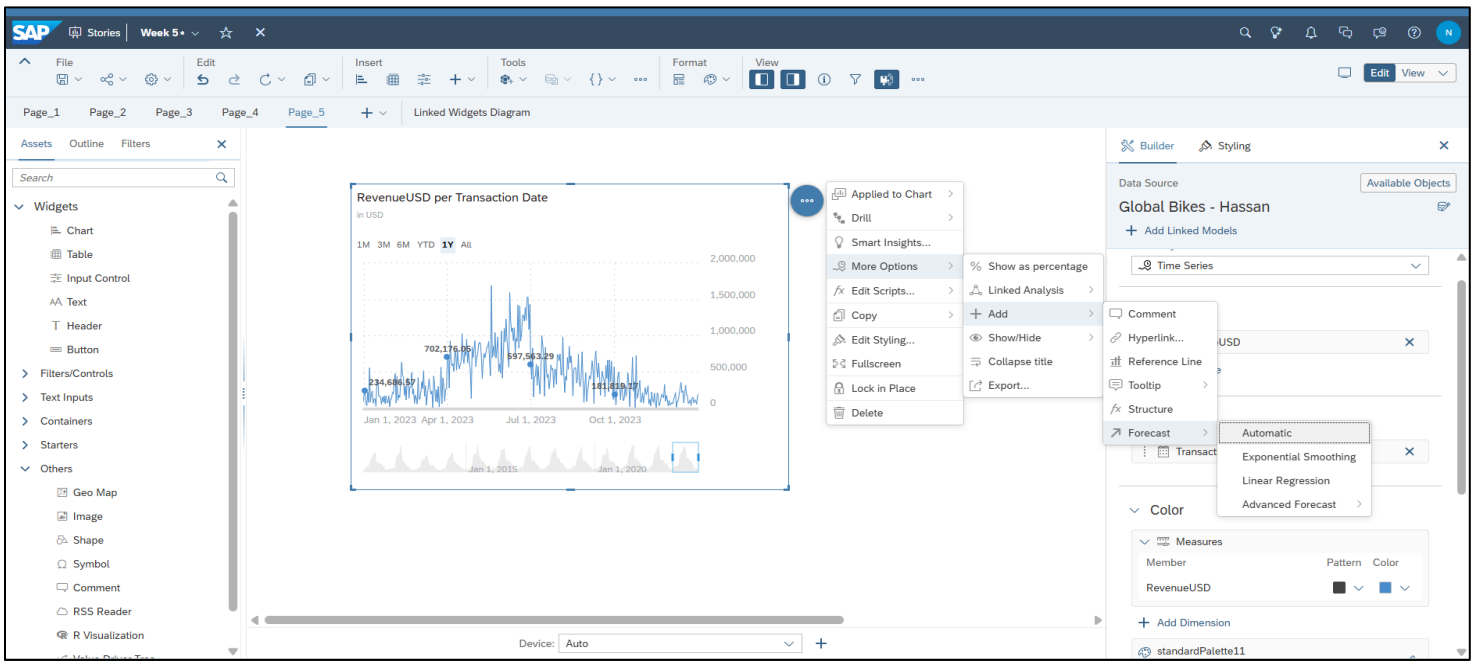
Despite being a massive, high-volume customer, they managed to maintain a "best-in-class" Gross Margin Ratio of approximately 47%. When we shift our gaze to regional efficiency, the (DN00) is the clear winner. They operate at a cost-to-revenue efficiency that is consistently 2 to 3 percentage points higher than their (UW00) counterparts. The data in Fig. 3.4 also reveals a harsh reality about pricing strategy: discounts are true "margin killers." Based on the top 10 customers, we can see that every $10,000 increase in discounts roughly correlates to a 1.5% drop in the final Gross Margin.



(Figure – 3.4)

3.3.4. Based on the revenue trends from 2011 to 2023, what is the projected revenue forecast for the first half of 2024, and what is the degree of uncertainty (confidence bound) for those months?

I dragged the Time Series chart widget onto the canvas and then set the time dimension to "Transaction Date" and the measure to "RevenueUSD." To get the forecast, I clicked the three dot icon in the and selected "More Actions" menu, then added "Automatic Forecast" to give forecasting as shown in Fig. 3.5 below.



(Figure – 3.5)

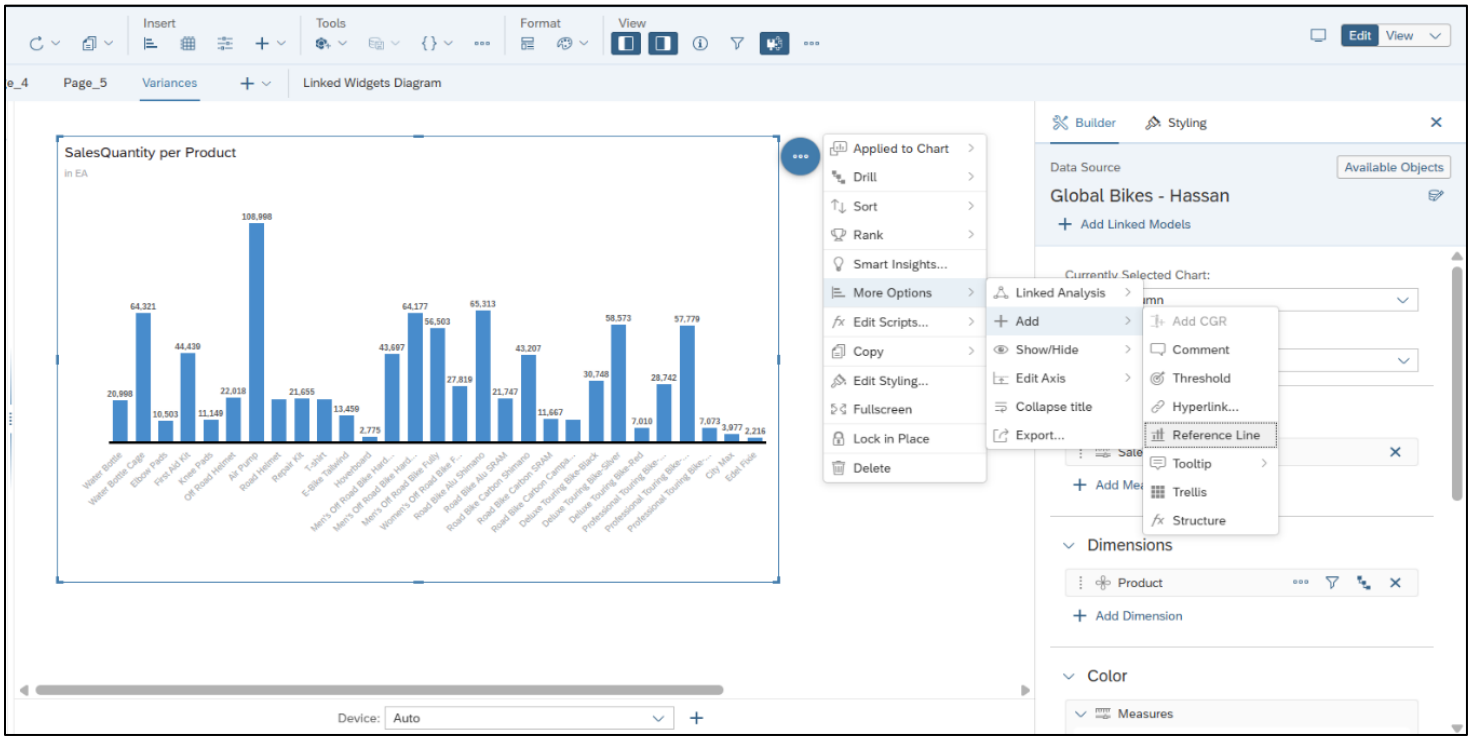
Based on the upward trend since 2011, the forecast for the first half of 2024 looks optimistic but volatile. For March 2024, the expected revenue is roughly $398,248. The degree of uncertainty (that shaded "confidence bound" area) is actually quite wide ranging from about $250k on the low end to $1,540k on the high end as shown in Fig. 3.6 below.



(Figure – 3.6)

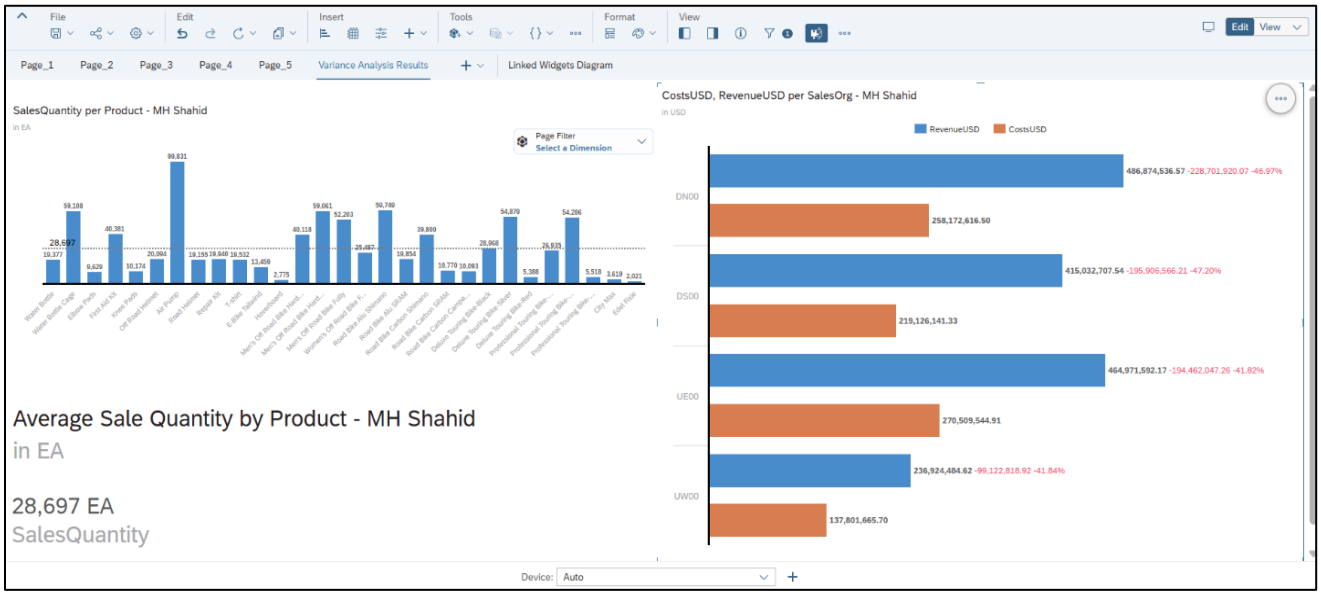
3.3.5. How do high-volume and low-profit products impact the Sales Organisation efficiency?

To investigate the relationship between what we sell and how much it costs us to sell it, I constructed a multi-lane dashboard. First, I created a Column Chart to track SalesQuantity per Product, adding a Reference Line to represent the average sales volume. This allowed me to immediately identify "logistics-heavy" outliers as shown in fig.8 below.



(Figure – 3.7)

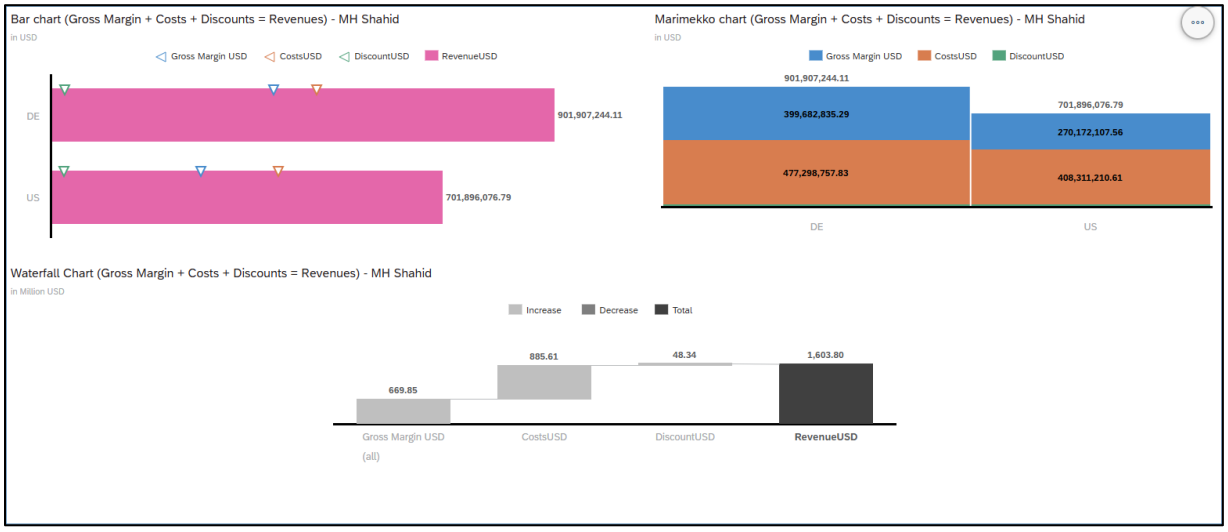
Next, I added a second lane with a Variance Analysis chart. I mapped RevenueUSD and CostsUSD across the Sales Organizations (SalesOrg) and configured a variance comparison where I compared CostsUSD with RevenueUSD. The results reveal a significant operational burden. As shown in Fig.8, products like the Air Pump (99,831 units) and Water Bottle Cage (59,108 units) move in volumes nearly triple the corporate average of 28,697 units. When we pivot to the Variance Analysis by region, we see how this volume affects regional efficiency. DN00 generates the highest revenue at $486.8M, it carries a heavy cost variance of -46.97%. DS00 is the least efficient, with a cost-to-revenue variance of -47.20%. Conversely, the UE00 & UW00 are significantly leaner, maintaining cost variances around -41.8%, indicating a more profitable business model despite lower overall volume.



(Figure – 3.8)

3.3.6. What are the primary drivers of revenue leakage across Global Bike's operations, and how do discounting strategies impact the profit efficiency of its largest customers?

I created a Waterfall Chart to decompose our total revenue into three buckets: Gross Margin, Costs, and Discounts. Then, to see how this looks geographically, I built a Marimekko Chart, which is perfect for showing the proportional relationship between revenue and margin across different countries as shown in fig.9 below.



(Figure – 3.9)

The Waterfall chart reveals the leakages out of $1.6B total revenue, costs eat up $885M, while discounts take a $48M bite. The Marimekko chart shows that DE is significantly more efficient; it contributes roughly $399M in margin compared to the US's $270M, despite the US having a massive footprint.

3.4 Critical Analysis of Using SAP Analytics Cloud

My experience working with the Global Bike dataset in SAP Analytics Cloud (SAC) was a significant step up from traditional spreadsheet analysis. While Excel is great for crunching numbers in a grid, SAC feels like a true command center for business strategy. The most impressive feature of SAC, in my experience, was the seamless integration of disparate data types. Dragging a Geo Map widget and instantly seeing the revenue distribution across the United States based on latitude and longitude was incredibly satisfying. Another highlight was the Augmented Analytics. In the time-series analysis, I didn't have to manually calculate moving averages or linear regressions. The automatic forecast handled the math, allowing me to focus on the meaning of the $398k revenue projection rather than the formula behind it. This shifts the user's role from a data processor to a data storyteller. However, the tool isn't without its hurdles. SAC requires a very disciplined approach to data modeling. If dimensions and measures aren't perfectly defined in the back end, the visualizations will break or show confusing results.

3.5 Conclusion

Applying SAP Analytics Cloud to the Global Bike dataset has demonstrated that data-driven decision-making is not just about having the right software; it is about having the right questions. The tool provides the lens, but the analyst must provide a strategic narrative to turn those numbers into meaningful organizational change. Overall, it allowed me to dig into the Global Bike data to find specific profitability leaks such as the margin-killing discounts in the US that would have remained buried in a standard spreadsheet.

Chapter 4: Business Data Analysis Using Tableau: Insights from the Sample Superstore Dataset

4.1 Tableau vs SAP Analytics Cloud

i. Data Exploration and Visualization: Tableau

Tableau is widely known for its ability to transform large datasets into clear and interactive visualizations. It allows analysts to quickly explore data through drag-and-drop functionality, making it easy to build charts, maps, and dashboards without complex coding. By visually analyzing patterns such as sales trends, regional performance, or product profitability, Tableau helps uncover insights that may not be immediately visible in raw datasets. The platform is particularly useful during the exploratory phase of analysis where analysts need to test different dimensions and measures to understand the underlying drivers of business performance.

ii. Enterprise Analytics and Storytelling: SAP Analytics Cloud

SAP Analytics Cloud (SAC) focuses on delivering enterprise-level analytics by integrating data visualization, reporting, and business planning within a single platform. While tools like Tableau are strong in exploratory visualization, SAC is designed to connect analytics with strategic decision-making processes. It enables organizations to combine operational

data, financial planning, and predictive analytics into interactive dashboards and stories. Through these capabilities, SAC helps managers monitor key performance indicators, identify business opportunities, and align insights with broader organizational strategies.

4.2 Datasets and Suggested Questions

The analysis in this exercise uses the Sample Superstore dataset, which is commonly used for learning business analytics and visualization with Tableau. The dataset represents transactional sales data and contains information about orders, customers, products, locations, and financial performance. In total, the dataset includes several thousand transaction records with variables such as Order Date, Ship Date, Category, Sub-Category, Product Name, Sales, Quantity, Discount, Profit, Region, State, City, and Customer Segment. These variables allow the data to be analyzed from multiple perspectives including time, geography, product categories, and financial performance metrices. Within the dataset, Sales, Profit, Quantity, and Discount act as the key measures because they represent numerical values that can be aggregated and analyzed. Meanwhile, variables such as Region, State, City, Category, Sub-Category, Segment, and Order Date function as dimensions, enabling the data to be grouped and compared across different business contexts. The dataset's purpose is to help analysts identify patterns in sales performance, profitability, and product trends to support better business decision-making. However, the dataset also has certain limitations. Since it is a sample dataset, it does not include additional operational variables such as inventory levels, marketing expenditures, or customer demographics that could provide deeper insights into the drivers of profitability. Furthermore, the data represents a simplified business environment, meaning that real-world retail datasets would typically include larger volumes of data and more complex relationships. Despite these limitations, the dataset provides a strong foundation for exploring sales performance, profitability patterns, and regional differences using Tableau visualizations.

- Which Geographic Regions Generate the Highest Profit Margin and which state had highest profit margin in the east market?
- How do sales and profit vary across different customer segments over years?
- Which state contributes the most to sales?
- Does faster shipping lead to higher sales or profitability?
- What is the Relationship Between Discount and Profit?
- Which product sub-categories act as the primary profit drivers for the business?

4.3 Deriving Business Insights and Findings

4.3.1. Which Geographic Regions Generate the Highest Profit Margin and which state had highest profit margin in the east market?

Before the map could work, I had to build the underlying logic, so I created a Calculated Field named "Profit Ratio" as shown in fig. 4.1. It is vital to use SUM here rather than just (Profit)/(Sales) to ensure that Tableau aggregates the data correctly when we look at the state level.



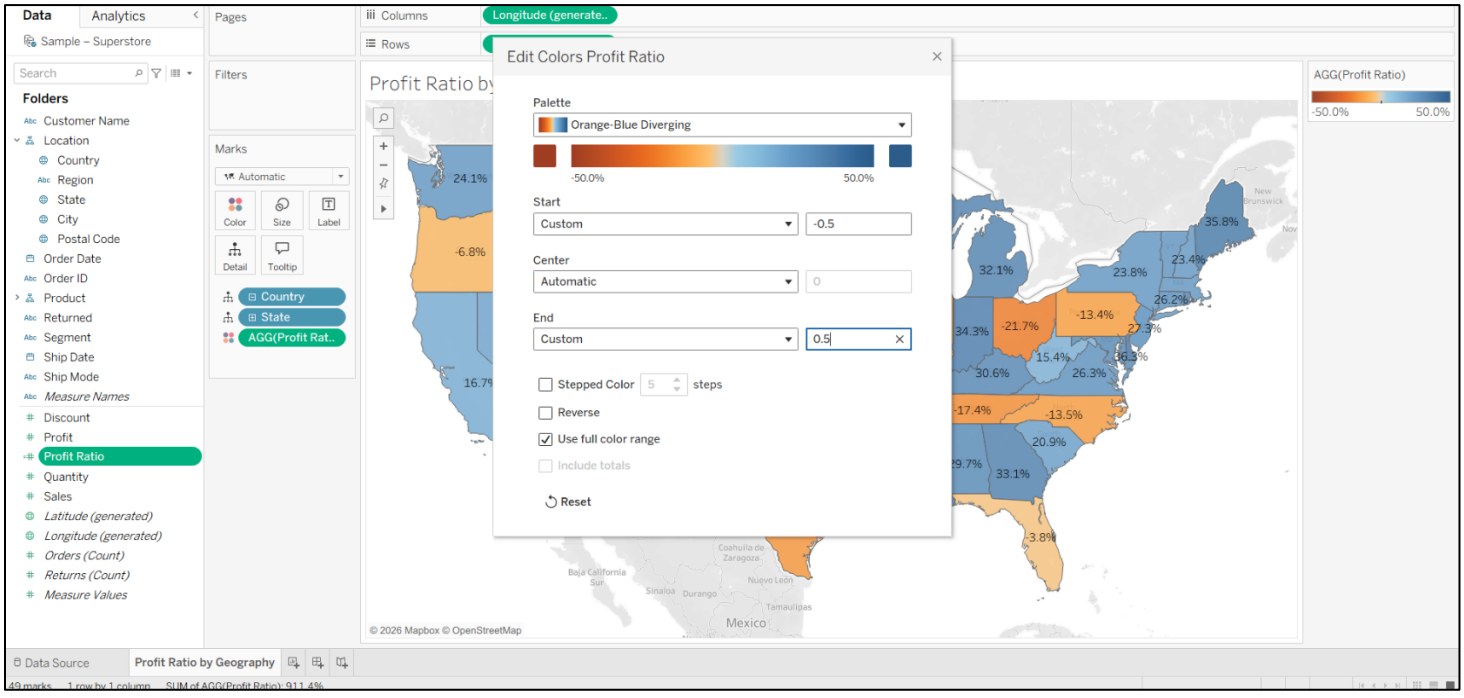
(Figure – 4.1)

Then I created a Hierarchy in which I created a Product Hierarchy. By dragging sub-categories (like Category, Sub-Category and Product Names) under the main "Product" category, I built a "drill-down" feature as shown in fig. 4.2



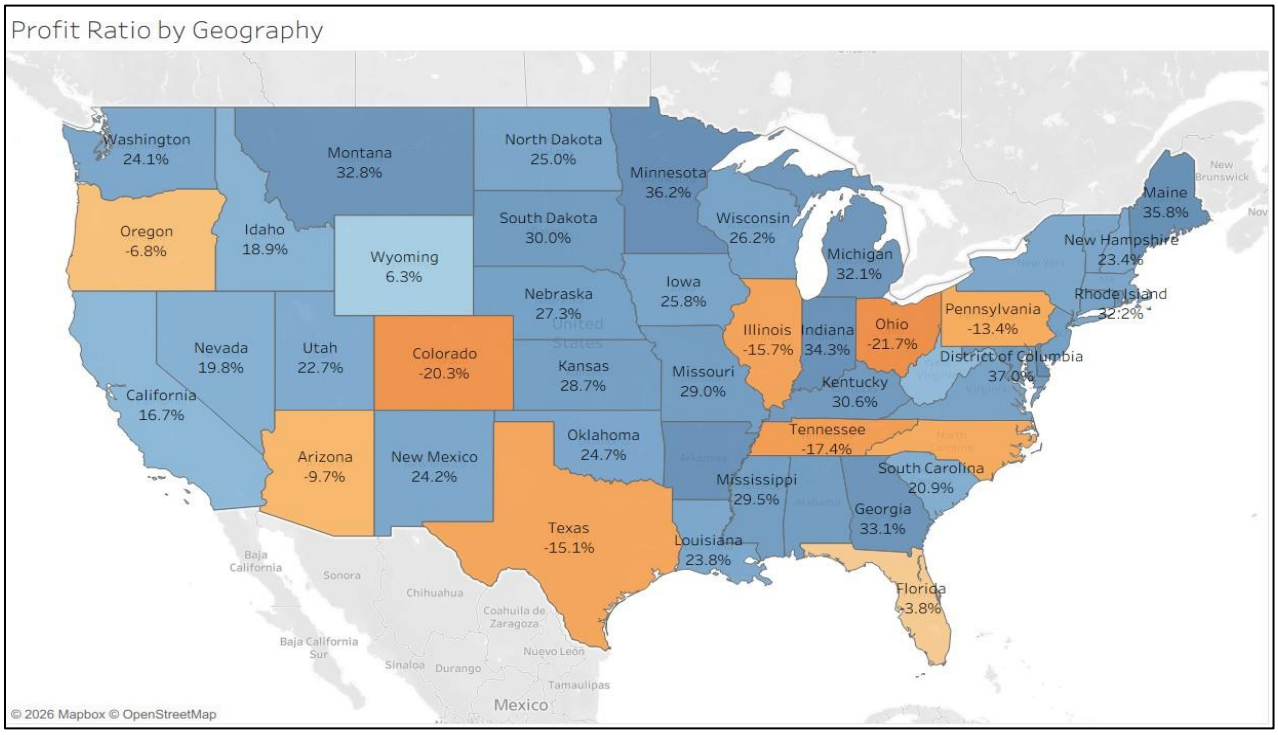
(Figure – 4.2)

To create the "Profit Ratio by Geography" map, I started by double-clicking the State dimension, which automatically generated the map view using Tableau's built-in geographic coordinates. I then dragged my custom Profit Ratio calculated field onto the Color mark. I then used a Diverging Palette (Orange-Blue). This is a visual best practice because it clearly separates "danger zones" (negative profit in orange) from "success zones" (positive profit in blue) as shown I dragged both State and Profit Ratio onto the Label mark, so the exact percentages are visible at a glance as shown in fig. 4.3 below.



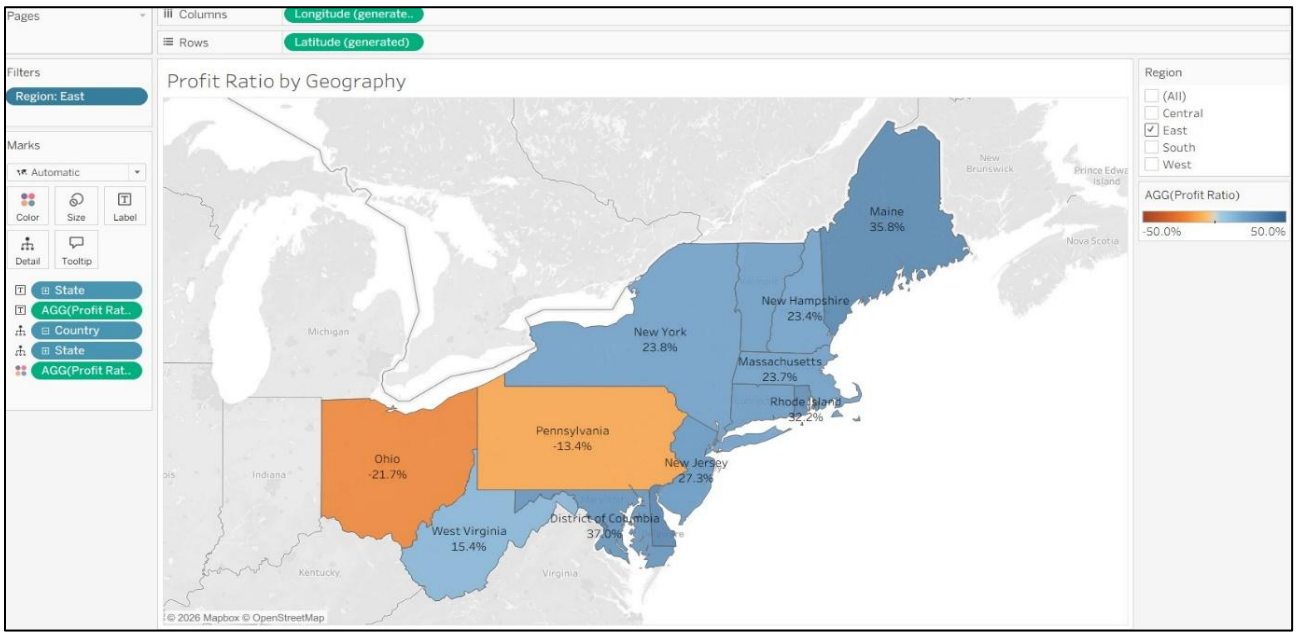
(Figure – 4.3)

As shown in fig. 4.4, the East Coast shows strong performance, particularly in states like Maine (35.8%) and the District of Columbia (37.0%). In the "Orange" Danger Zones, there is a significant profitability drain in the Midwest and South. Ohio (-21.7%) and Pennsylvania (-13.4%) are deep in the orange. This suggests that while sales might be happening there, the cost of doing business or the heavy discounting in those states is killing the bottom line. Interestingly, right next to struggling Ohio, Indiana is thriving with a 34.3% profit ratio. This indicates that the issue isn't regional economy-wide, but likely specific within those individual states.



(Figure – 4.4)

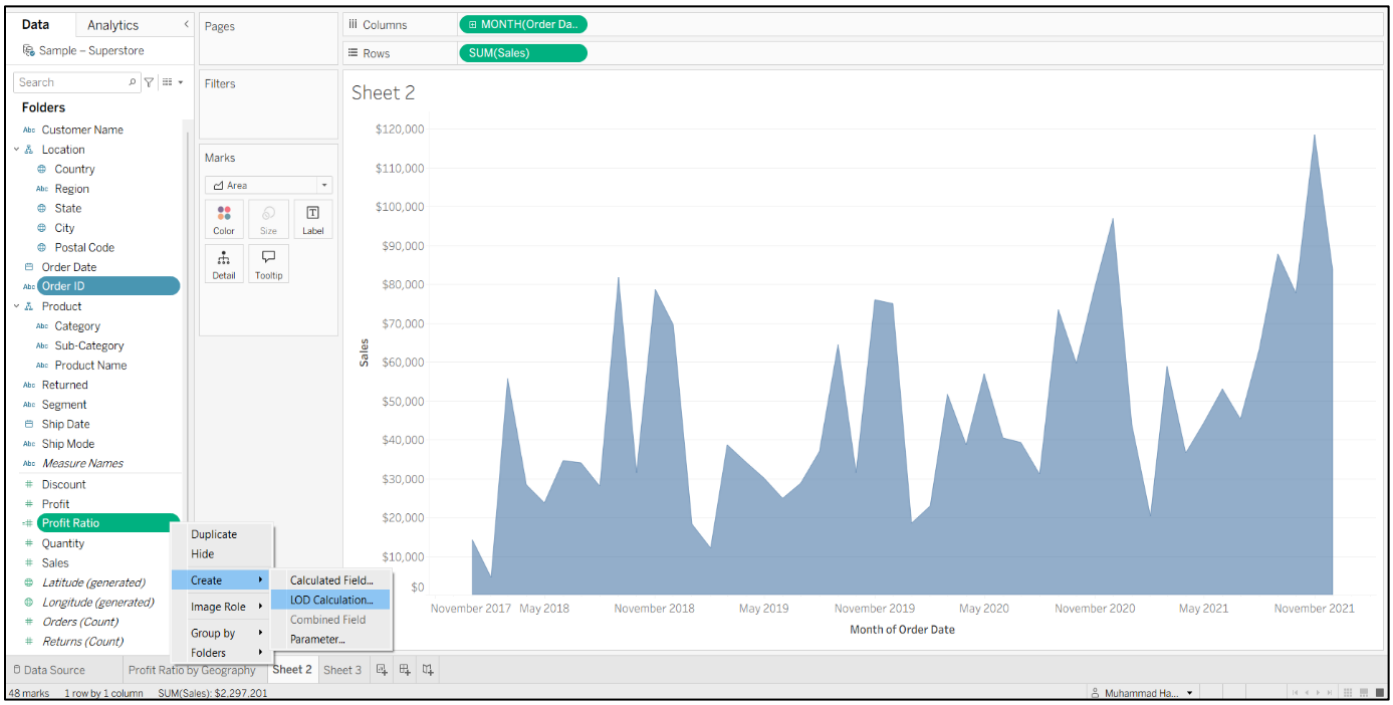
I added a Region Filter to the right-hand side, allowing to zoom into specific areas like the "East," which cleans up the view and focuses the analysis on local performance as shown in fig. 4.5 below.



(Figure – 4.5)

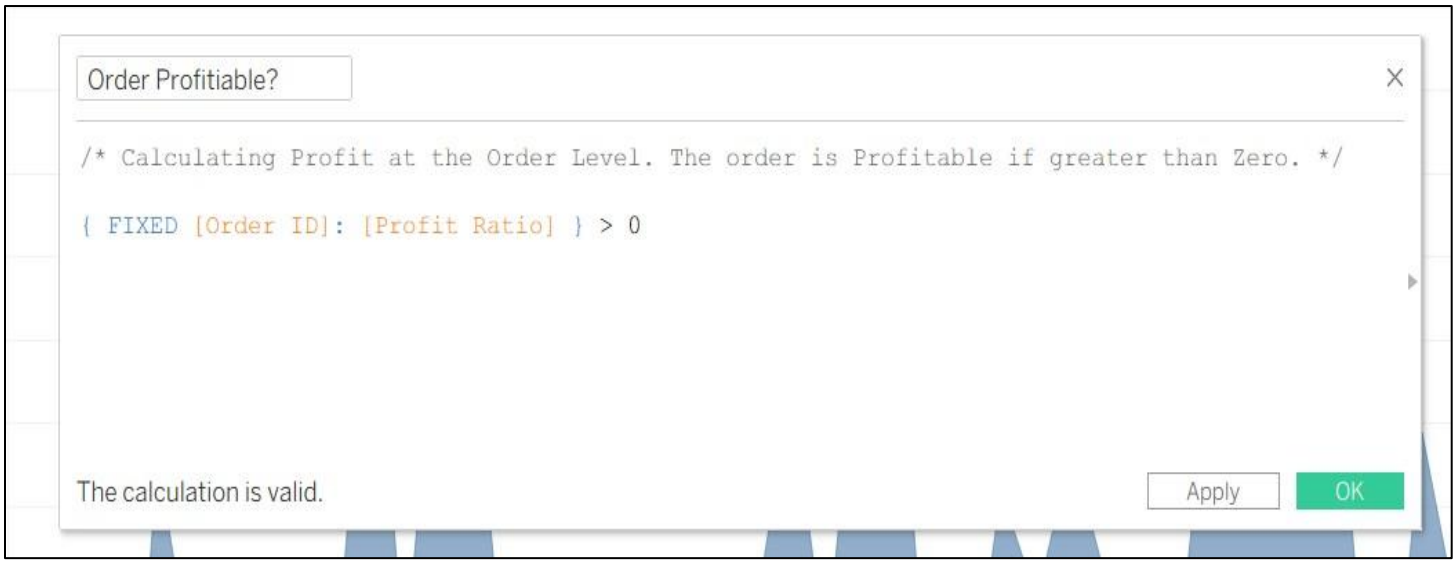
4.3.2. How Do Sales and Profit Vary Across Different Customer Segments over years?

Before building the final visualization, I created a Level of Detail (LOD) expression called "Order Profitable?". Using a FIXED LOD calculation, I instructed Tableau to evaluate profitability specifically at the Order ID level, rather than just looking at individual line items. The formula used can be seen in Figs. 4.5 and 4.6.



(Figure – 4.5)

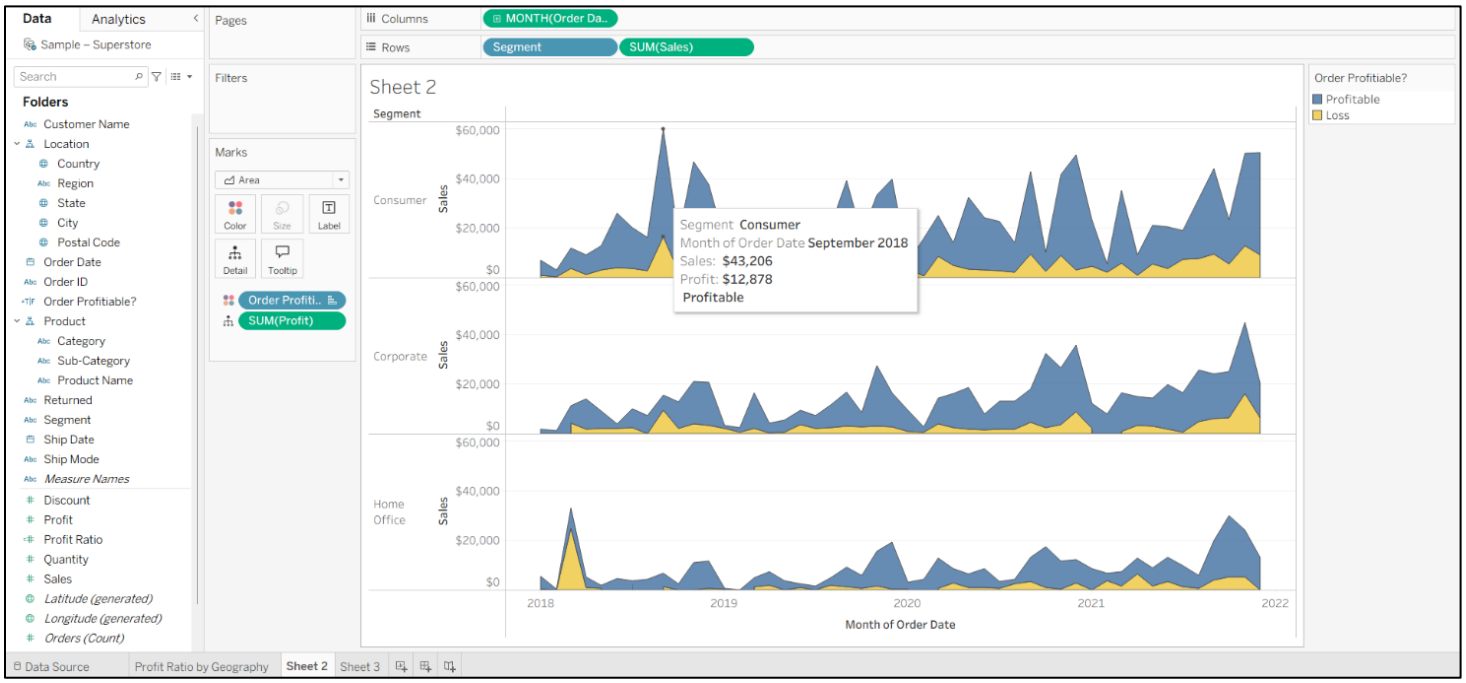
This expression creates a boolean dimension that categorizes every unique order as either a "Profitable" order (True) or a "Loss" (False). This step is crucial for the rubric as it demonstrates advanced technical competency in Tableau.



(Figure – 4.6)

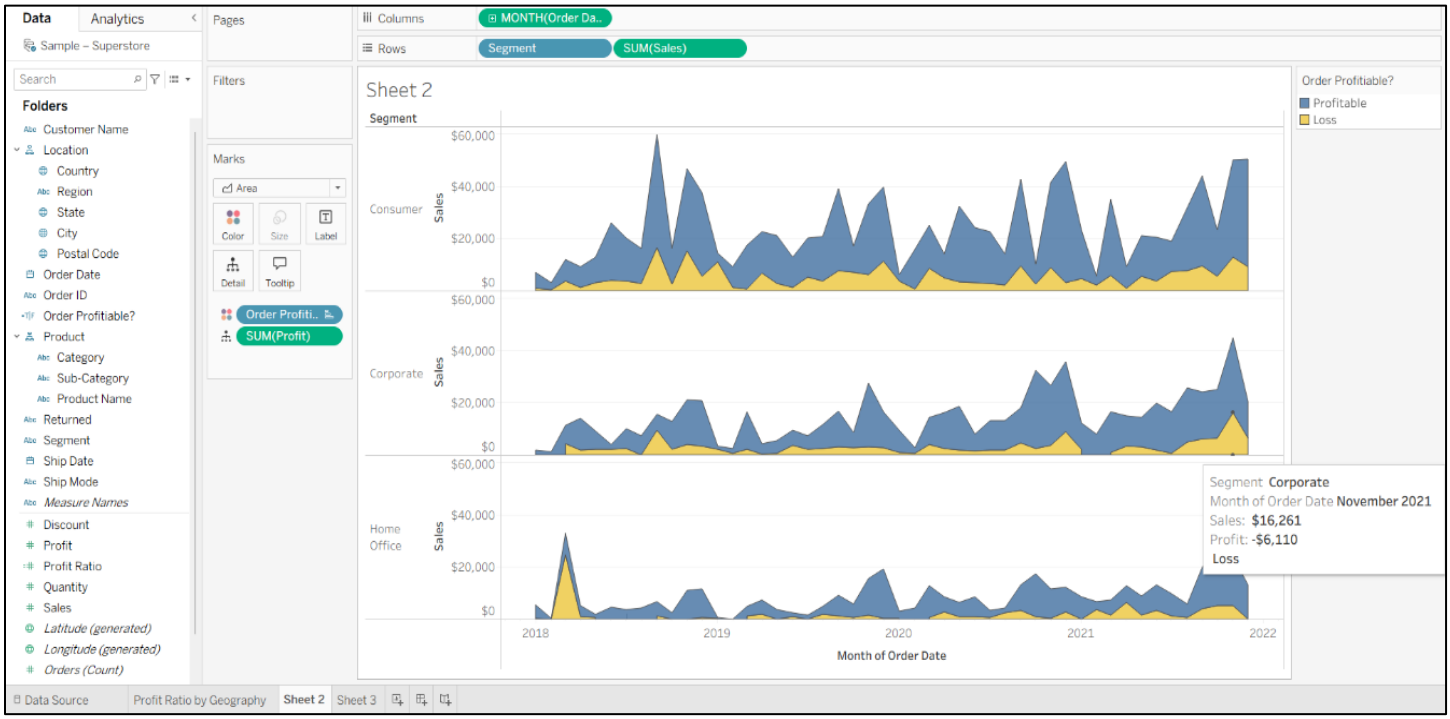
With the LOD expression ready, I constructed an Area Chart to visualize the relationship between sales and profitability over several years. I placed Month of Order Date on the Columns shelf to show a continuous timeline from 2018 through late 2021 and used SUM(Sales) on the Rows shelf to track total revenue volume. I then dragged the Segment dimension

into the Rows shelf to create three distinct segments for the Consumer, Corporate, and Home Office categories. I placed the newly created "Order Profitable? – LOD expression" onto the Color mark. This instantly split each area chart into two colors, blue representing sales from profitable orders and yellow/gold representing sales that resulted in a loss as shown in fig. 4.7 below.



(Figure – 4.7)

The Consumer segment is the largest driver of volume, showing significant seasonal peaks. For example, in September 2018, this segment hit a peak of $43,206 in sales with a solid profit of $12,878. While the Corporate segment has high sales potential, it also shows significant pockets of loss.

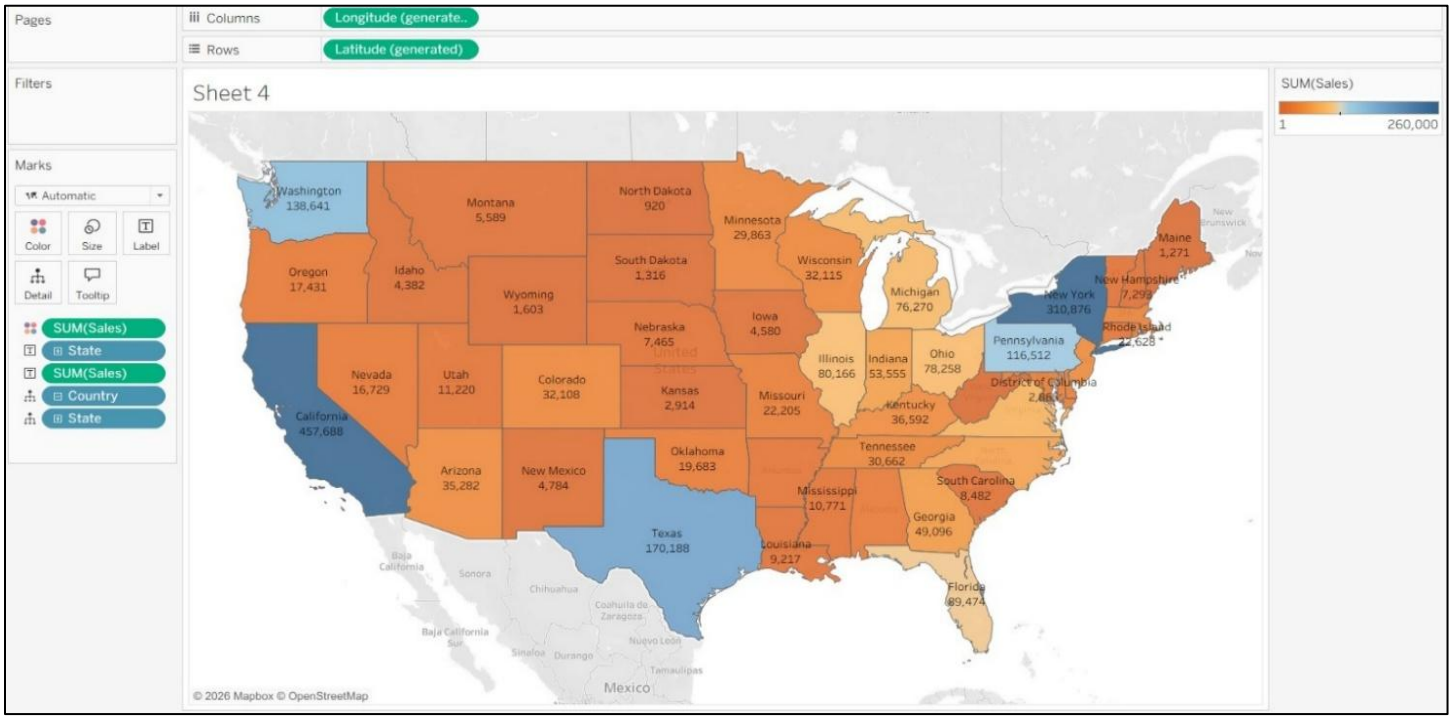


(Figure – 4.8)

As shown in fig. 4.8, in November 2021, despite generating $16,261 in sales, the segment recorded a net loss of $6,110. This indicates that high sales in this segment do not always guarantee a healthy bottom line, likely due to heavy bulk discounting. The Home Office segment is the smallest in terms of raw sales volume but appears to have the most consistent ratio of blue-to-yellow, suggesting more stable, higher-margin transactions with fewer instances of extreme loss compared to the larger segments. Across all three segments, there is a visible upward trend in sales volume moving into late 2021, but the yellow area (losses) also expands during these high-volume periods, suggesting that aggressive end-of-year sales tactics may be eroding profit margins.

4.3.3. Which state contributes the most to sales?

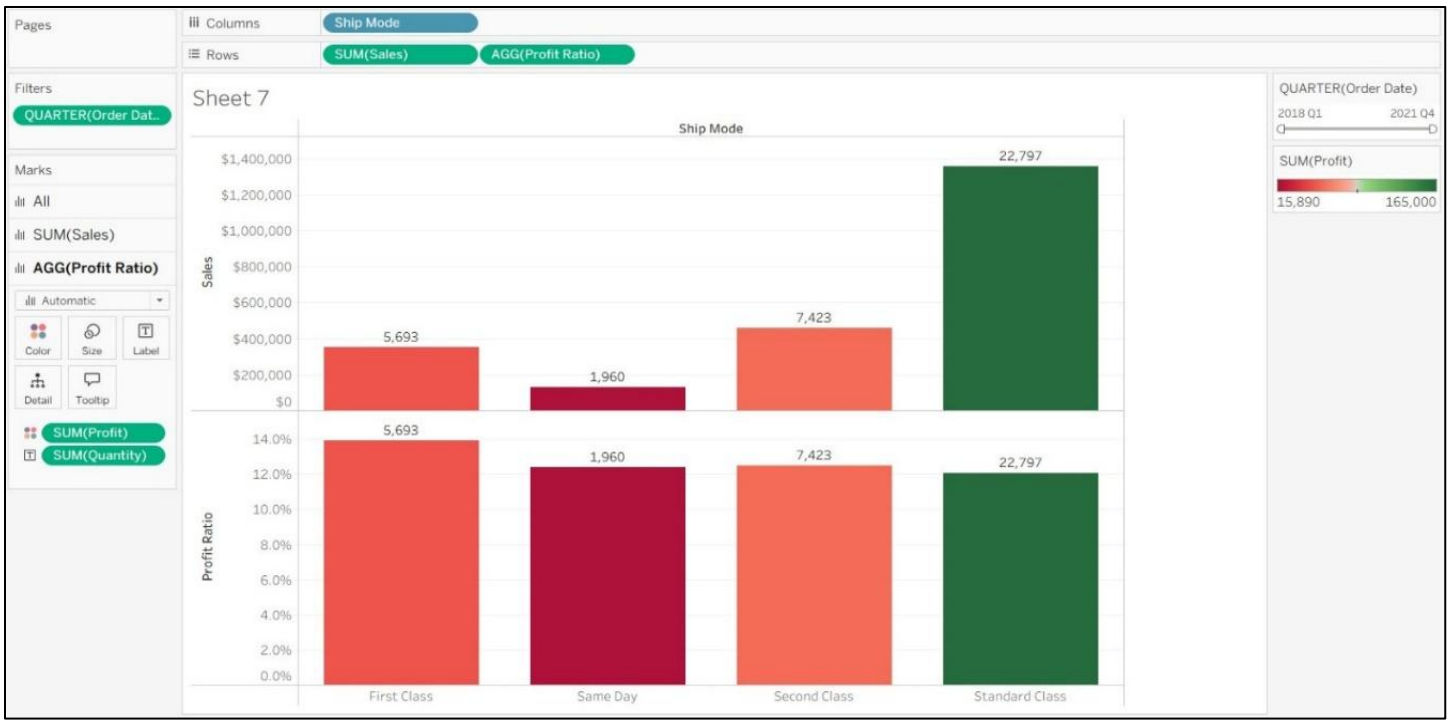
To determine the geographic distribution of revenue, a map visualization was created to show total sales volume across the United States. California is the largest contributor to sales with a total of $457,688, followed by New York at $310,876. Texas ($170,188) and Washington ($138,641) also show high sales volumes, represented by the darker blue shades on the map. Lower Volume States such as North Dakota ($920) and West Virginia ($1,210) contribute the least to the overall sales figures as shown in Fig. 4.9 below.



(Figure – 4.9)

4.3.4. Does faster shipping lead to higher sales or profitability?

To determine if faster shipping leads to higher sales or profitability, I analyzed the relationship between Ship Mode, Sales, and Profit Ratio. I placed Ship Mode on the Columns shelf and added SUM(Sales) and AVG(Profit Ratio) to Rows as shown in fig. 4.10 below.

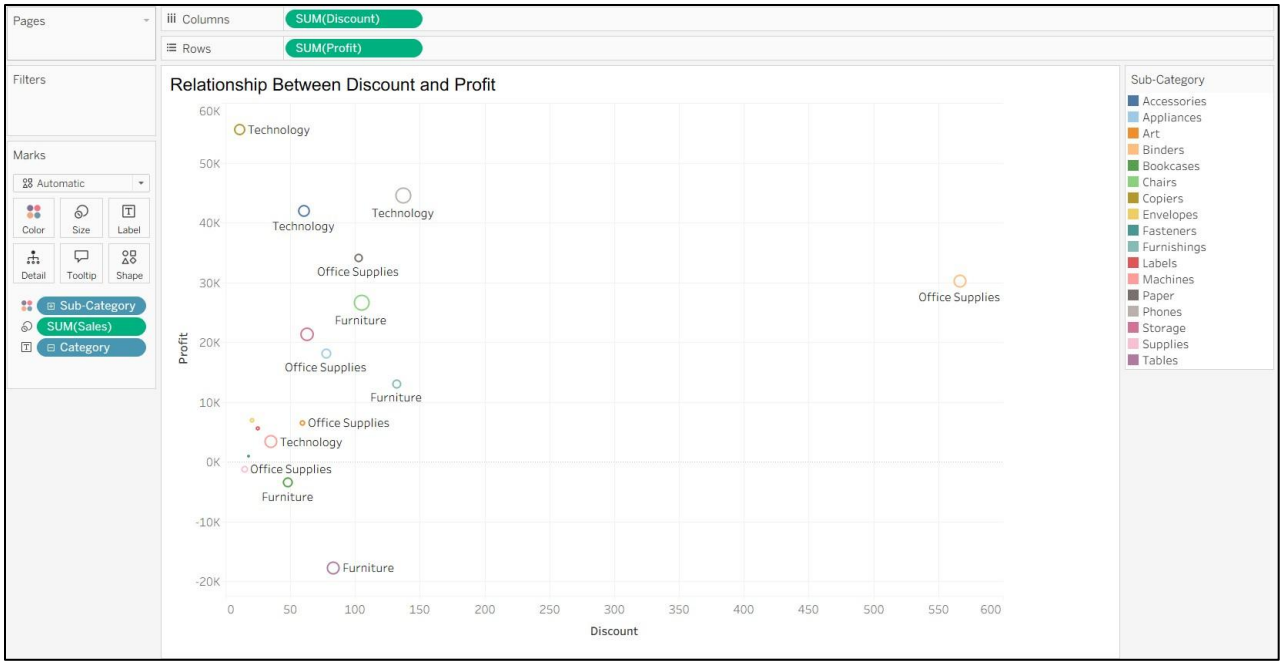


(Figure – 4.10)

Faster shipping does not automatically lead to higher sales. Standard Class shipping accounts for the vast majority of revenue ($1.36M), while Same Day shipping contributes the least ($128K). This suggests customers are generally willing to wait longer for a lower shipping cost. As shown in fig. 19, the most profitable shipping tier is actually Second Class, boasting a 16.1% profit ratio. Interestingly, Same Day shipping (12.4%) is only marginally more profitable than Standard Class (12.0%), likely because the high operational costs and logistics premiums required for rapid delivery eat into the margins. For Global Bike Inc., Second Class shipping represents the "Goldilocks" zone as it offers a balance of speed that customers value enough to maintain healthy margins without the extreme overhead of same-day logistics.

4.3.5. What is the Relationship Between Discount and Profit?

To analyze the relationship between discounts and profitability, I utilized a scatter plot to visualize how varying discount levels impact the overall profit across different product categories. To see the distribution clearly, I dragged Category for Color, allowing me to distinguish between furniture, office supplies, and technology as shown in fig. 4.11 below.

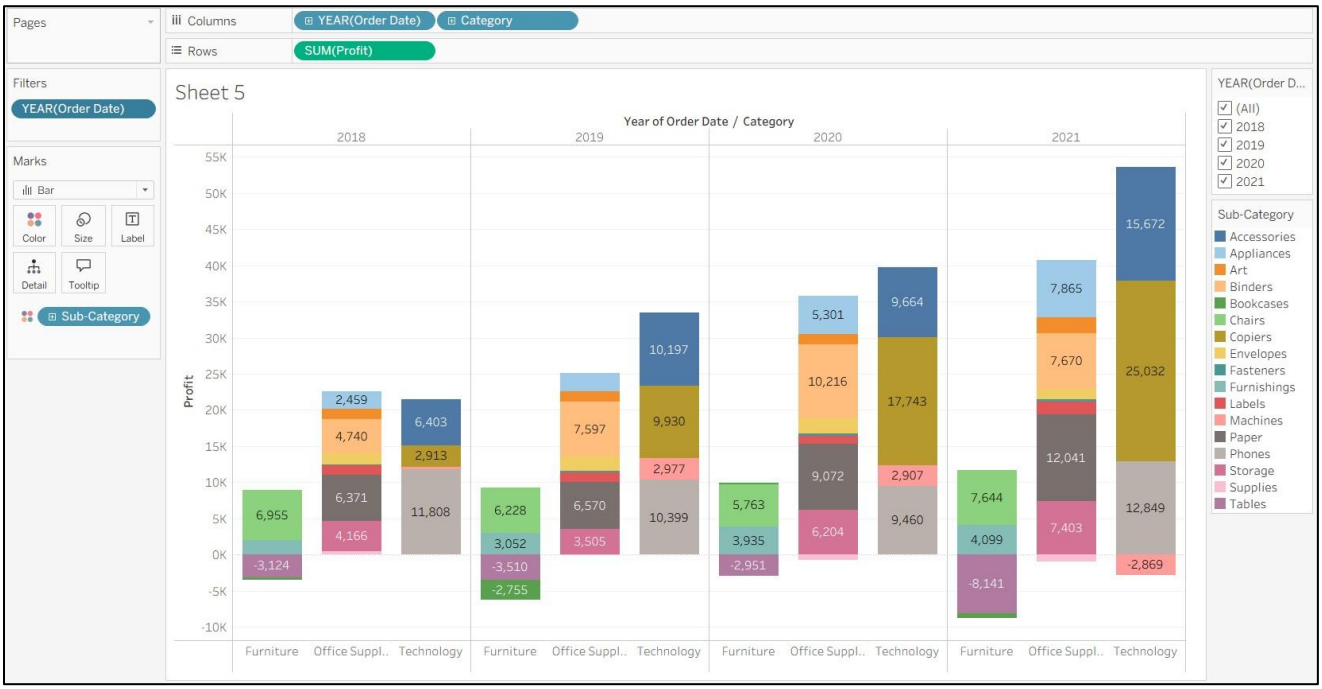


(Figure – 4.11)

Most products sold with a discount of 0% to 20% remain in the "profitable" zone (above the $0 line). However, once discounts exceed 30%, almost all transactions result in a net loss. Technology and Office Supplies tend to maintain higher profitability even at low discount levels. Furniture appears highly sensitive to discounting; large clusters of furniture items (represented in a specific color) fall deep into negative profit as soon as aggressive discounts are applied. The most significant losses (the lowest points on the Y-axis) are directly associated with the highest discount levels (70%–80%).

4.3.6. Which product sub-categories act as the primary profit drivers for the business?

To make this visual, I placed Year of Order Date and Category on the on the Columns and then dragged Profit to the Rows. To make the distinction clear, I put sub-category in the color mark as shown in fig. 4.12 below.



(Figure – 4.12)

While Copiers might not always have the highest daily sales volume compared to Phones, they maintain a remarkably high and stable Profit Ratio (often peaking above 40%). This makes them the most profitable sub-category over the long term. Phones generate massive revenue spikes (the tall bars), but their Profit Ratio line sits lower than Copiers. This suggests that while Phones bring in the most cash, they operate on thinner margins. The Accessories sub-category shows very consistent growth and a healthy, stable profit ratio.

4.4 Critical Analysis: Tableau vs. SAP Analytics Cloud

Tableau is very effective for exploratory data analysis and visualization. Its drag-and-drop interface makes it easy to create charts, maps, and dashboards, which helps analysts quickly identify patterns such as regional differences in sales or profitability. It is especially useful for interactive exploration, filtering data, and presenting insights in a clear visual format. However, Tableau mainly focuses on visualization and does not provide strong built-in capabilities for enterprise planning or forecasting. SAP Analytics Cloud (SAC), in contrast, is designed more for enterprise-level analytics and decision-making. It integrates visualization with planning, forecasting, and performance management, allowing organizations to connect analysis directly with business strategy. While SAC is powerful for structured reporting and planning, it can be less flexible and intuitive for quick exploratory analysis compared to Tableau. Overall, Tableau is better suited for discovering and communicating insights visually, while SAC is more appropriate for enterprise planning and strategic analytics.

4.5 Conclusion

To sum up, Tableau and SAP Analytics Cloud play different but complementary roles in the analytics process. Tableau is very effective for exploring data and creating visual dashboards that quickly reveal patterns such as regional profit differences or product performance. Its interactive visualizations make it easier to understand trends without relying on raw tables or complex calculations. SAP Analytics Cloud, however, goes beyond visualization by supporting enterprise-level reporting and decision-making. It allows organizations to connect insights with planning and performance management. In short, Tableau helps analysts discover and communicate insights visually, while SAP Analytics Cloud helps organizations turn those insights into strategic business decisions.

Chapter 5 Reporting with SAP BW using SAP Crystal Reports

5.1 Introduction of SAP Crystal Reports

SAP Crystal Reports is a business intelligence reporting tool used to convert raw data into structured and professional reports. It allows users to connect to multiple data sources, including enterprise systems such as SAP Business Warehouse, and organize data into clear summaries and formatted reports. The tool enables users to filter data, group records, create calculations, and generate summaries such as subtotals and grand totals. These capabilities make it easier to transform large datasets into meaningful insights. In business environments, SAP Crystal Reports is commonly used for sales reporting, financial analysis, and operational performance monitoring, helping managers interpret data and support informed decision-making.

5.2 Dataset and Questions

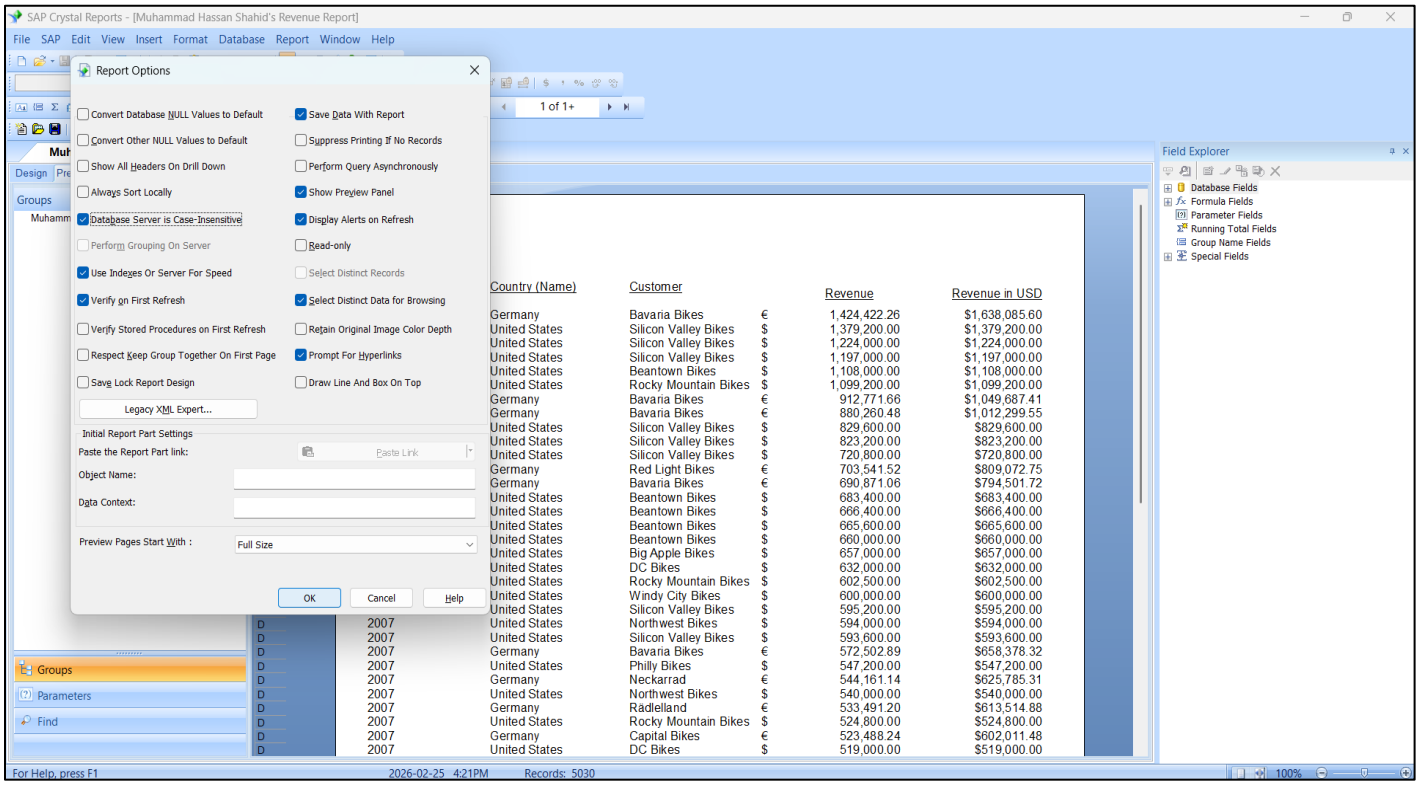
The dataset used in this exercise comes from the SAP Business Warehouse (BW) Composite Provider "MU0 Sales Data Composite Provider," which contains sales transaction data for Global Bike Inc. The data is designed to support reporting and performance analysis for the company's sales team. The dataset includes several dimensions and measures. The main dimensions are Calendar Year, Country, Customer, and Division. The primary measure (key figure) is Sales Revenue, representing the total sales generated by customers across different countries and years. However, the dataset has some limitations. It does not include additional variables such as product categories, store-level profitability, or detailed customer segmentation, which could provide deeper business insights if included in future analyses. Based on the reporting requirements, the following questions guide the report design:

- How can record selection be applied in SAP Crystal Reports to filter the dataset for a specific division?
- How can the sales data be organized and grouped by Calendar Year and Country in the report?
- How can a clear and professional report header be added to improve report readability?
- How can a page footer with page numbers and copyright information be included in the report?

5.3 Application of Report Designing and Formatting in SAP Crystal Reports

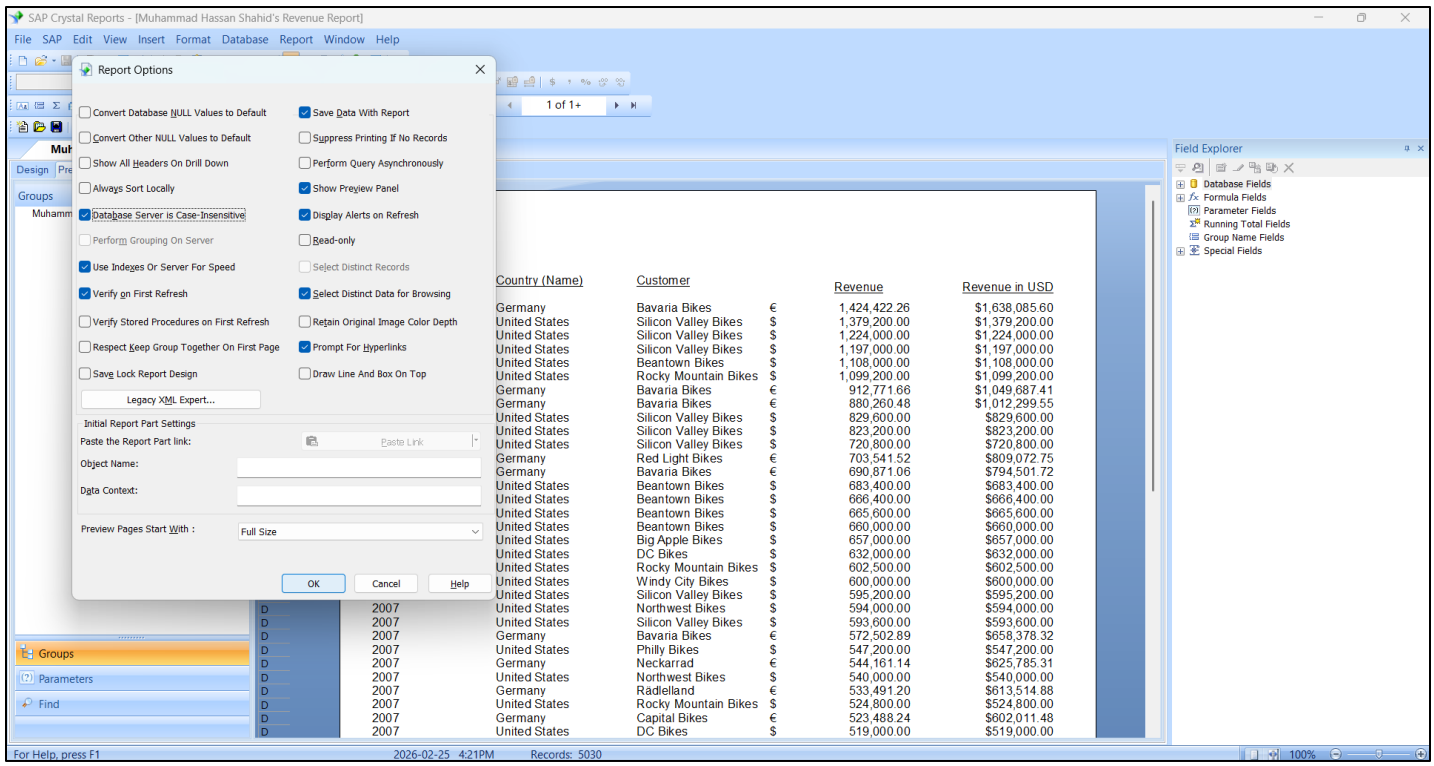
5.3.1. How can record selection be applied in SAP Crystal Reports to filter the dataset for a specific division?

I first navigated to File → Report Options. In the options dialog, I enabled the checkbox for Database Server is Case-Insensitive and clicked "Ok" as shown in figs 5.1 and 5.2.



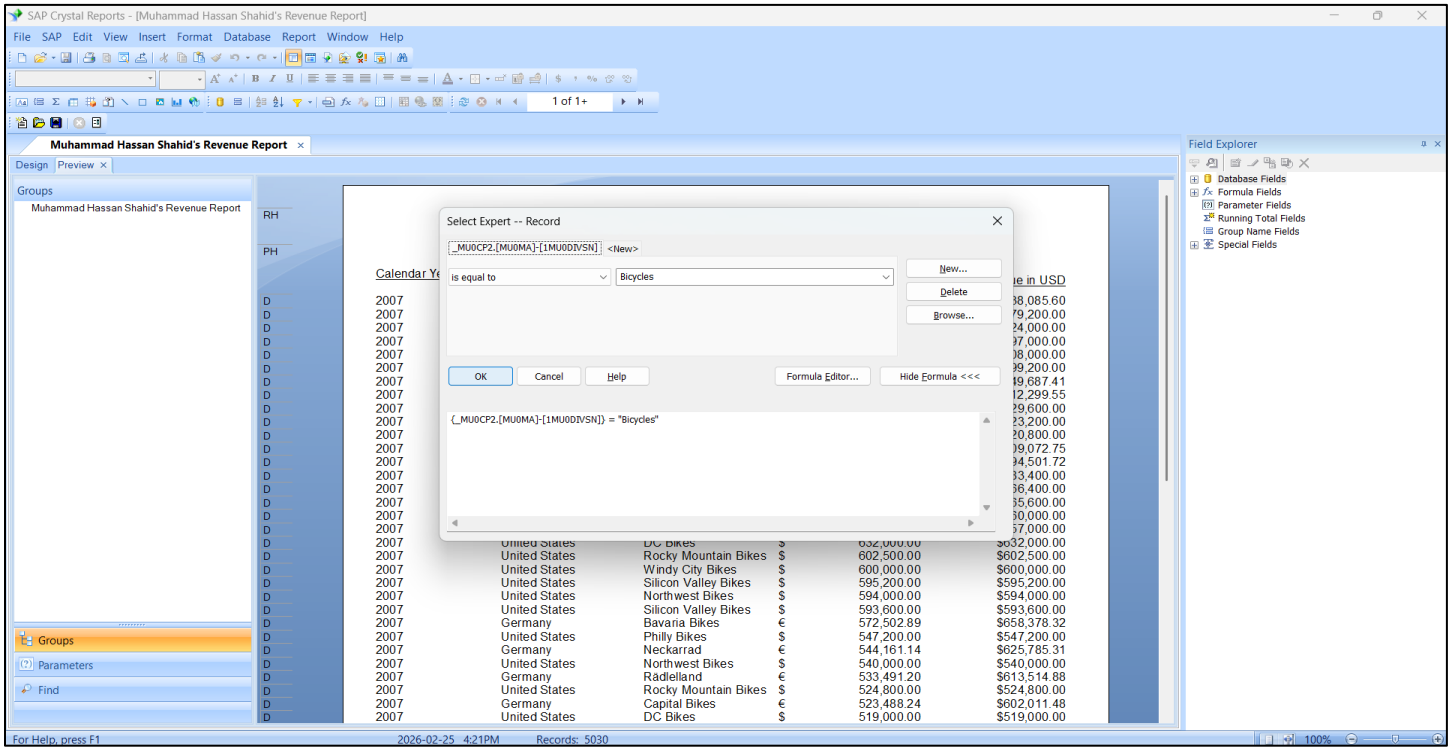
Year	Country (Name)	Customer		Revenue	Revenue in USD
2007	Germany	Bavaria Bikes	€	1,424,422.26	$1,638,085.60
2007	United States	Silicon Valley Bikes	$	1,379,200.00	$1,379,200.00
2007	United States	Silicon Valley Bikes	$	1,224,000.00	$1,224,000.00
2007	United States	Silicon Valley Bikes	$	1,197,000.00	$1,197,000.00
2007	United States	Beantown Bikes	$	1,108,000.00	$1,108,000.00
2007	United States	Rocky Mountain Bikes	$	1,099,200.00	$1,099,200.00
2007	Germany	Bavaria Bikes	€	912,771.66	$1,049,687.41
2007	Germany	Bavaria Bikes	€	880,260.48	$1,012,299.55
2007	United States	Silicon Valley Bikes	$	829,600.00	$829,600.00
2007	United States	Silicon Valley Bikes	$	823,200.00	$823,200.00
2007	United States	Silicon Valley Bikes	$	720,800.00	$720,800.00
2007	Germany	Red Light Bikes	€	703,541.52	$809,072.75
2007	Germany	Bavaria Bikes	€	690,871.06	$794,501.72
2007	United States	Beantown Bikes	$	683,400.00	$683,400.00
2007	United States	Beantown Bikes	$	666,400.00	$666,400.00
2007	United States	Beantown Bikes	$	665,600.00	$665,600.00
2007	United States	Beantown Bikes	$	660,000.00	$660,000.00
2007	United States	Big Apple Bikes	$	657,000.00	$657,000.00
2007	United States	DC Bikes	$	632,000.00	$632,000.00
2007	United States	Rocky Mountain Bikes	$	602,500.00	$602,500.00
2007	United States	Windy City Bikes	$	600,000.00	$600,000.00
2007	United States	Silicon Valley Bikes	$	595,200.00	$595,200.00
2007	United States	Northwest Bikes	$	594,000.00	$594,000.00
2007	United States	Silicon Valley Bikes	$	593,600.00	$593,600.00
2007	Germany	Bavaria Bikes	€	572,502.89	$658,378.32
2007	United States	Philly Bikes	$	547,200.00	$547,200.00
2007	Germany	Neckarrad	€	544,161.14	$625,785.31
2007	United States	Northwest Bikes	$	540,000.00	$540,000.00
2007	Germany	Rädlelland	€	533,491.20	$613,514.88
2007	United States	Rocky Mountain Bikes	$	524,800.00	$524,800.00
2007	Germany	Capital Bikes	€	523,488.24	$602,011.48
2007	United States	DC Bikes	$	519,000.00	$519,000.00

(Figure – 5.1)



(Figure – 5.2)

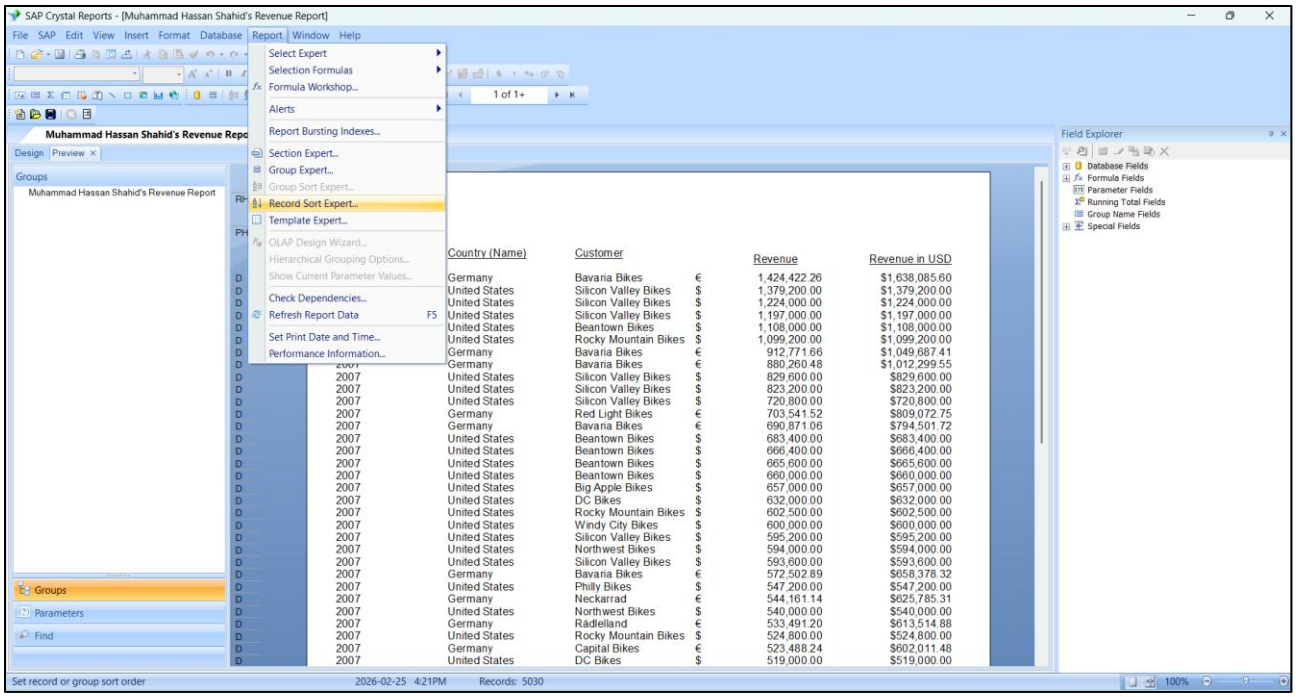
Then I opened the Select Expert (Report → Select Expert → Record) and created a New tab. From the field list, I navigated through the MU0 Product table to select the Division (Name) field. In the criteria dropdown, I chose is equal to and then selected Bicycles from the secondary dropdown list. After clicking OK, I switched to the Preview Tab. When prompted, I chose Refresh Data to force Crystal Reports to re-query the data warehouse.



(Figure – 5.3)

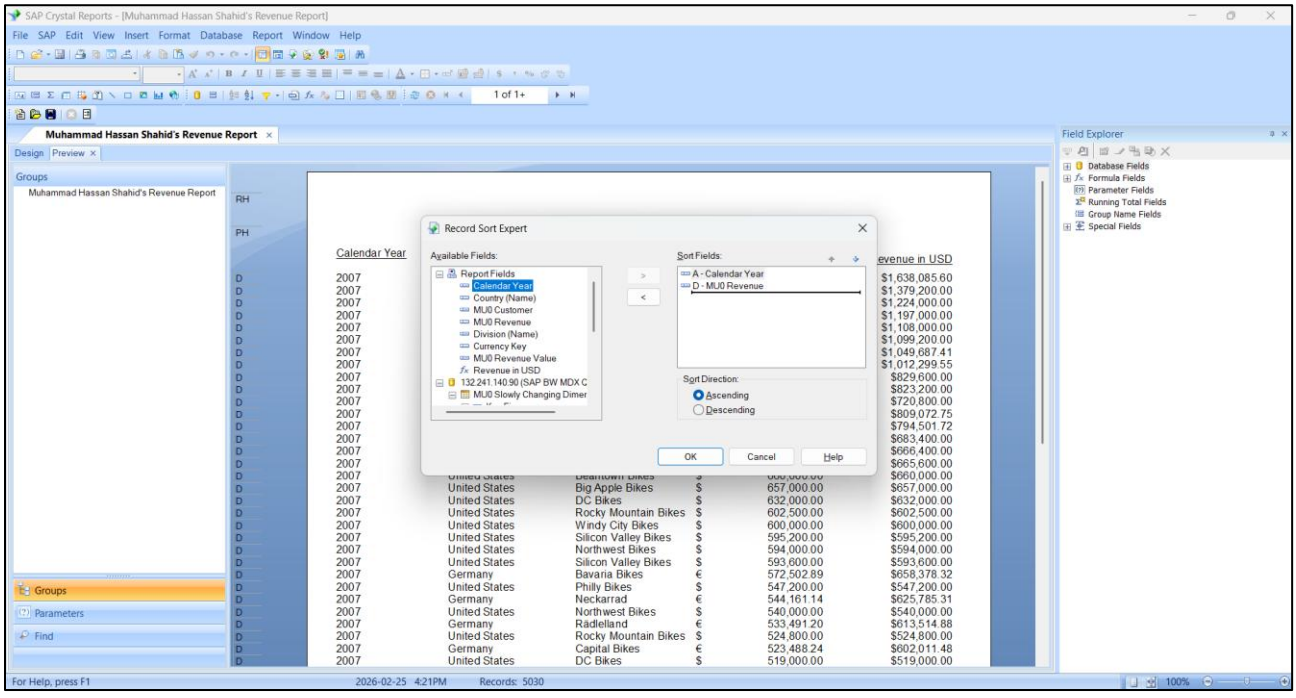
5.3.2. How can the sales data be organized and grouped by Calendar Year and Country in the report?

I navigated to the top menu and selected Report → Record Sort Expert. Fig. 5.4 shows the exact moment I accessed the Record Sort Expert from the main Report menu. In the Record Sort Expert dialog box, I located the Calendar Year field under "Available Fields" and moved it to the "Sort Fields" list using the arrow button. I set the Sort Direction to Ascending, ensuring the report starts with the earliest year.



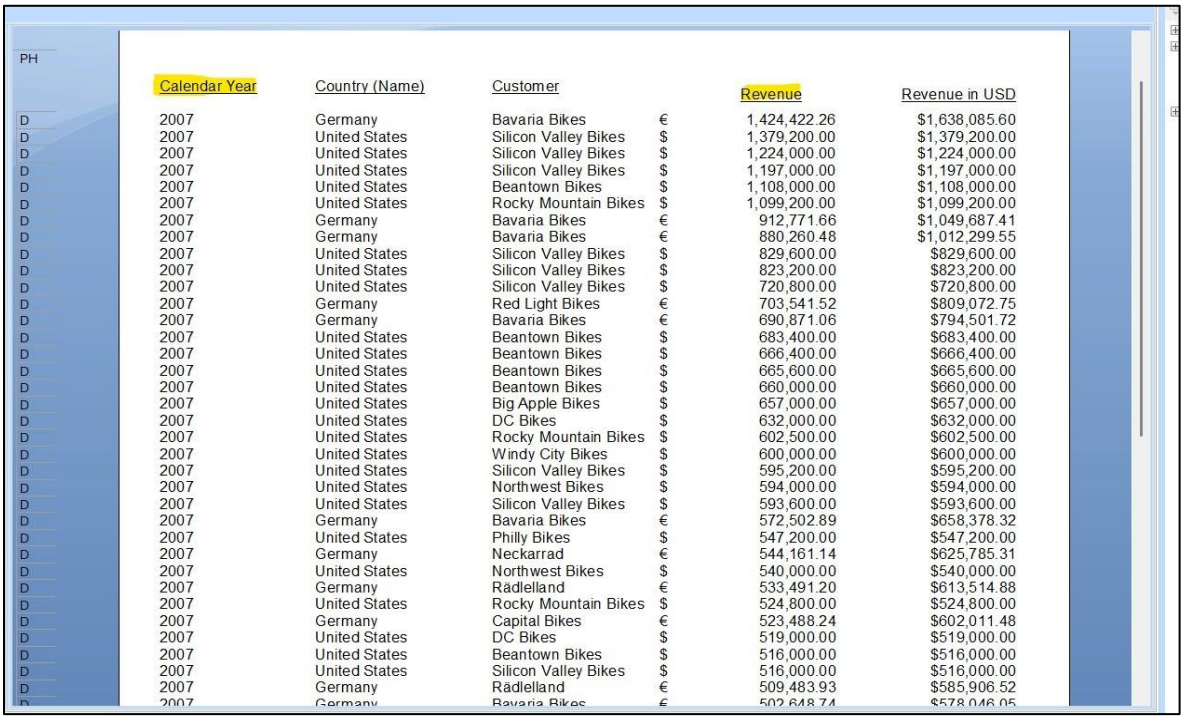
(Figure – 5.4)

Fig. 5.5 is the most important "technical" visual; it confirms the logic I applied; sorting first by Year (A) and then by Revenue (D). I then added the MU0 Revenue field to the sort list. Crucially, I changed its Sort Direction to Descending. This ensures that within each calendar year, the highest-grossing transactions appear at the top of the list.



(Figure – 5.5)

Once I clicked OK, the report layout updated immediately. As seen in Figure 28, the data is now perfectly organized: all records are grouped under 2007 (Ascending), but the Revenue column shows values decreasing from the highest amount ($1,424,422.26) downward (Descending).

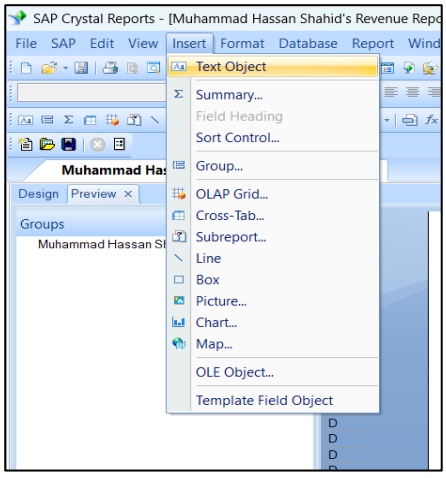


Calendar Year	Country (Name)	Customer		Revenue	Revenue in USD
2007	Germany	Bavaria Bikes	€	1,424,422.26	$1,638,085.60
2007	United States	Silicon Valley Bikes	$	1,379,200.00	$1,379,200.00
2007	United States	Silicon Valley Bikes	$	1,224,000.00	$1,224,000.00
2007	United States	Silicon Valley Bikes	$	1,197,000.00	$1,197,000.00
2007	United States	Beantown Bikes	$	1,108,000.00	$1,108,000.00
2007	United States	Rocky Mountain Bikes	$	1,099,200.00	$1,099,200.00
2007	Germany	Bavaria Bikes	€	912,771.66	$1,049,687.41
2007	Germany	Bavaria Bikes	€	880,260.48	$1,012,299.55
2007	United States	Silicon Valley Bikes	$	829,600.00	$829,600.00
2007	United States	Silicon Valley Bikes	$	823,200.00	$823,200.00
2007	United States	Silicon Valley Bikes	$	720,800.00	$720,800.00
2007	Germany	Red Light Bikes	€	703,541.52	$809,072.75
2007	Germany	Bavaria Bikes	€	690,871.06	$794,501.72
2007	United States	Beantown Bikes	$	683,400.00	$683,400.00
2007	United States	Beantown Bikes	$	666,400.00	$666,400.00
2007	United States	Beantown Bikes	$	665,600.00	$665,600.00
2007	United States	Beantown Bikes	$	660,000.00	$660,000.00
2007	United States	Big Apple Bikes	$	657,000.00	$657,000.00
2007	United States	DC Bikes	$	632,000.00	$632,000.00
2007	United States	Rocky Mountain Bikes	$	602,500.00	$602,500.00
2007	United States	Windy City Bikes	$	600,000.00	$600,000.00
2007	United States	Silicon Valley Bikes	$	595,200.00	$595,200.00
2007	United States	Northwest Bikes	$	594,000.00	$594,000.00
2007	United States	Silicon Valley Bikes	$	593,600.00	$593,600.00
2007	Germany	Bavaria Bikes	€	572,502.89	$658,378.32
2007	United States	Philly Bikes	$	547,200.00	$547,200.00
2007	Germany	Neckarrad	€	544,161.14	$625,785.31
2007	United States	Northwest Bikes	$	540,000.00	$540,000.00
2007	Germany	Rädlelland	€	533,491.20	$613,514.88
2007	United States	Rocky Mountain Bikes	$	524,800.00	$524,800.00
2007	Germany	Capital Bikes	€	523,488.24	$602,011.48
2007	United States	DC Bikes	$	519,000.00	$519,000.00
2007	United States	Beantown Bikes	$	516,000.00	$516,000.00
2007	United States	Silicon Valley Bikes	$	516,000.00	$516,000.00
2007	Germany	Rädlelland	€	509,483.93	$585,906.52
2007	Germany	Bavaria Bikes	€	502,648.74	$578,046.05

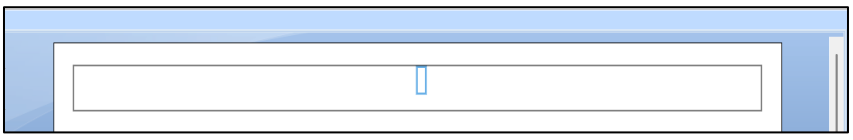
(Figure – 5.6)

5.3.3. How can a clear and professional report header be added to improve report readability?

As shown in fig. 5.7, I navigated to Insert → Text Object and placed the new object in the center of the Report Header (RH) section. I entered the title "Hassan's Report" into the text box. To ensure it looked like a formal header, I used the sizing handles to stretch the text box across the entire width of the report header section as shown in fig. 5.8. Using the alignment icons in the toolbar, I centered the text within the object to maintain visual balance.

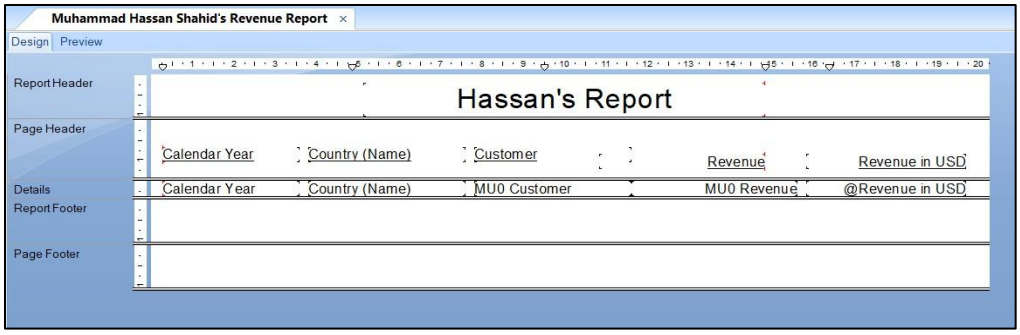


(Figure – 5.7)



(Figure – 5.8)

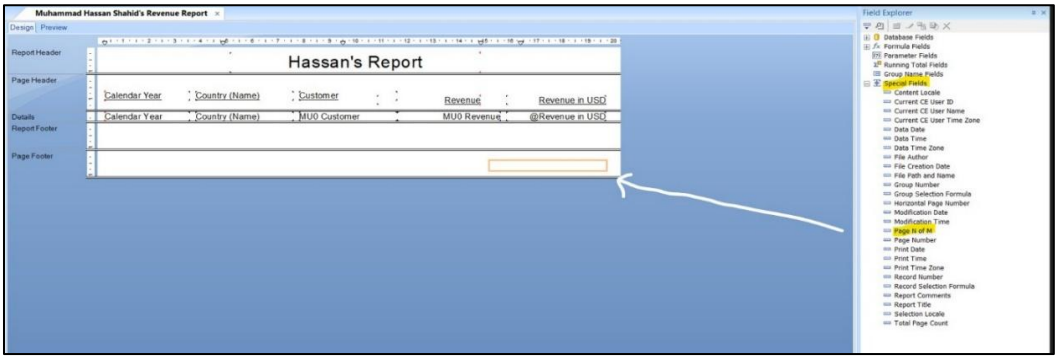
I highlighted the text, accessed Text Formatting, and increased the font size to 20pt to make the title prominent as shown in fig. 5.9 below.



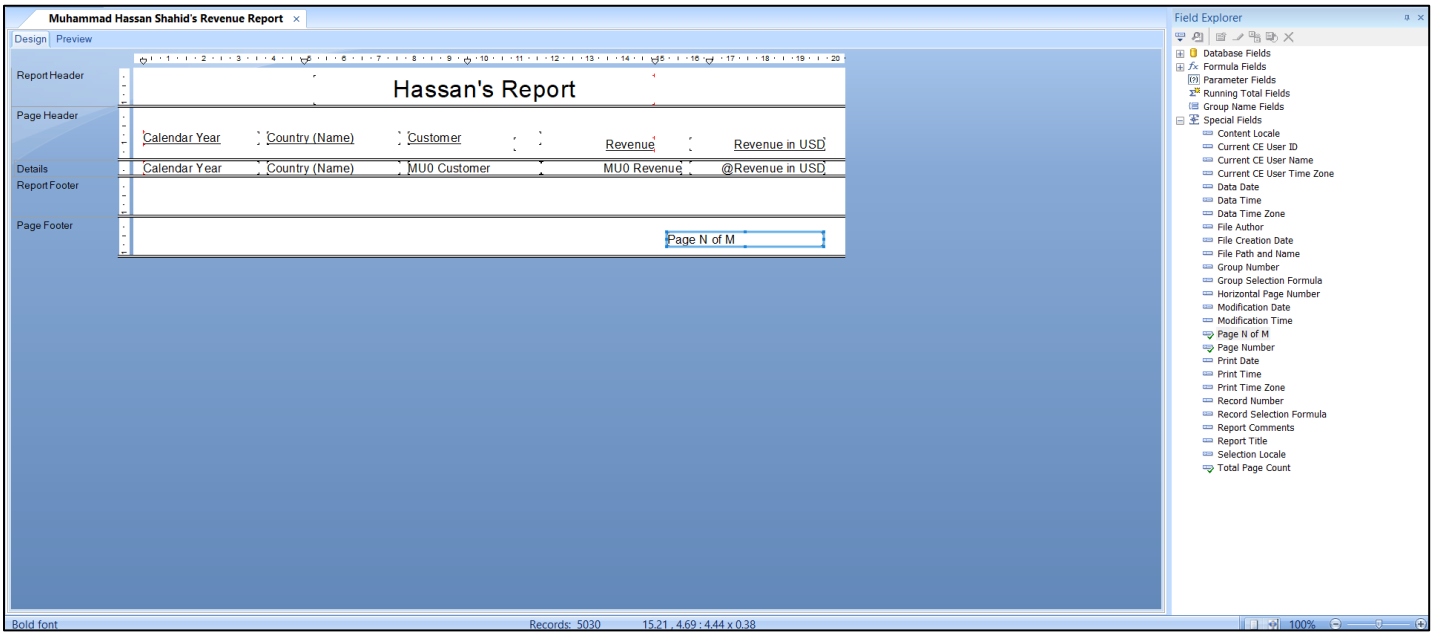
(Figure – 5.9)

5.3.4. How can a page footer with page numbers and copyright information be included in the report?

In the Field Explorer, I navigated to the Special Fields node. I selected the Page N of M field, which automatically calculates current and total pages and dragged it into the right-hand side of the Page Footer (PF) section as shown in figs. 5.10 & 5.11.

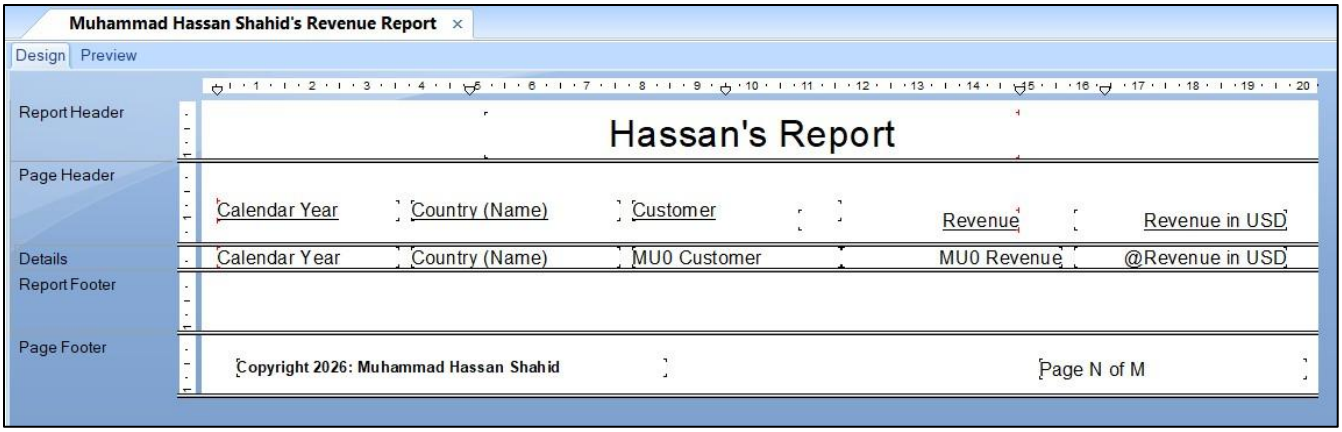


(Figure – 5.10)



(Figure – 5.11)

I then inserted a new Text Object into the center of the Page Footer. I entered the text "Copyright 2026: Muhammad Hassan Shahid," centered the object using the drag handles, and applied bold to match the styling of the page numbers. I also toggled between the Design and Preview panes to verify that the footer objects were correctly aligned and visible across different pages before saving the final report as shown in fig. 5.12.



(Figure – 5.12)

5.4 Critical Analysis: SAP Crystal Reports

SAP Crystal Reports is primarily designed for structured business reporting rather than exploratory data analysis. It allows report designers to connect directly to enterprise data sources such as SAP Business Warehouse and generate highly formatted reports with grouping, sorting, and summarized calculations. One of the strengths of SAP Crystal Reports is its ability to produce professional, well-formatted reports that follow organizational reporting standards. However, compared to modern visualization tools, Crystal Reports can feel less interactive and more rigid when exploring data. It is most effective for generating standardized operational reports, but it is less suitable for quick exploratory analysis or interactive dashboards.

5.5 Conclusion

Overall, SAP Crystal Reports is a powerful tool for creating structured and professional business reports. In this exercise, it helped transform raw sales data from the SAP Business Warehouse into a well-organized report that highlights revenue performance across years and countries. Features such as grouping, sorting, and summarized totals made it easier to interpret the data and present it in a clear reporting format. While the tool may not provide the same level of interactivity as modern visualization platforms, it remains highly valuable for producing reliable and standardized enterprise reports used by management for monitoring business performance.

Chapter 6: Tableau Desktop

6.1 Introduction of Tableau for Textual and Sentiment Analysis

Tableau is widely known for its strong data visualization capabilities, but it can also be effectively used for analyzing textual and sentiment-based data when combined with pre-processed inputs. In this exercise, Tableau is used to visualize sentiment scores derived from text data, allowing patterns in customer opinions to be explored in a structured and visual way. Rather than performing text processing directly, Tableau works with outputs such as sentiment scores (e.g., polarity or compound sentiment), which can then be analyzed across different dimensions like country, continent, or product category. This makes it easier to identify trends such as which regions have more positive or negative reviews, or whether certain characteristics (like price) influence customer sentiment. Overall, Tableau serves as a powerful tool for transforming unstructured text insights into clear, interactive visualizations that support better understanding and decision-making.

6.2 Dataset and Data Enrichment Workflow

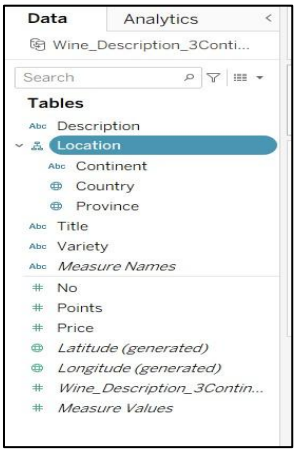
The analysis in this exercise is based on a wine reviews dataset, originally sourced from Kaggle and later reduced to a smaller sample for practical use. The dataset includes key attributes such as country, province, price, rating points, and textual descriptions of wines. Since the original dataset contains unstructured text in the form of wine descriptions, a data enrichment step was required. This was done using Communalytic, where sentiment analysis techniques were applied to convert textual data into numerical sentiment scores. These scores (such as polarity and compound sentiment) make it possible to analyze customer opinions quantitatively. In this dataset, variables like price and rating points act as measures, while country, province, and continent serve as dimensions. The enriched dataset allows for both numerical and sentiment-based analysis, making it possible to explore relationships between price, ratings, and customer perceptions. However, there are some limitations to consider. The dataset is based on a sampled subset, so it may not fully represent the global wine market. Additionally, sentiment scores are generated using automated tools, which may not always capture the full context or tone of a review. Despite these limitations, the dataset provides a useful foundation for exploring patterns in pricing, ratings, and customer sentiment in the form of the questions listed below:

- How do average wine prices and rating scores compare across different continents and countries?
- What is the relationship between wine price and rating points across countries and provinces of origin?
- How do average wine prices vary geographically across countries based on a choropleth map visualization?
- How does the distribution of sentiment scores differ across continents?
- Do higher-priced wines tend to receive more positive sentiment in reviews across different countries?
- Which country has the lowest average sentiment score, indicating the most negative reviews?

6.3 Deriving Business Insights and Findings

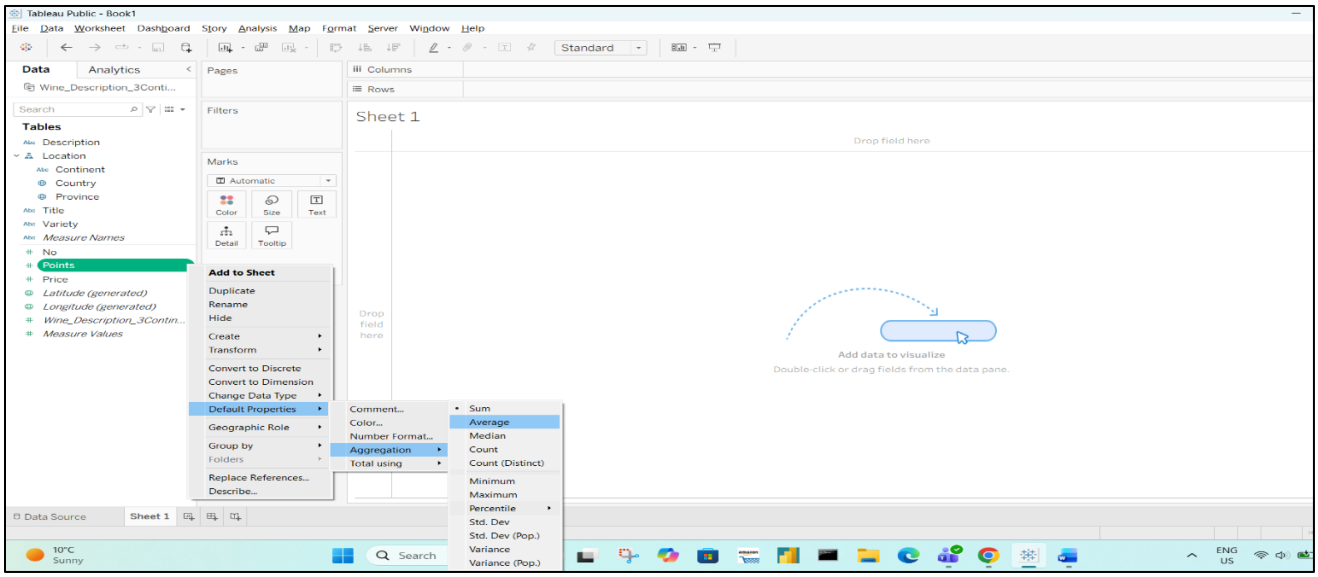
6.3.1. How do average wine prices and rating scores compare across different continents and countries?

The investigation began by establishing a structured foundation, where Continent, Country, and Province were merged into a single "Location Hierarchy." This was a crucial first step because it transformed a flat list of names into a dynamic "drill-down" system, allowing for a seamless transition from a global overview to specific regional details. By organizing the data this way, the analysis moved beyond simple categories and began to reflect the actual geographical layers of the wine industry as shown in Fig. 6.1 below.



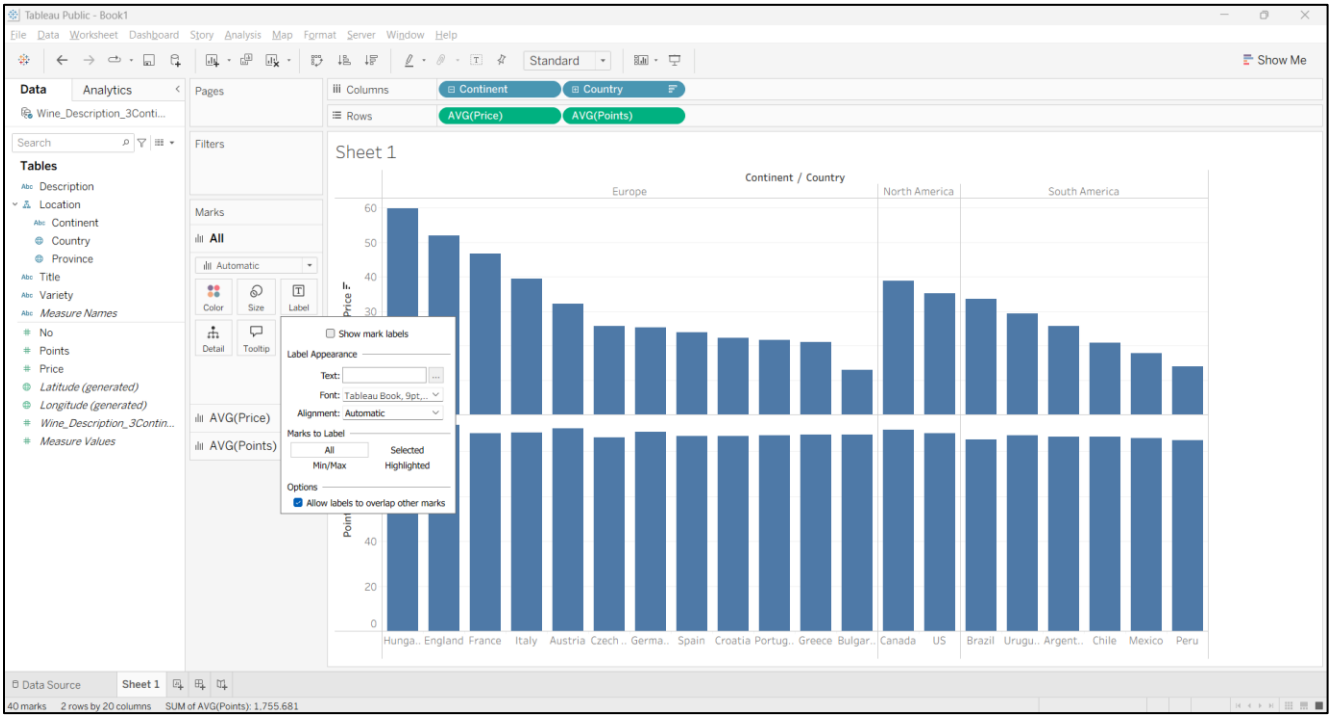
(Fig. 6.1)

Once the structure was in place, the focus shifted to ensuring the mathematical integrity of the comparison. By default, tableau attempted to calculate the total sum of prices and points, which would have unfairly favored high-volume producers; to correct this, the aggregation was manually shifted to average for both metrics. This shift in logic was the turning point of the analysis, as it leveled the playing field between large and small markets, allowing for an authentic comparison of the "typical" bottle's quality and cost regardless of a country's total output as shown in Fig. 6.2 below.



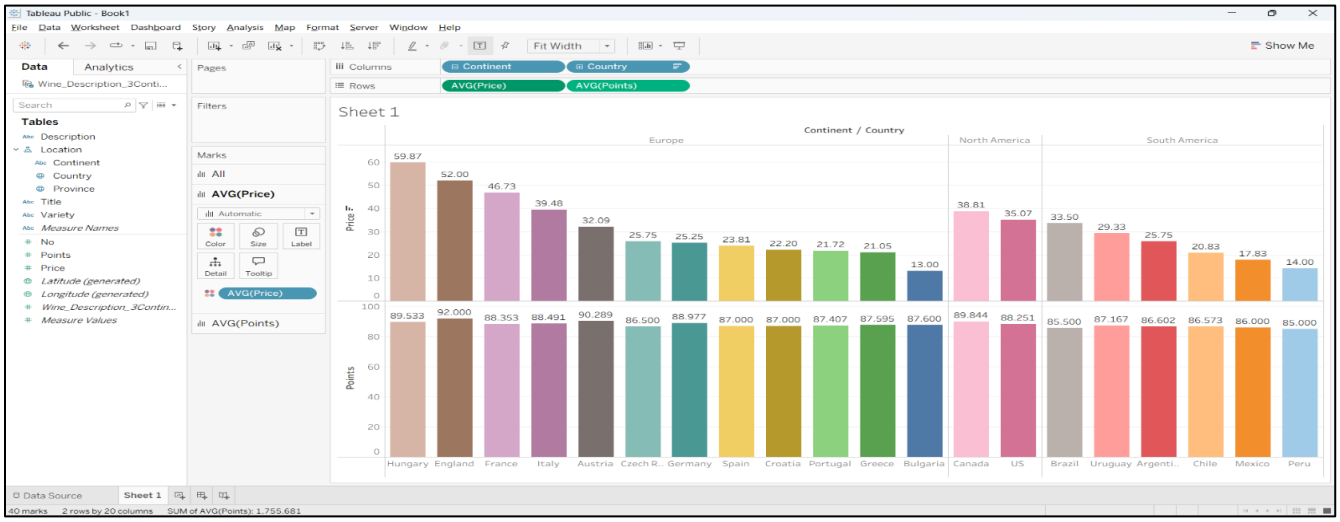
(Fig. 6.2)

To bring these numbers to meaning, the visual presentation was enhanced through the strategic use of labels and color scales. By enabling Mark Labels, the specific dollar values and rating points became immediately scannable. Simultaneously, dragging the Average Price onto the Color Mark transformed the bar chart's color. This made the data not just readable, but intuitive, allowing the eye to catch trends before even reading the axis as shown in Fig. 6.3 below.



(Fig. 6.3)

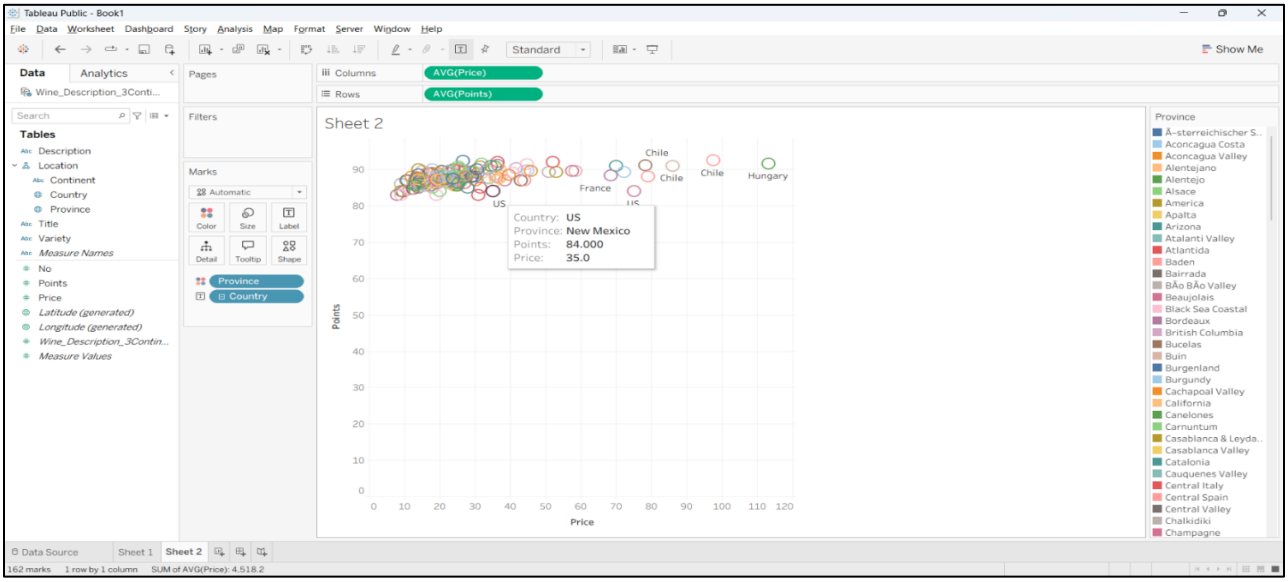
The culmination of this workflow revealed a compelling story about global wine value. The final visualization clearly identified Europe specifically Hungary and England as the leaders in premium pricing. While European prices were often double or triple those of South American counterparts like Chile or Peru, the average Rating Points remained remarkably consistent across all three continents. This final in the (Fig. 6.4) insight proved that while price is highly dependent on regional prestige and location, quality (as measured by points) is much more democratically distributed across the global market.



(Fig. 6.4)

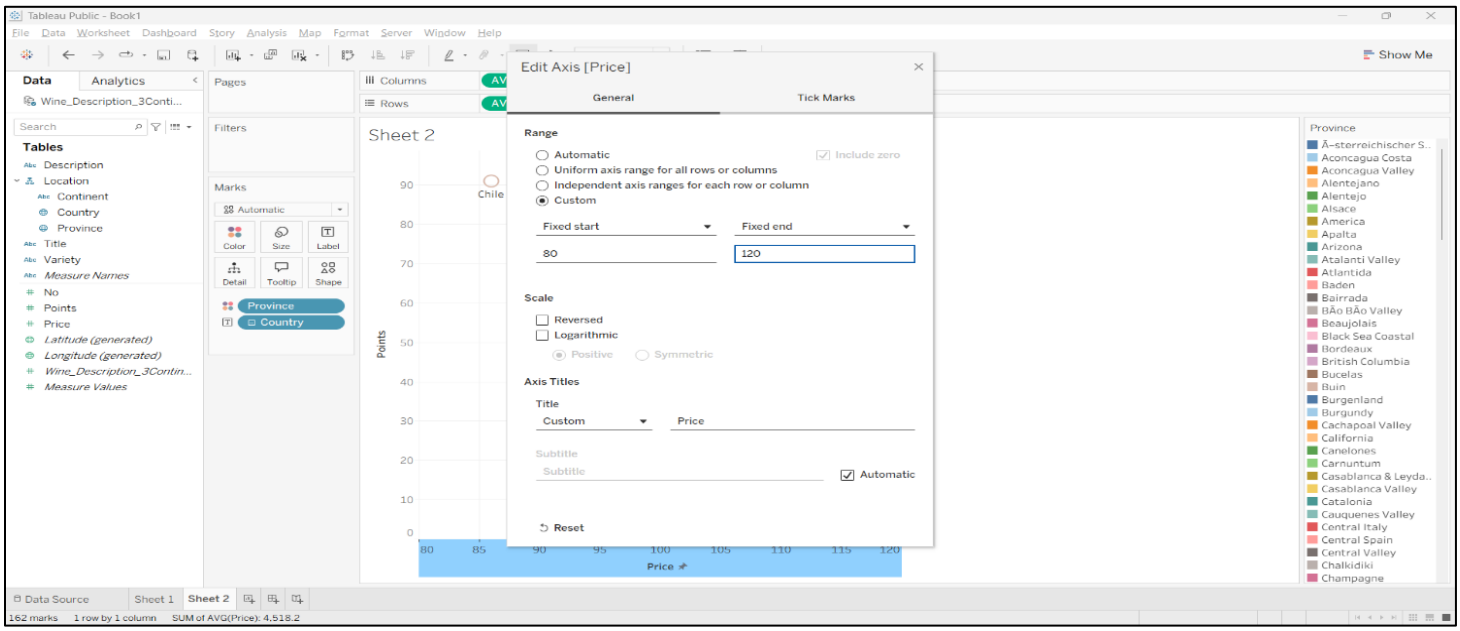
6.3.2. What is the relationship between wine price and rating points across countries and provinces of origin?

To understand the relationship between cost and quality, the analysis transitioned from basic bar charts to a more granular scatter plot investigation. This phase of the analysis was designed to determine if a higher price tag consistently results in a superior wine rating across different global provinces. The process began by plotting average price on the X-axis against average points on the Y-axis. To see the "fine print" of the data, Province was added to the Color mark and Country was added to the Detail mark (Fig. 6.5) below. This initially created a dense cluster of data points, showing a general upward trend where higher quality usually commands a higher price. However, with many wines retailing at lower price points, the "luxury" segment of the market was compressed and difficult to analyze individually.



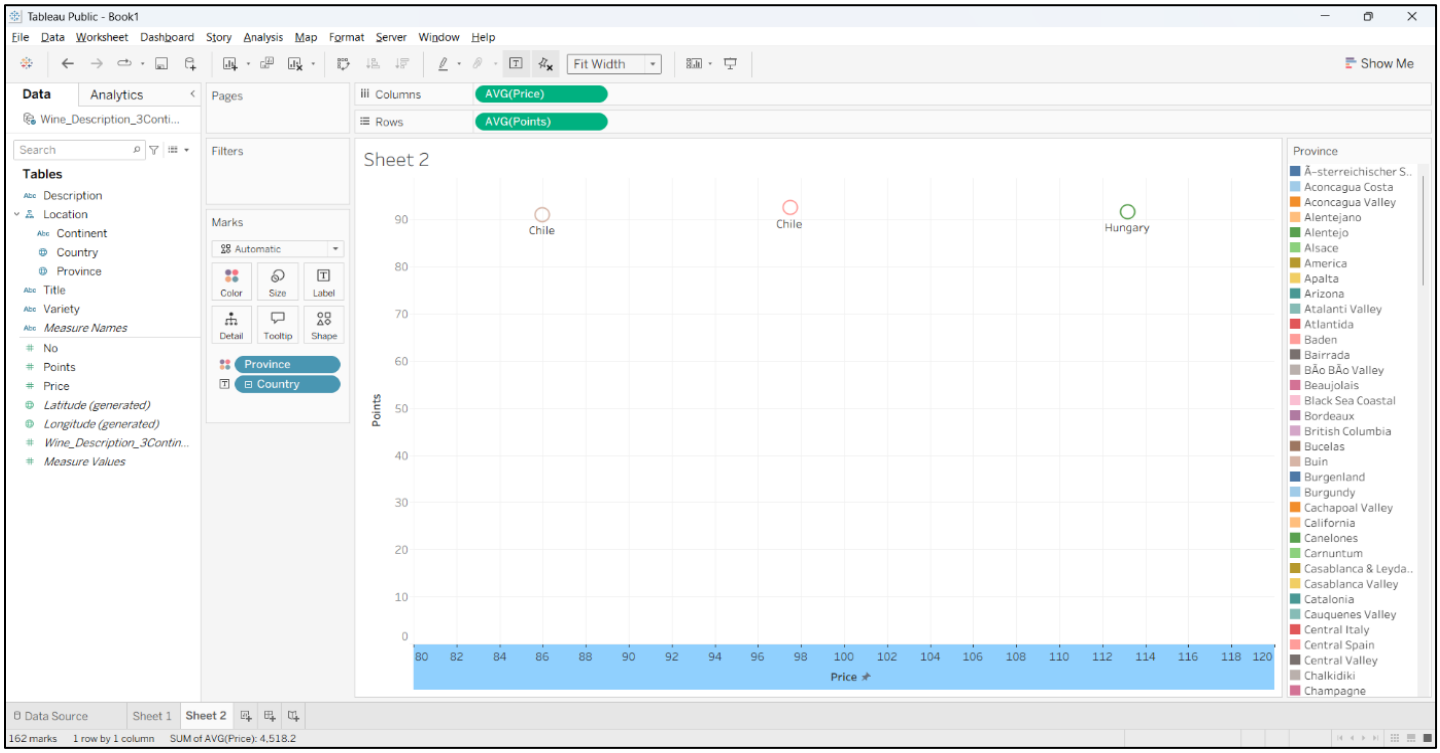
(Fig. 6.5)

To resolve this and "zoom in" on the premium market, the analytical focus was sharpened by modifying the scale of the visualization. By right clicking the price axis and selecting edit axis, the range was changed from automatic to a custom fixed start of 80 and an end of 120 as shown in the (Fig. 6.6) below.



(Fig. 6.6)

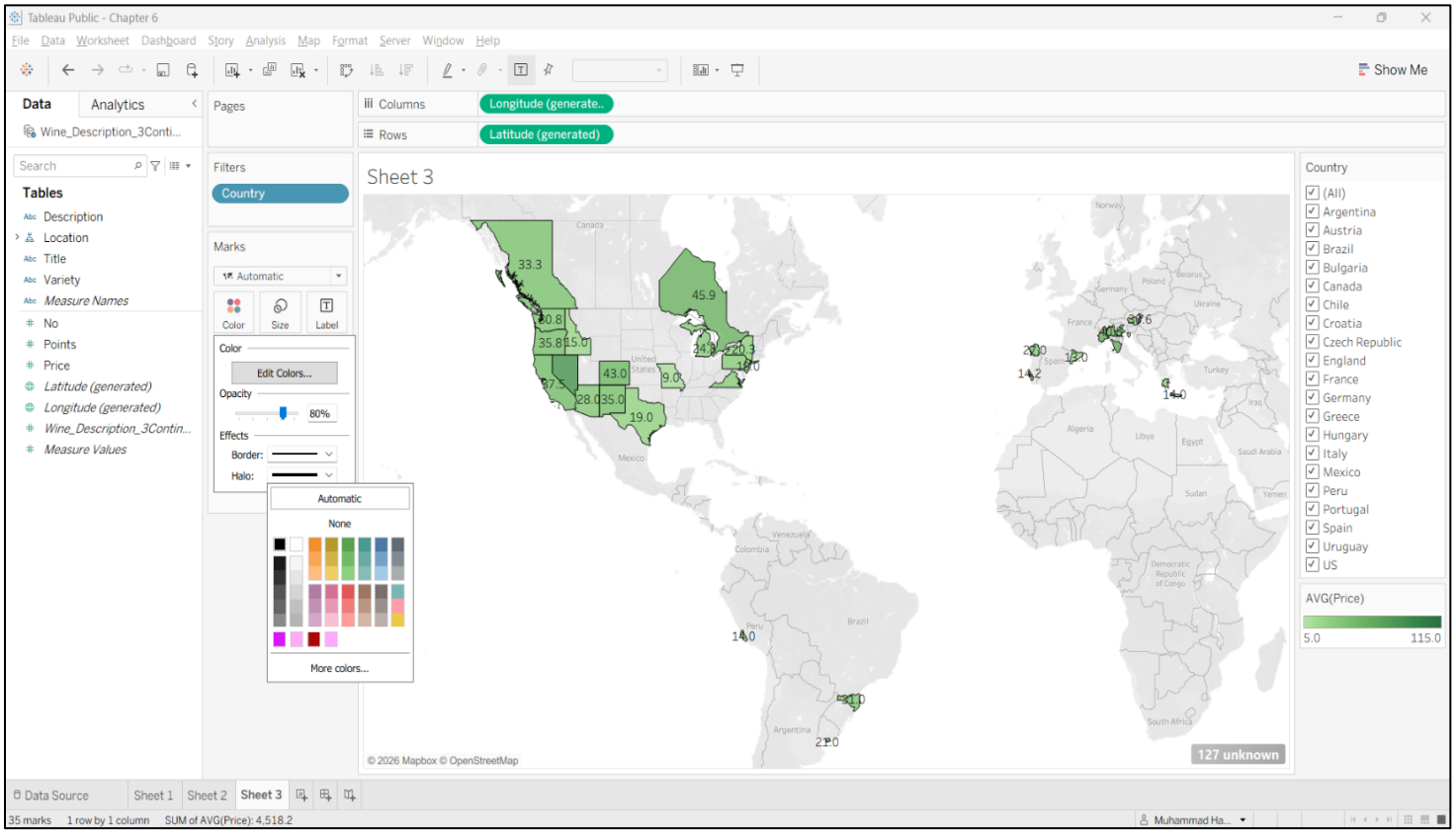
The resulting focused visualization (Fig. 6.7) provided the final clarity needed for the investigation. By isolating the $80 – $120 price range, it became clear how specific provinces like those in Hungary and Chile compete at the top tier. The final insight confirmed a positive correlation in the luxury segment as the investment per bottle increases, the consistency of high ratings (90+ points) becomes much more evident. This step-by-step refinement moved the analysis from a broad observation to a specific, actionable understanding of the premium wine market.



(Fig. 6.7)

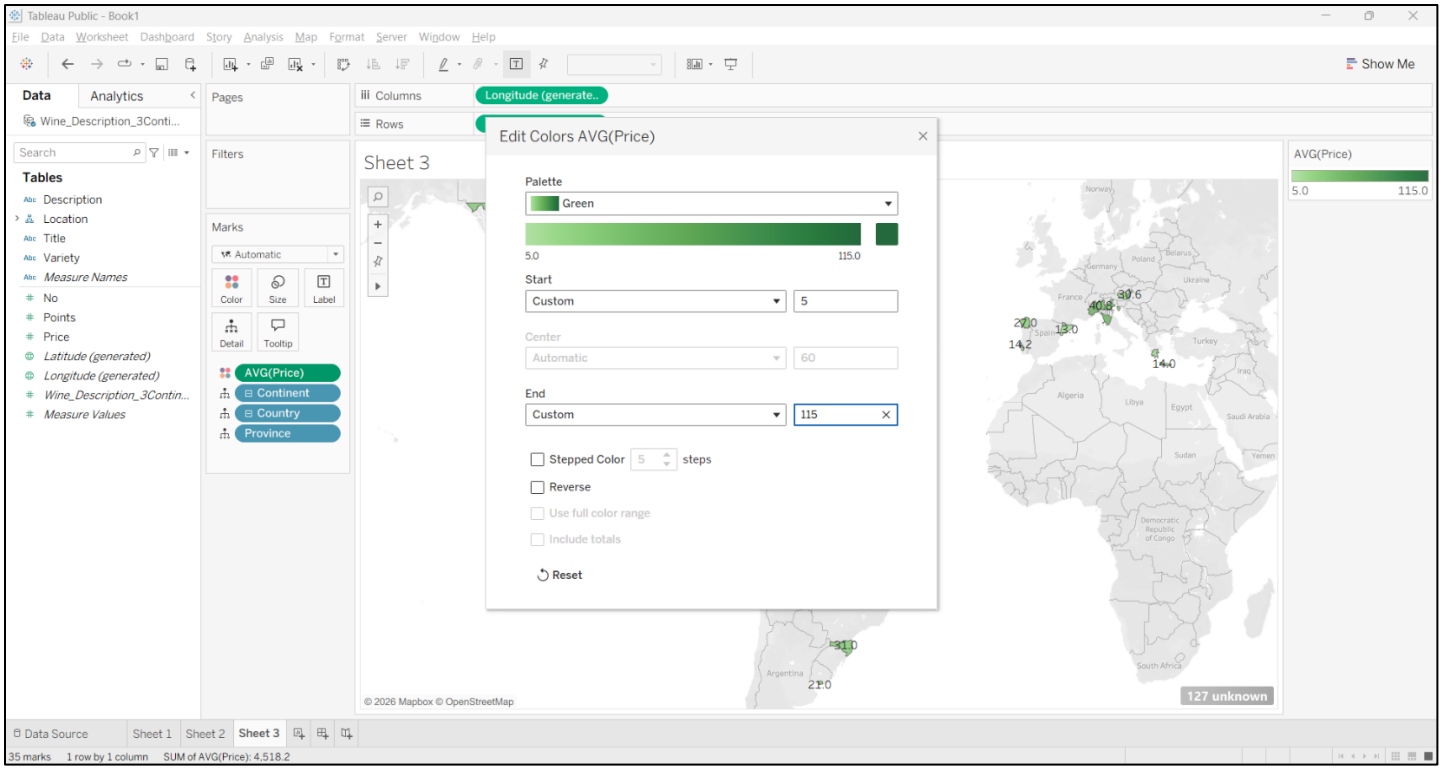
6.3.3. How do average wine prices vary geographically across countries based on a choropleth map visualization?

For this final part of this analysis, I looked at the global landscape through a Choropleth Map. This visualization allows to see how wine prices are distributed across the world, turning the data into a color-coded story of regional markets. The geographical analysis was built to identify which parts of the world are the "hotspots" for high-end wines and where the best value regions are located. The process began by dragging the Country dimension into the center of the workspace. Tableau automatically recognized the geographic names and generated a global map. To show the price differences, the average price was dropped onto the color mark as shown in the (Fig. 6.8) below. This immediately filled the map with different shades, where each country's color intensity represented its average wine price. At this stage, we could see that wine production is clustered in specific regions of North America, South America, and Europe.



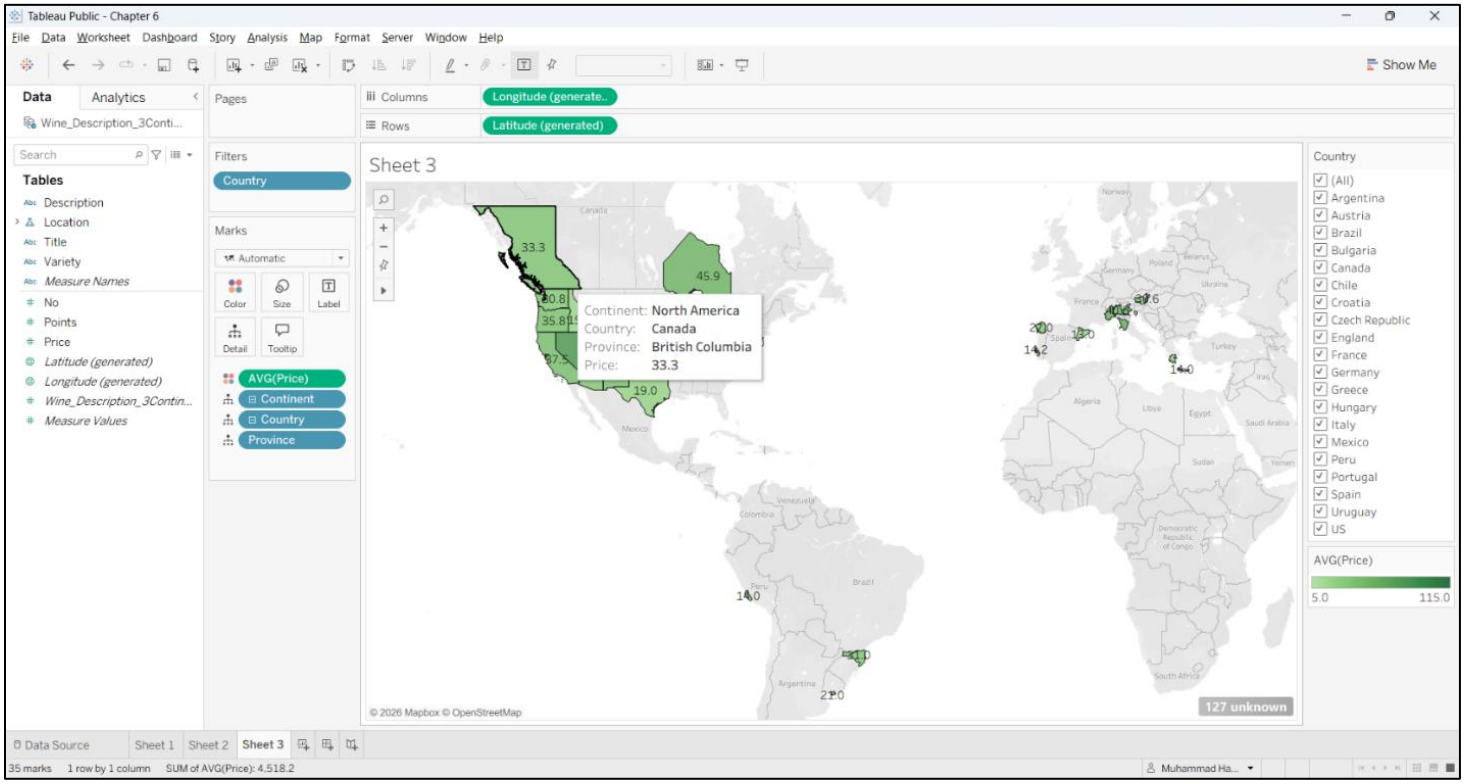
(Fig. 6.8)

To make the price differences more dramatic and easier to read, the color scale was customized. By going into the Edit Colors menu (Fig. 6.9), a "Green" palette was selected, and a Custom range was set with a start at 5 and an end at 115. By narrowing the range, the map became much more sensitive to price changes, allowing even small differences between neighboring countries to become visible.



(Fig. 6.9)

To ensure the map was informative at a glance, Border and Halo effects were added through the Color marks menu to make country outlines sharper (Fig. 6.10). Average price labels were also enabled so the exact dollar amount was visible over each region. This transformed the map from a simple drawing into a professional analytical tool where the user can hover over a province, like British Columbia in Canada, and see the specific price ($33.3) and rating context.

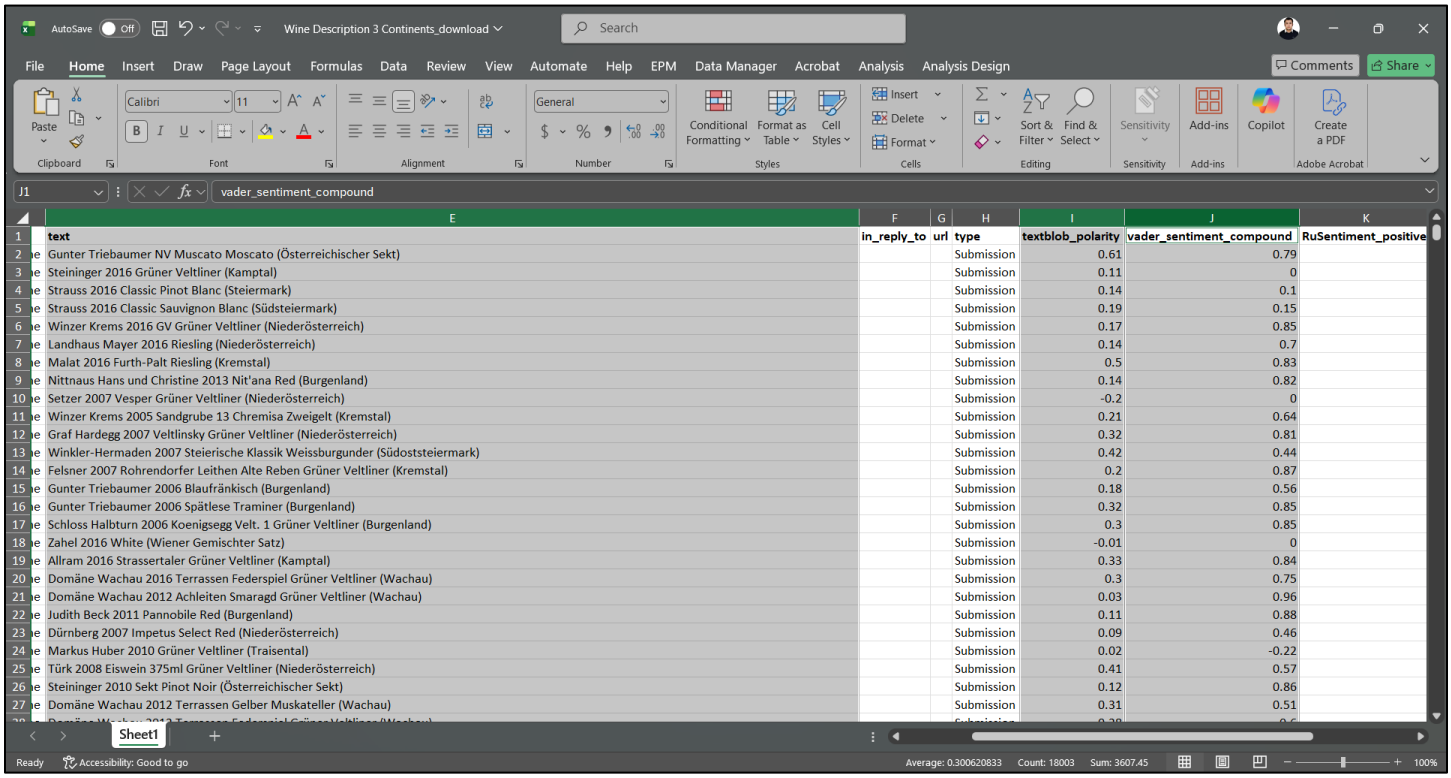


(Fig. 6.10)

The choropleth map reveals a clear geographic trend. Europe acts as the high-price epicenter, with the darkest green shades indicating premium markets. In contrast, South America serves as a "value hub" with lighter shades, while North America represents a middle-ground market. This map proves that wine pricing is not just about the bottle; it is heavily influenced by the prestige and economic factors of the geography it comes from.

6.3.4. How does the distribution of sentiment scores differ across continents?

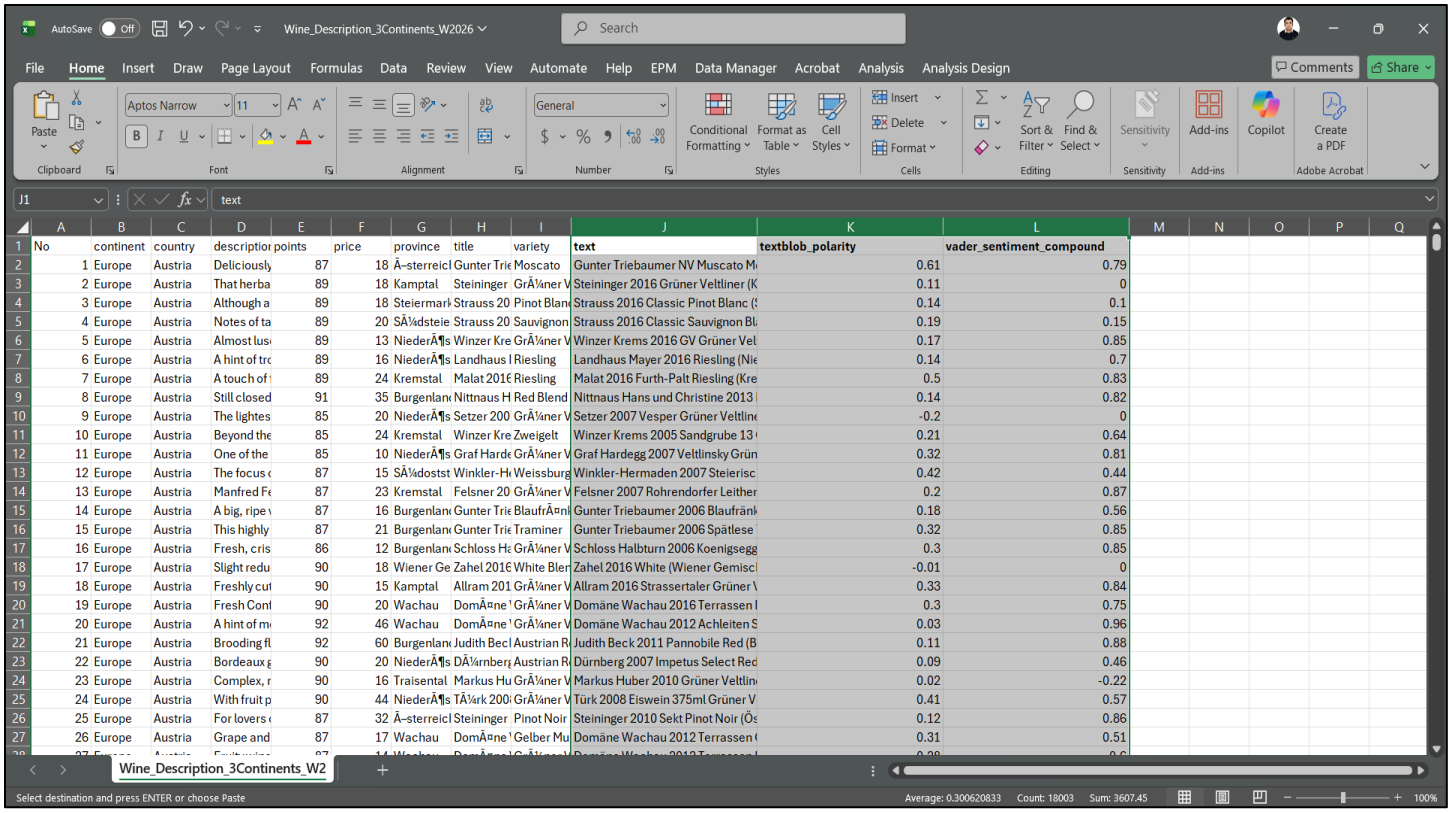
To analyze how sentiment is distributed across different regions, the study shifted from financial metrics to transition the data through communalytic to extract emotional context from the wine descriptions. The investigation began with the realization that a wine's "quality" isn't just a numerical score, it is also reflected in the language used by reviewers. Before any visualization could happen in Tableau, the raw wine descriptions had to be processed. The dataset was imported into communalytic to perform a sentiment analysis. This process generated three new columns in the Excel file, specifically the vader_sentiment_compound and textblob_polarity scores as shown in the (Fig. 6.11) below.



	text	in_reply_to	url	type	textblob_polarity	vader_sentiment_compound	RuSentiment_positive
2	Gunter Triebaumer NV Muscato Moscato (Österreichischer Sekt)			Submission	0.61	0.79	
3	Steininger 2016 Grüner Veltliner (Kamptal)			Submission	0.11	0	
4	Strauss 2016 Classic Pinot Blanc (Steiermark)			Submission	0.14	0.1	
5	Strauss 2016 Classic Sauvignon Blanc (Südsteiermark)			Submission	0.19	0.15	
6	Winzer Krems 2016 GV Grüner Veltliner (Niederösterreich)			Submission	0.17	0.85	
7	Landhaus Mayer 2016 Riesling (Niederösterreich)			Submission	0.14	0.7	
8	Malat 2016 Furth-Palt Riesling (Kremstal)			Submission	0.5	0.83	
9	Nittnaus Hans und Christine 2013 Nit'ana Red (Burgenland)			Submission	0.14	0.82	
10	Setzer 2007 Vesper Grüner Veltliner (Niederösterreich)			Submission	-0.2	0	
11	Winzer Krems 2005 Sandgrube 13 Chremisa Zweigelt (Kremstal)			Submission	0.21	0.64	
12	Graf Hardegg 2007 Veltlinsky Grüner Veltliner (Niederösterreich)			Submission	0.32	0.81	
13	Winkler-Hermaden 2007 Steierische Klassik Weissburgunder (Südoststeiermark)			Submission	0.42	0.44	
14	Felsner 2007 Rohrendorfer Leithen Alte Reben Grüner Veltliner (Kremstal)			Submission	0.2	0.87	
15	Gunter Triebaumer 2006 Blaufränkisch (Burgenland)			Submission	0.18	0.56	
16	Gunter Triebaumer 2006 Spätlese Traminer (Burgenland)			Submission	0.32	0.85	
17	Schloss Halbturn 2006 Koenigsegg Velt. 1 Grüner Veltliner (Burgenland)			Submission	0.3	0.85	
18	Zahel 2016 White (Wiener Gemischter Satz)			Submission	-0.01	0	
19	Allram 2016 Strassertaler Grüner Veltliner (Kamptal)			Submission	0.33	0.84	
20	Domäne Wachau 2016 Terrassen Federspiel Grüner Veltliner (Wachau)			Submission	0.3	0.75	
21	Domäne Wachau 2012 Achleiten Smaragd Grüner Veltliner (Wachau)			Submission	0.03	0.96	
22	Judith Beck 2011 Pannobile Red (Burgenland)			Submission	0.11	0.88	
23	Dürnberg 2007 Impetus Select Red (Niederösterreich)			Submission	0.09	0.46	
24	Markus Huber 2010 Grüner Veltliner (Traisental)			Submission	0.02	-0.22	
25	Türk 2008 Eiswein 375ml Grüner Veltliner (Niederösterreich)			Submission	0.41	0.57	
26	Steininger 2010 Sekt Pinot Noir (Österreichischer Sekt)			Submission	0.12	0.86	
27	Domäne Wachau 2012 Terrassen Gelber Muskateller (Wachau)			Submission	0.31	0.51	

(Fig. 6.11)

However, because the original dataset was large and unstructured, a bit of data cleaning was required. The new sentiment columns had to be carefully aligned with the original geographical data (Continent, Country, Province) to ensure that every sentiment score was correctly linked to its place of origin (Fig. 6.12) as shown below.



No	continent	country	description	points	price	province	title	variety	text	textblob_polarity	vader_sentiment_compound
1	Europe	Austria	Deliciously	87	18	Ã–sterreich	Gunter Trie	Moscato	Gunter Triebaumer NV Muscato M	0.61	0.79
2	Europe	Austria	That herba	89	18	Kamptal	Steininger	GrÃ¼ner V	Steininger 2016 Grüner Veltliner (K	0.11	0
3	Europe	Austria	Although a	89	18	Steiermark	Strauss 20	Pinot Blan	Strauss 2016 Classic Pinot Blanc (S	0.14	0.1
4	Europe	Austria	Notes of ta	89	20	SÃ¼dsteie	Strauss 20	Sauvignon	Strauss 2016 Classic Sauvignon Bl	0.19	0.15
5	Europe	Austria	Almost lus	89	13	NiederÃ¶s	Winzer Kre	GrÃ¼ner V	Winzer Krems 2016 GV Grüner Vel	0.17	0.85
6	Europe	Austria	A hint of tro	89	16	NiederÃ¶s	Landhaus I	Riesling	Landhaus Mayer 2016 Riesling (Nie	0.14	0.7
7	Europe	Austria	A touch of	89	24	Kremstal	Malat 2016	Riesling	Malat 2016 Furth-Palt Riesling (Kre	0.5	0.83
8	Europe	Austria	Still closed	91	35	Burgenland	Nittnaus H	Red Blend	Nittnaus Hans und Christine 2013	0.14	0.82
9	Europe	Austria	The lightes	85	20	NiederÃ¶s	Setzer 200	GrÃ¼ner V	Setzer 2007 Vesper Grüner Veltline	-0.2	0
10	Europe	Austria	Beyond the	85	24	Kremstal	Winzer Kre	Zweigelt	Winzer Krems 2005 Sandgrube 13	0.21	0.64
11	Europe	Austria	One of the	85	10	NiederÃ¶s	Graf Harde	GrÃ¼ner V	Graf Hardegg 2007 Veltlinsky Grün	0.32	0.81
12	Europe	Austria	The focus	87	15	SÃ¼dsteir	Winkler-H	Weissburg	Winkler-Hermaden 2007 Steierisc	0.42	0.44
13	Europe	Austria	Manfred Fe	87	23	Kremstal	Felsner 20	GrÃ¼ner V	Felsner 2007 Rohrendorfer Leither	0.2	0.87
14	Europe	Austria	A big, ripe	87	16	Burgenland	Gunter Trie	BlaufrÃ¤nk	Gunter Triebaumer 2006 Blaufränk	0.18	0.56
15	Europe	Austria	This highly	87	21	Burgenland	Gunter Trie	Traminer	Gunter Triebaumer 2006 Spätlese	0.32	0.85
16	Europe	Austria	Fresh, cris	86	12	Burgenland	Schloss Hã	GrÃ¼ner V	Schloss Halbturn 2006 Koenigsegg	0.3	0.85
17	Europe	Austria	Slight redu	90	18	Wiener Ge	Zahel 2016	White Blen	Zahel 2016 White (Wiener Gemisch	-0.01	0
18	Europe	Austria	Freshly cut	90	15	Kamptal	Allram 201	GrÃ¼ner V	Allram 2016 Strassertaler Grüner \	0.33	0.84
19	Europe	Austria	Fresh Conl	90	20	Wachau	DomÃ¤ne '	GrÃ¼ner V	Domäne Wachau 2016 Terrassen I	0.3	0.75
20	Europe	Austria	A hint of m	92	46	Wachau	DomÃ¤ne '	GrÃ¼ner V	Domäne Wachau 2012 Achleiten S	0.03	0.96
21	Europe	Austria	Brooding fl	92	60	Burgenland	Judith Bec	Austrian R	Judith Beck 2011 Pannobile Red (B	0.11	0.88
22	Europe	Austria	Bordeaux g	90	20	NiederÃ¶s	DÃ¼rnberg	Austrian R	Dürnberg 2007 Impetus Select Red	0.09	0.46
23	Europe	Austria	Complex, r	90	16	Traisental	Markus Hu	GrÃ¼ner V	Markus Huber 2010 Grüner Veltlin	0.02	-0.22
24	Europe	Austria	With fruit p	90	44	NiederÃ¶s	TÃ¼rk 200	GrÃ¼ner V	Türk 2008 Eiswein 375ml Grüner V	0.41	0.57
25	Europe	Austria	For lovers	87	32	Ã–sterreich	Steininger	Pinot Noir	Steininger 2010 Sekt Pinot Noir (Ös	0.12	0.86
26	Europe	Austria	Grape and	87	17	Wachau	DomÃ¤ne '	Gelber Mu	Domäne Wachau 2012 Terrassen (	0.31	0.51

(Fig. 6.12)

Once the enriched file was brought back into Tableau, an area chart was created to visualize the distribution of these scores (Fig. 6.13). By placing Compound Sentiment (bin) on the Columns and the Count of Compound Sentiment on the Rows, the analysis revealed the "shape" of consumer emotion across Europe, North America, and South America.

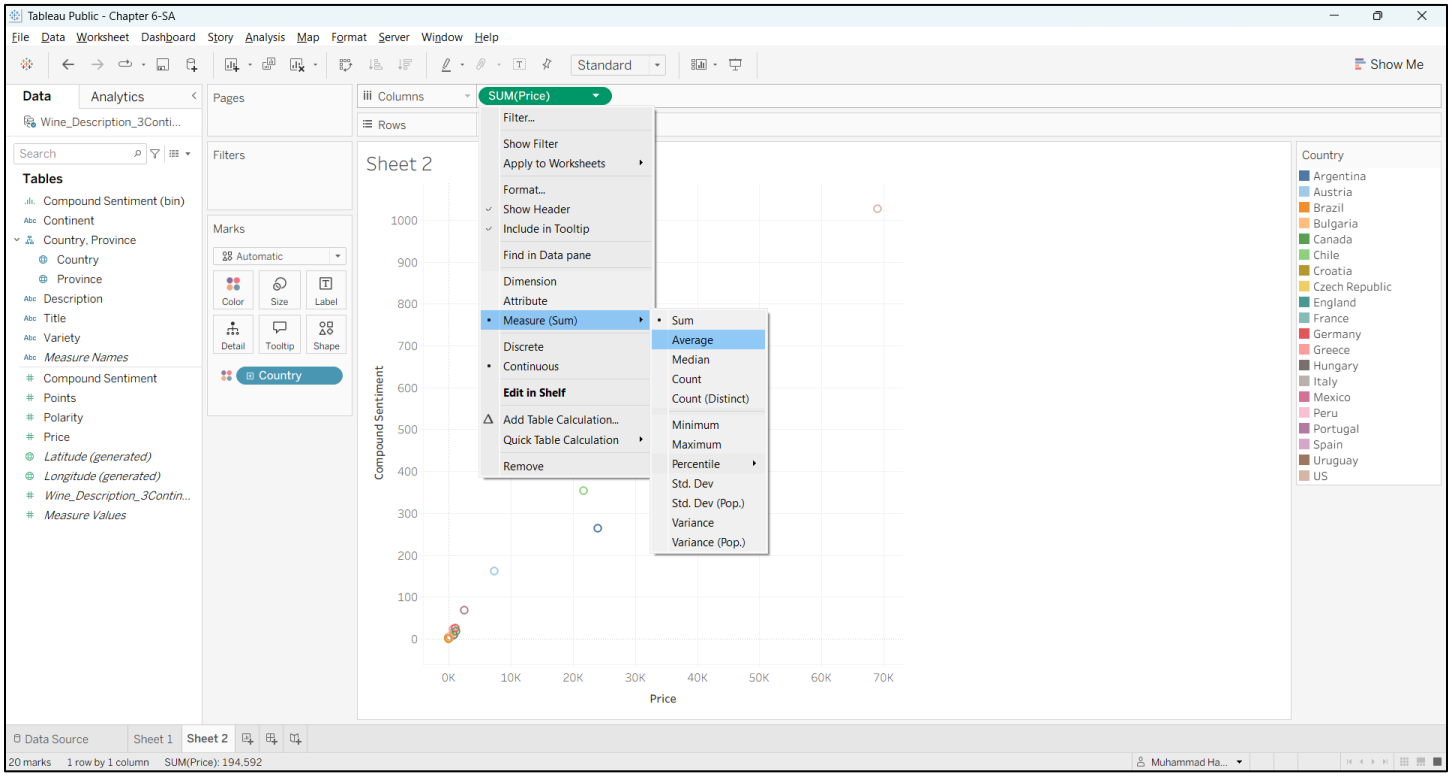


(Fig. 6.13)

The resulting visual provided a fascinating cross-continental comparison. Across all three continents, the distribution is heavily "right-skewed." This means that the vast majority of reviews cluster between 0.4 and 1.0, indicating that wine descriptions are overwhelmingly positive regardless of where the wine is produced. Europe shows the highest volume of reviews, with a massive peak of 357 reviews at the highest sentiment bin. North America and South America follow a nearly identical pattern, with peaks at 350 and 327 respectively. This distribution proves that the "language of wine" is globally consistent. Whether a reviewer is describing a luxury bottle, they tend to use highly favorable and evocative language. The similarity in these "mountain peaks" across the chart suggests that the emotional experience of wine is a universal constant, even if the price points vary significantly.

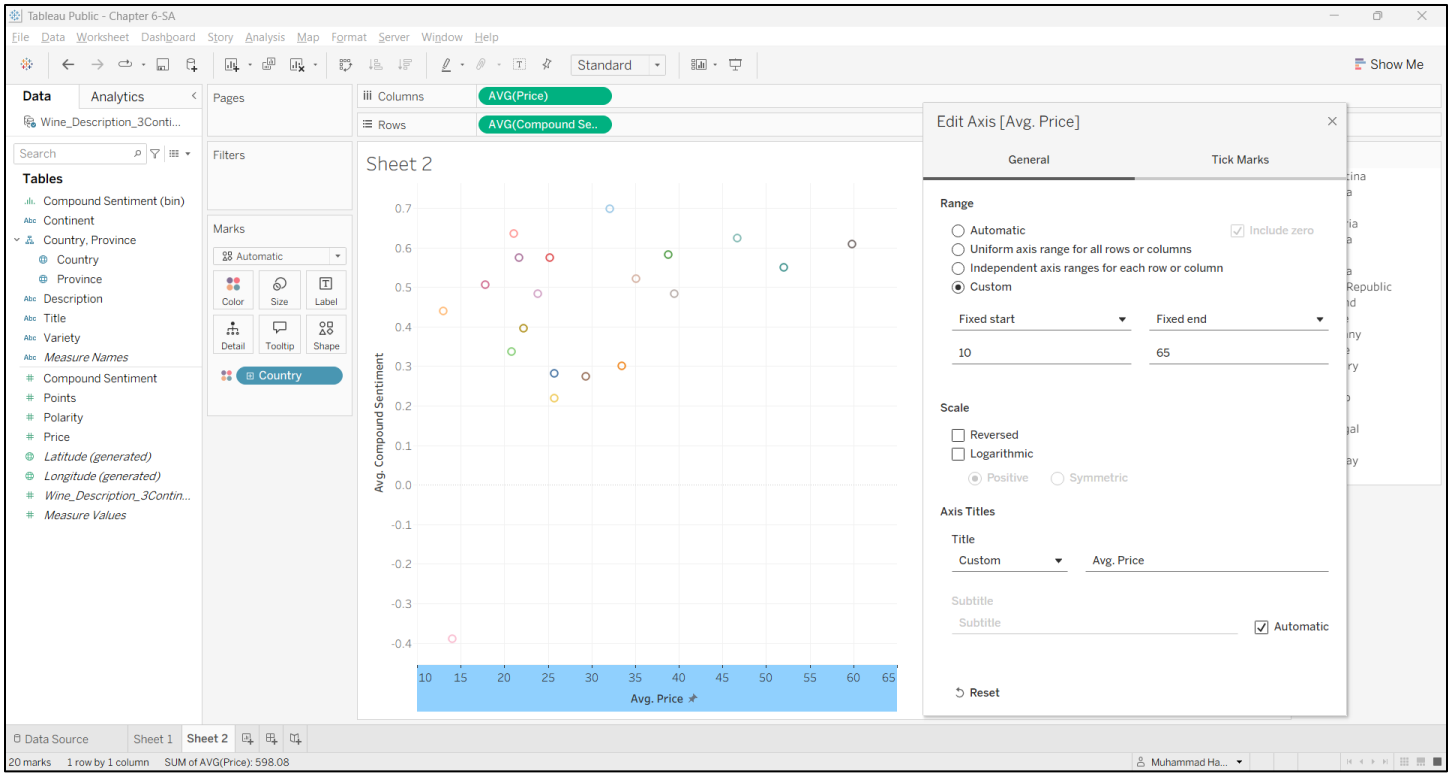
6.3.5. Do higher average-priced wines tend to receive more positive sentiment in reviews across different countries?

This investigation combined the financial metrics from the original dataset with the emotional insights derived from the Communalytic processing. The process began by dragging Price to the Columns shelf and the newly calculated Average Compound Sentiment to the Rows shelf. To fix this, I manually changed the measure for both Price and Sentiment from SUM to AVERAGE. This adjustment was vital; it ensured the analysis compared the typical sentiment of a bottle rather than just counting the total number of reviews in the (Fig. 6.14) below. To differentiate the data by region, Country was added to the color mark.



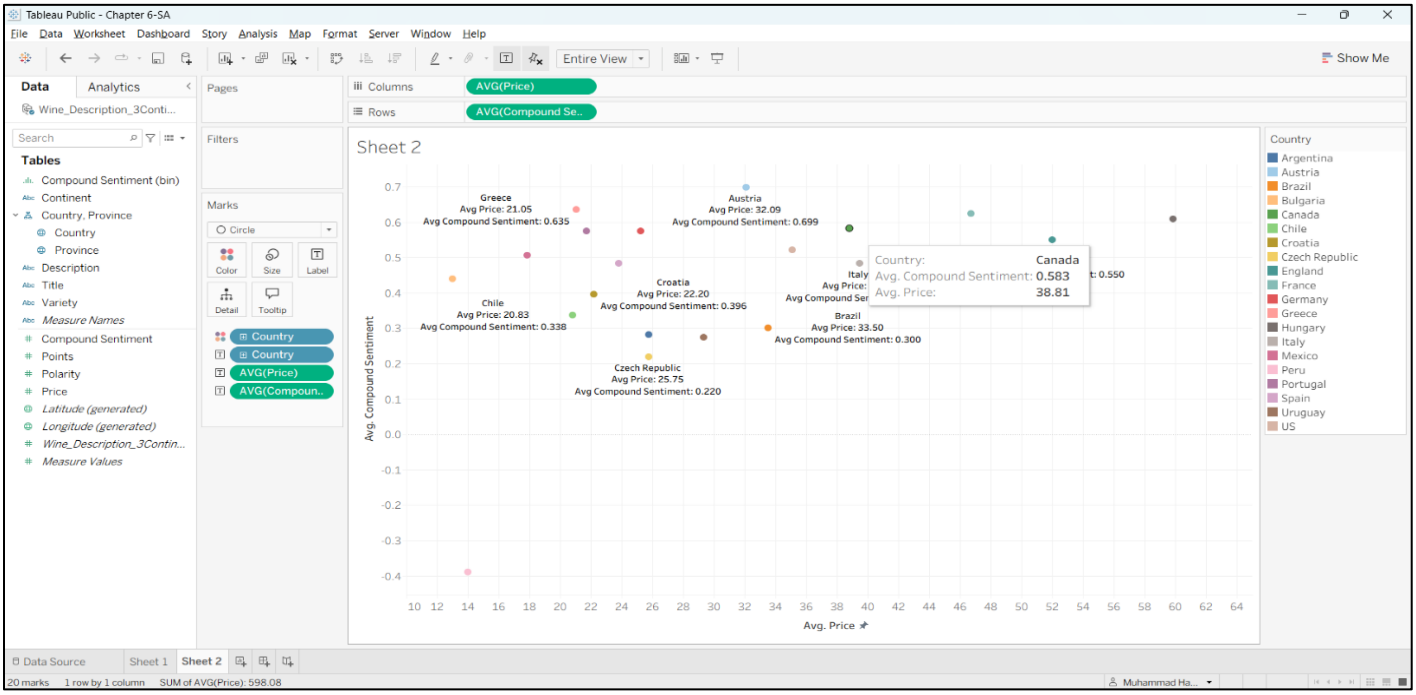
(Fig. 6.14)

Even with averages, the data was spread across a wide range that included several low-price outliers. To zoom in on the most relevant market activity, I utilized the Edit Axis feature for the average price (Fig. 6.15). By setting a Custom Range with a fixed start of 10 and a fixed end of 65, I removed the noise from the visualization.



(Fig. 6.15)

The final step involved adding Mark Labels to key data points to provide immediate context (Fig. 6.16). By hovering over a specific point, such as Canada, the analysis reveals an Average Price of $38.81 paired with a very high Average Compound Sentiment of 0.583. Similarly, Austria stands out as a high-performing outlier with a sentiment score of nearly 0.700 at a $32.09 price point.

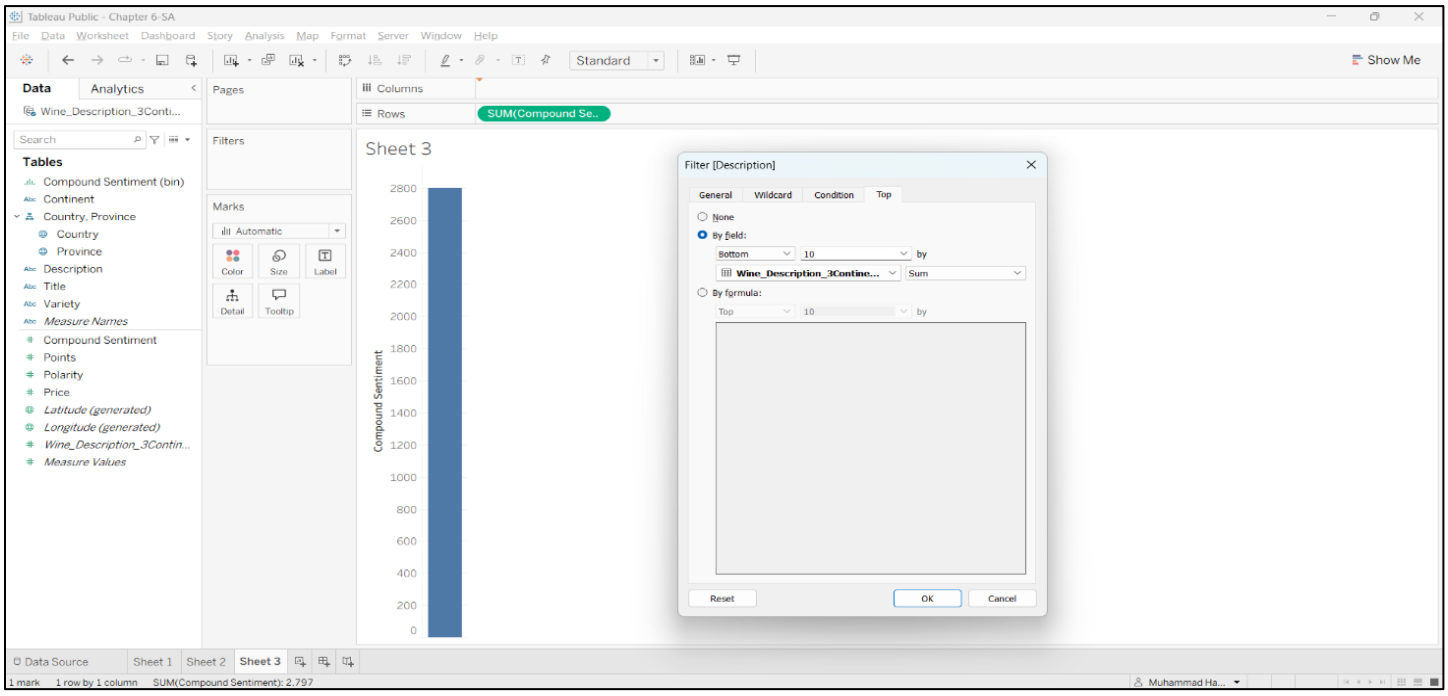


(Fig. 6.16)

The completed visualization confirms a general positive trend: as the average price of wine increases, the average sentiment tends to follow an upward trajectory. However, the data also reveals "Value Champions", countries like Greece and Austria that achieve exceptionally high sentiment scores despite having mid-range prices. This proves that while higher-priced wines generally receive more positive reviews, price is not the only driver of consumer happiness; regional quality and reviewer expectations also play a significant role in the emotional language of a review.

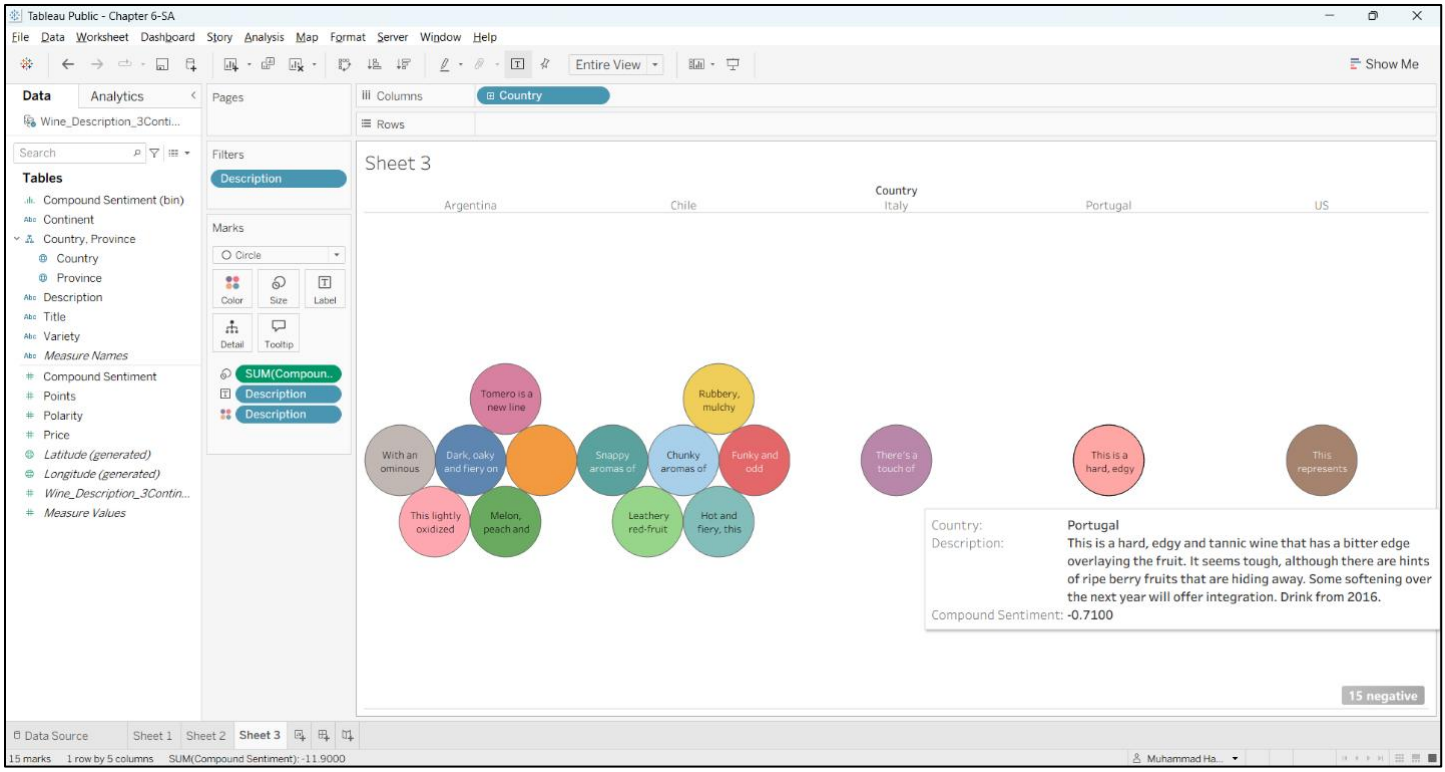
6.3.6. Which country has the lowest average sentiment score, indicating the most negative reviews?

The investigation began by applying a Top Filter to the Description field, specifically selecting the "Bottom 10" records based on the Sum of Compound Sentiment. This technical step was necessary to strip away the thousands of positive reviews and focus exclusively on the instances where reviewers were genuinely dissatisfied as shown in the (Fig. 6.17) below.



(Fig. 6.17)

By visualizing these bottom results in a Circle View, several key patterns emerged regarding the specific countries and vocabulary associated with low sentiment scores. The analysis highlighted that negative sentiment is not restricted to one region; instead, it appears as "pockets" of dissatisfaction across Portugal, the US, Italy, Chile, and Argentina. While the average sentiment across all countries is positive, these filtered outliers show Compound Sentiment scores reaching as low as -0.7100 as shown in the (Fig. 6.18) below.



(Fig. 6.18)

A review of the descriptions within these circles reveals a recurring use of harsh, industrial, or unappealing terminology. For example, a highly negative review from Portugal described the wine as having a "hard, edgy and tannic" character with a "bitter edge". Other low-scoring reviews featured descriptors like "Rubbery, mulchy," "Funky and odd," and "with an ominous". This deep dive confirms that while the global wine market is overwhelmingly positive, the lowest sentiment scores are often driven by specific sensory failures; typically bitterness, excessive tannins, or "off" aromas (like rubber or mulch). By isolating these records, the analysis moves beyond broad averages to identify the exact "red flags" that lead to poor consumer reception in international markets.

6.4 Critical Analysis of Using Tableau for Unstructured Data Critical Analysis

Using Tableau for unstructured data analysis has both strengths and limitations. One of its main advantages is its ability to present complex information in a simple and visually appealing way. Once text data has been converted into structured formats like sentiment scores, Tableau makes it easy to identify trends, compare regions, and explore relationships between variables. For example, it becomes straightforward to analyze how sentiment varies across continents or whether higher-priced wines receive better reviews. The interactive nature of Tableau dashboards also allows users to filter and drill down into specific areas, which improves the depth of analysis. However, Tableau itself does not perform text processing or sentiment analysis. It relies on external tools, such as Communalytic, to convert unstructured text into usable data. This means that the quality of analysis in Tableau depends heavily on the accuracy of the preprocessing step. If sentiment scores are inaccurate or biased, the visualizations may lead to misleading conclusions. In addition, Tableau may not capture deeper linguistic patterns such as sarcasm, context, or nuanced opinions, since it works with simplified numerical representations of text. Therefore, while Tableau is excellent for visualization and interpretation, it should ideally be used alongside specialized text analysis tools for more comprehensive insights.

6.5 Conclusion

In conclusion, Tableau proves to be a highly effective tool for visualizing and interpreting both structured and sentiment-enriched data. It allows users to explore patterns in wine prices, ratings, and customer opinions in a clear and interactive manner. Through different visualizations such as bar charts, scatter plots, and maps, it becomes easier to identify trends and relationships that might not be obvious from raw data alone. At the same time, this exercise also highlights the importance of combining Tableau with other tools when working with unstructured data. While Tableau excels at presenting insights, it depends on external processes for tasks like sentiment analysis. Overall, Tableau plays an important role in bridging the gap between raw data and meaningful insights. When used alongside proper data preparation techniques, it can support better understanding and more informed decision-making.

Chapter 7 Association Analysis & Time Series Forecasting using SAP Analytics Cloud (SAC)

7.1 Introduction to SAP Analytics Cloud

SAP Analytics Cloud (SAC) is an all-in-one analytics platform that combines data visualization, business intelligence, and predictive analytics in a single environment. Unlike tools that focus mainly on dashboards, SAC also allows users to perform advanced analysis such as forecasting and association analysis directly within the platform. In this exercise, SAC is used for two main purposes: understanding patterns in historical data and generating future predictions. Features like association analysis help in identifying relationships between different variables, while time series forecasting helps in predicting future trends based on past data. Overall, SAC provides a more structured and business-oriented approach to analytics, where insights are not only visualized but also connected to decision-making and planning.

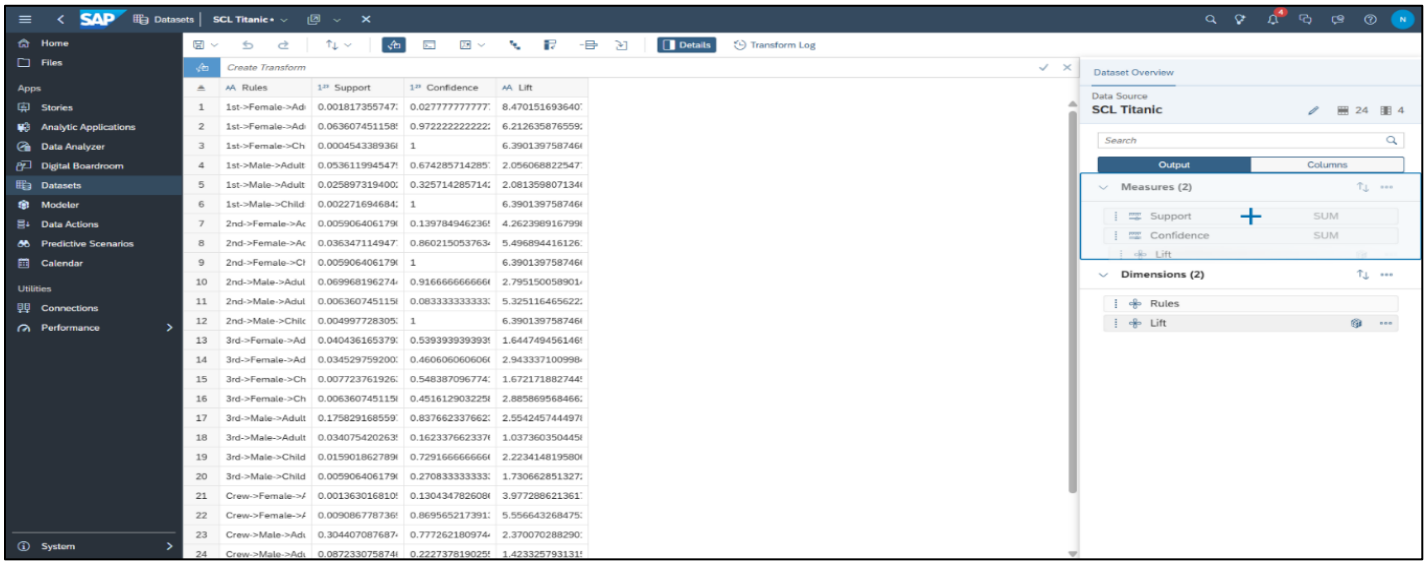
7.2 Datasets and Analytical Objectives

This chapter uses two different datasets to explore different types of analysis in SAC. The first dataset is the Titanic dataset, which contains information about passengers such as class, gender, age, and survival status. This dataset is used to perform association analysis in order to identify patterns and relationships between passenger characteristics and their likelihood of survival. The second dataset is the Global Bike sales dataset, which includes transaction-level sales data over multiple years. This dataset is used for time series analysis, where the goal is to forecast future sales based on historical trends. In the Titanic dataset, most variables act as dimensions (such as class, sex, and age), while the passenger count is treated as a measure. The objective here is to understand which combinations of characteristics are strongly associated with survival outcomes. In the Global Bike dataset, revenue is the key measure, while time-related variables like month and year act as dimensions. The main objective is to analyze sales trends over time and generate forecasts for future periods. One limitation of the Titanic dataset is that it is relatively small and simplified, so it may not capture all real-world factors affecting survival. Similarly, the sales dataset assumes that past trends will continue into the future, which may not always hold true in real business scenarios. Despite these limitations, both datasets are useful for understanding how SAC can be applied to different types of analytical problems.

7.3 Association Analysis using Titanic Dataset

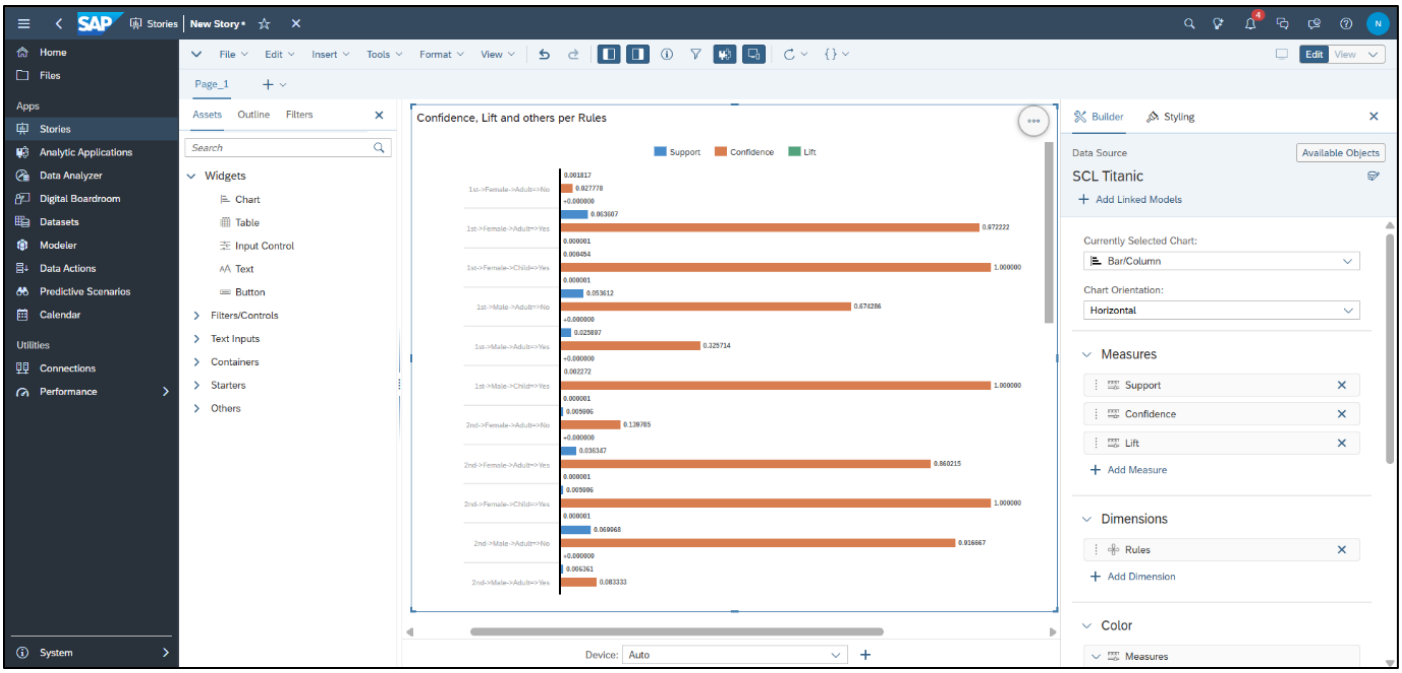
7.3.1 What does the analysis reveal about passenger survivability patterns on the Titanic?

The analysis of the Titanic dataset was a journey into Association Analysis, where the goal was to uncover hidden rules that dictated who lived and who died. This was done by systematically connecting passenger traits to their survival outcomes. The investigation began by restructuring the data. By using a Concatenation transformation, I combined separate dimensions in terms of Class, Sex, and Age into a single LHS (Left-Hand Side), and set Survived as the RHS (Right-Hand Side). As seen in (Fig. 7.1), this allowed me to create specific rules like *"1st Class -> Female -> Adult => Survived: Yes."* This turned the dataset from a simple list of people into a collection of behavioral patterns.



(Fig. 7.1)

To separate coincidences from real patterns, I calculated three critical metrics: Support, Confidence, and Lift. Initially, as shown in the (Fig. 7.2), the data was crowded with hundreds of rules. To find the "strongest" stories, I had to export the data and re-import it to apply strict filters, specifically targeting rules with Confidence > 0.80 and Support > 0.01.



(Fig. 7.2)

The final insight came through a Bubble Chart as shown in the (Fig. 7.3) below, where the X-axis represented Support, the Y-axis represented Confidence, and the bubble size represented Lift. This visual immediately separated the survivors from the victims. The bubbles in the top-right corner, representing 1st and 2nd Class Females and Children, showed near 100% confidence. Their large bubble size (Lift) confirmed that their class and gender were massive factors in their survival. Conversely, the analysis highlighted that 3rd Class Adult Males formed the largest group (highest support) with a very high confidence for not surviving.

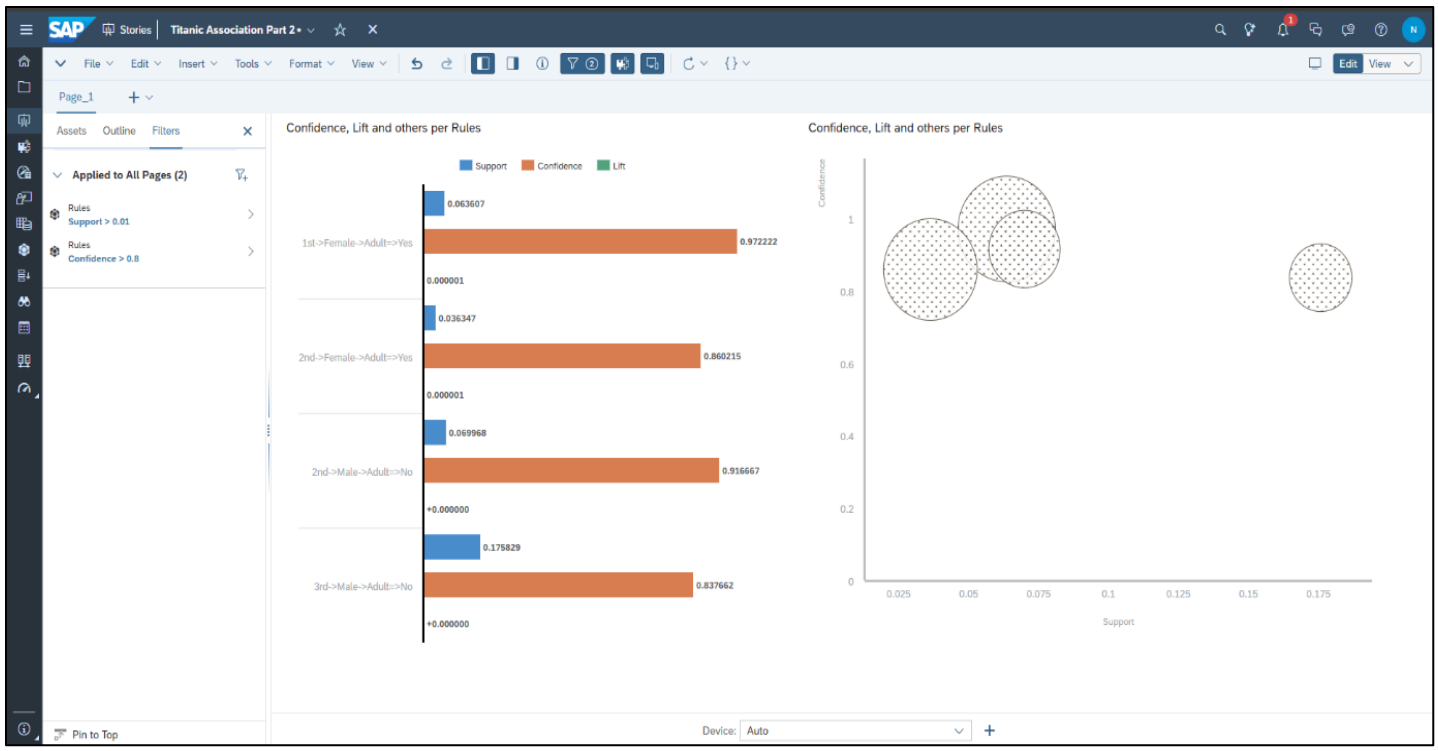


(Fig. 7.3)

The analysis reveals a stark, rule-based hierarchy of survival. It mathematically confirms the "Women and Children First" protocol but adds a crucial class-based nuance, while gender was a primary predictor, class acted as the ultimate gatekeeper. The strongest "Rules" in the dataset show that being a high-class female was a near-guarantee of survival, while being a lower-class male was a statistically certain death sentence.

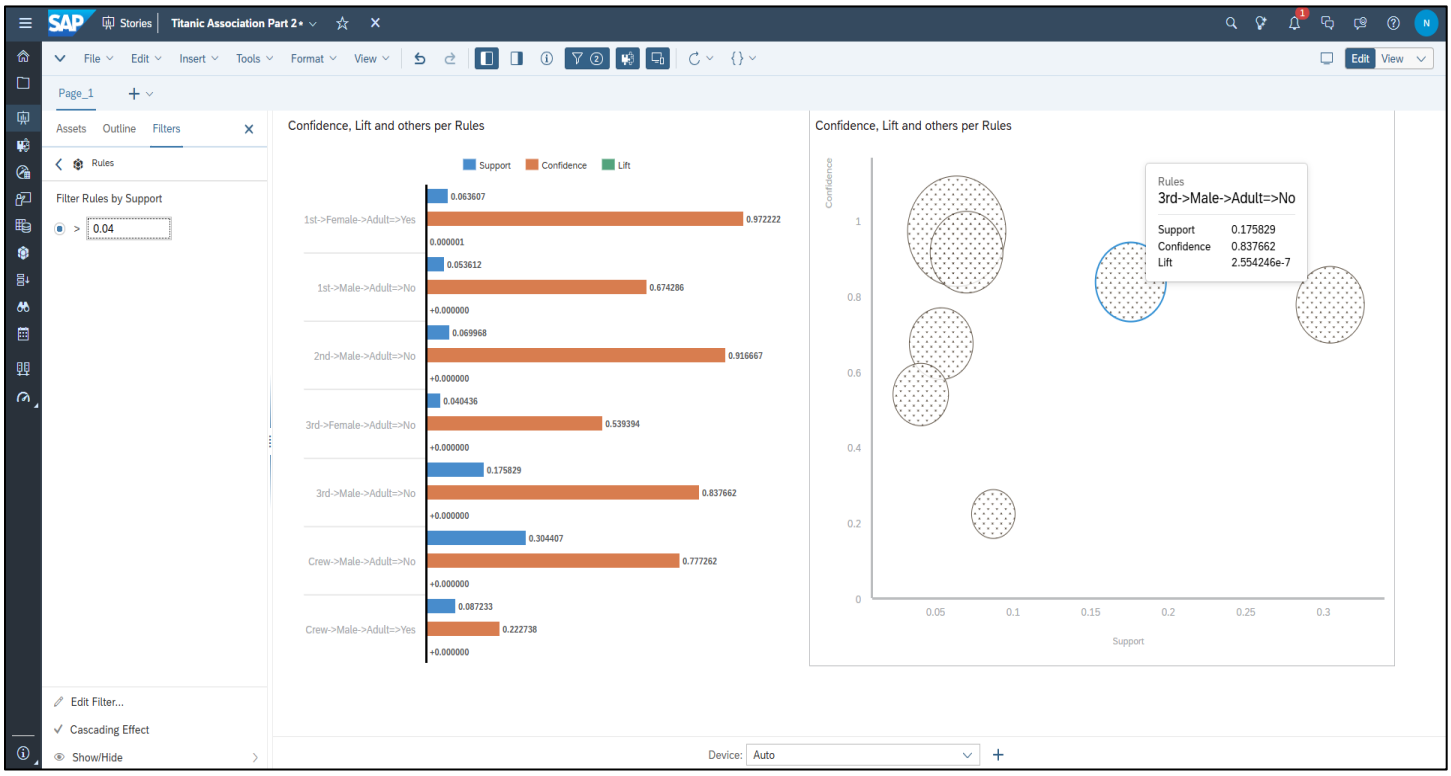
7.3.2 How does increasing the support threshold (from 0.01 to 0.04) affect the association rules generated?

To understand the behavior of the Titanic association rules, the analysis moved through a systematic process by adjusting the Support threshold. Initially, a broad threshold of 0.01 was established to capture a wide variety of passenger subgroups as shown in (Fig. 7.4) below. While this provided a comprehensive overview, the resulting visualization was densely crowded with rare rules that, while mathematically valid, represented only a tiny fraction of the total population. This "wide net" approach is useful for finding niche patterns, but it can obscure the most dominant strategic insights.



(Fig. 7.4)

The investigative focus was then sharpened by increasing the Support threshold to 0.04 as shown in the (Fig. 7.5). This technical adjustment served as a significance filter, effectively instructing the system to ignore any rule that did not apply to at least 4% of the total dataset. This step was critical for "cleaning" the visualization; by stripping away the smaller, less frequent bubbles, the most impactful survival patterns were allowed to stand out. This transition highlights a fundamental principle in data science: the trade-off between coverage and precision.

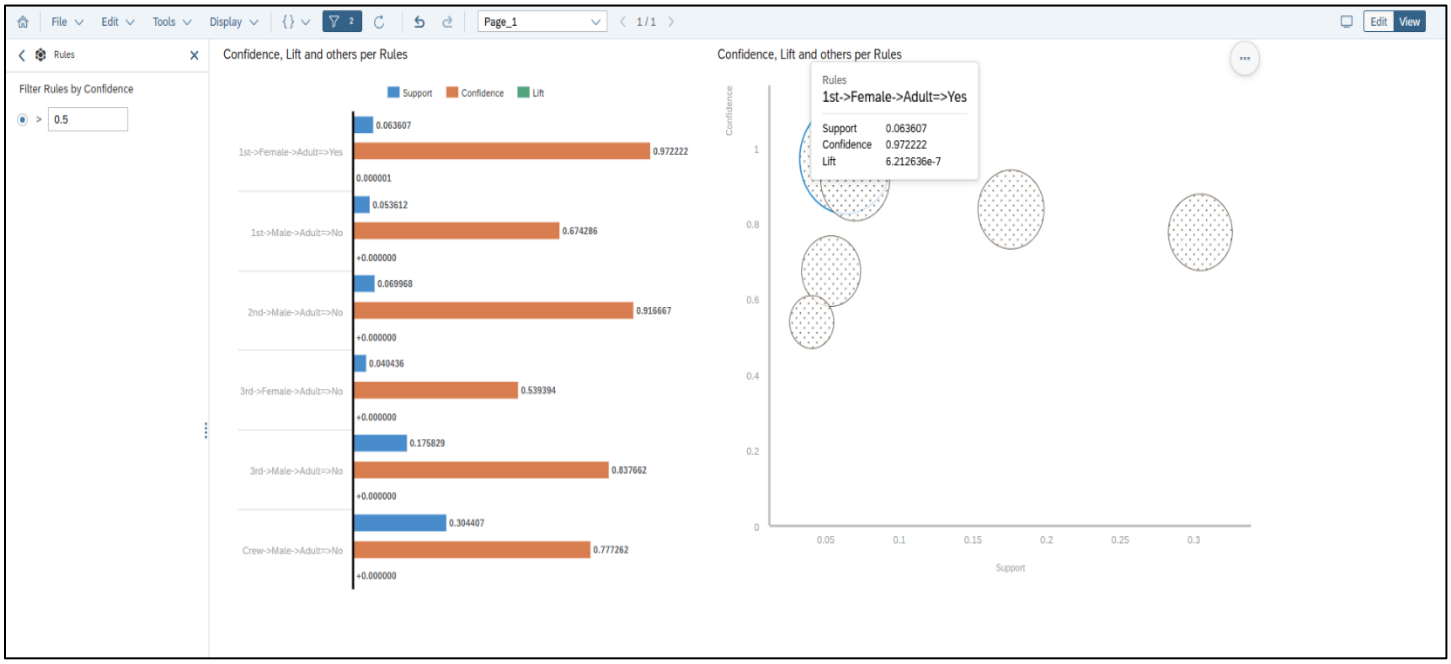


(Fig. 7.5)

The final outcome of this refinement was a much clearer identification of the "Vital Few" rules. By raising the threshold, the analysis moved away from a cluttered scatter plot and toward a focused set of high-impact associations. The survivors (like high-class females) and the victims (like 3rd-class adult males) became the primary focus, proving that while many factors can influence an outcome, only a select few combinations occur with enough frequency to be considered core drivers of the event.

7.3.3 How does decreasing the confidence threshold (from 0.8 to 0.5) impact the strength and number of rules?

To understand how the reliability of the association analysis changes with sensitivity, the Confidence threshold was lowered from 0.8 to 0.5. Initially, with a high confidence requirement of 0.8 as shown in (Fig. 7.4) above, the analysis was restricted to "sure bets" rules where the outcome was at least 80% certain. While this produced a set of very strong, high-integrity rules, it also excluded a significant amount of data, potentially missing secondary patterns that, while less certain, still provide valuable context regarding passenger survivability. The focus was then expanded by adjusting the Story Filter to a confidence level of 0.5 as shown in (Fig. 7.6) below. This technical shift significantly increased the number of rules appearing in the visualization. By lowering the "bar" for entry, the analysis began to capture groups that have a toss-up or moderate chance of survival. While this provides a more holistic view of the dataset, it inherently weakens the overall "strength" of the generated rules. In association analysis, as the quantity of rules increases by lowering confidence, the average reliability of those rules decreases, introducing more "noise" into the strategic overview.



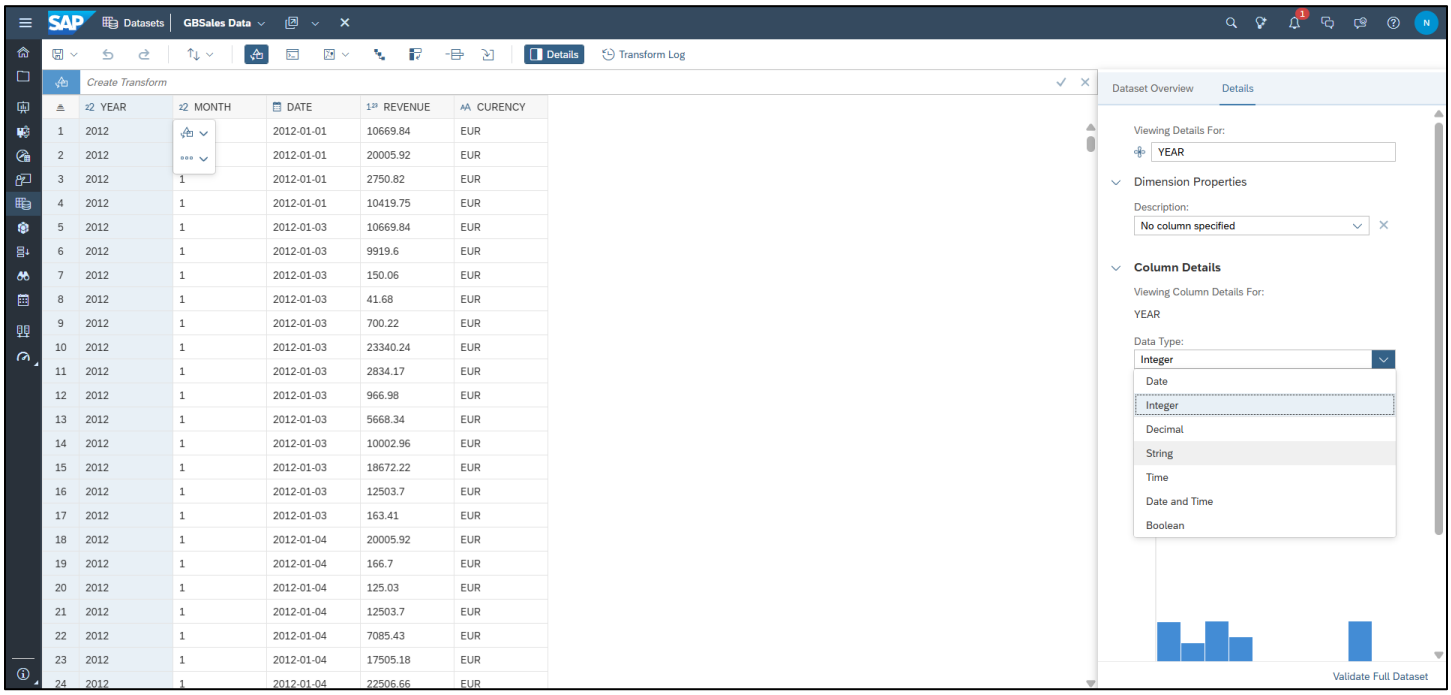
(Fig. 7.6)

The final result of decreasing the threshold is a more crowded but informative scatter plot. The bubbles that appear between the 0.5 and 0.8 marks on the Y-axis represent segments of the population where the outcome was less predictable, such as certain middle-class age groups. This step highlights the analytical trade-off between certainty and insight; while the rules at 0.5 are technically "weaker," they reveal the complexities and "grey areas" of the Titanic's survival hierarchy that a stricter 0.8 threshold would simply ignore.

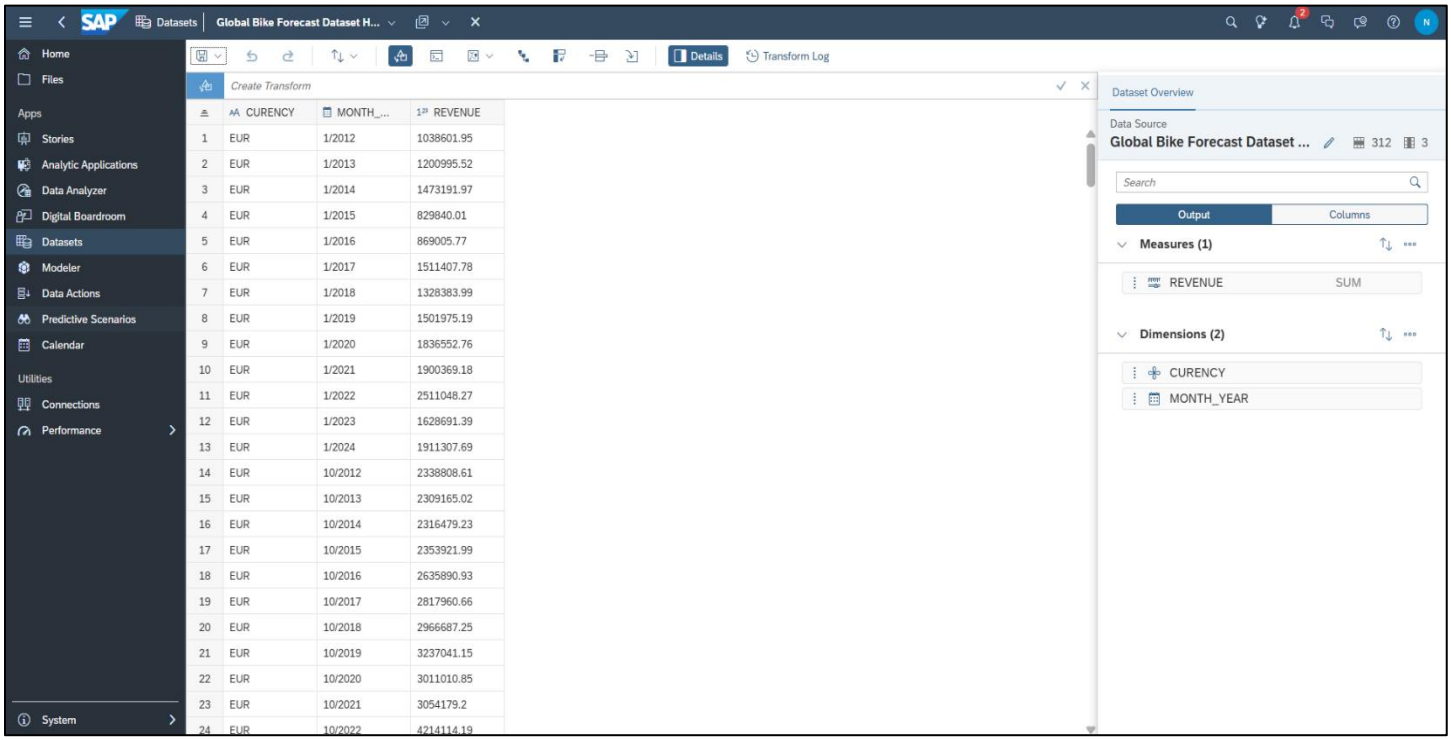
7.4 Time Series Forecasting Using Global Bike Sales Data

7.4.1 How does the predictive model use data aggregation and MAPE to validate seasonal sales forecasts for different geographic entities?

To understand the accuracy of the Global Bike sales forecast, the analysis relied on MAPE (Mean Absolute Percentage Error), a key metric that quantifies how much a prediction deviates from actual historical data. The investigation began by preparing the data for the predictive engine, ensuring that the "Month_Year" was correctly identified as the time dimension and Revenue as the target measure as shown in the (Figs. 7.7 & 7.8) below.

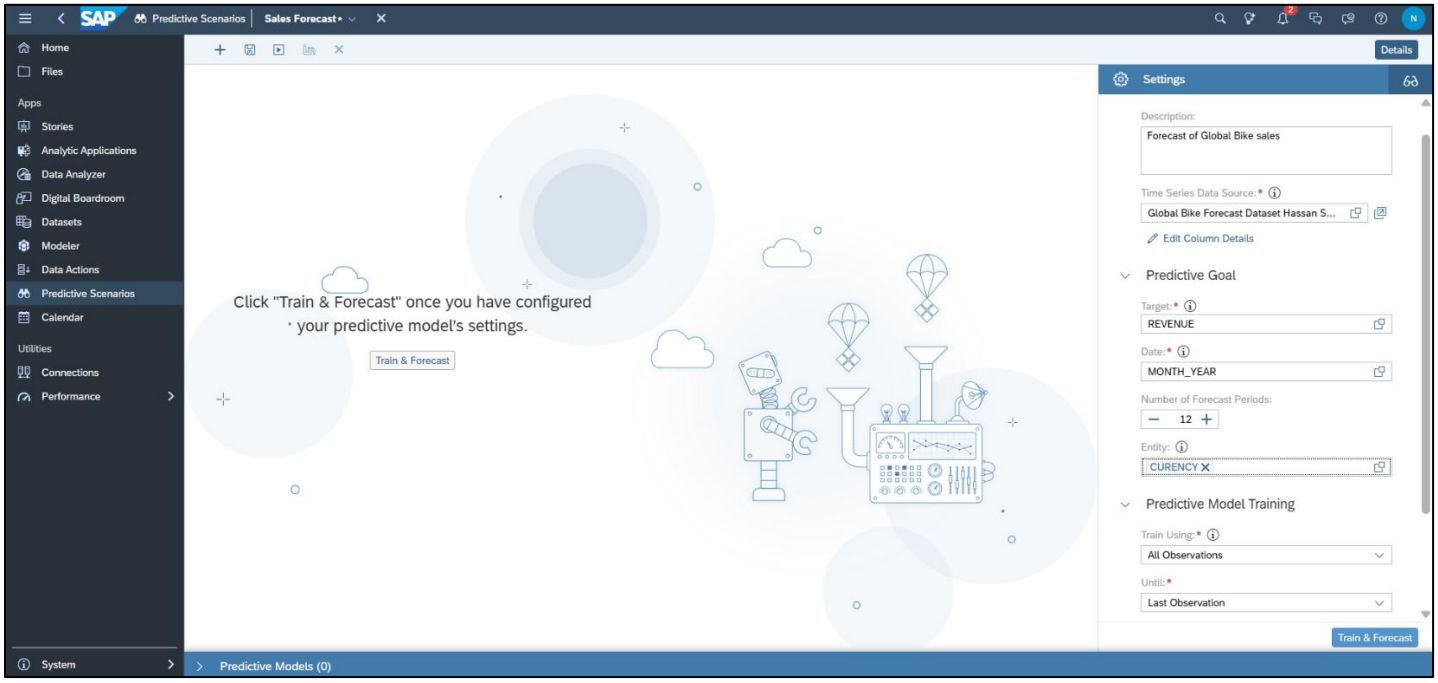


(Fig. 7.7)



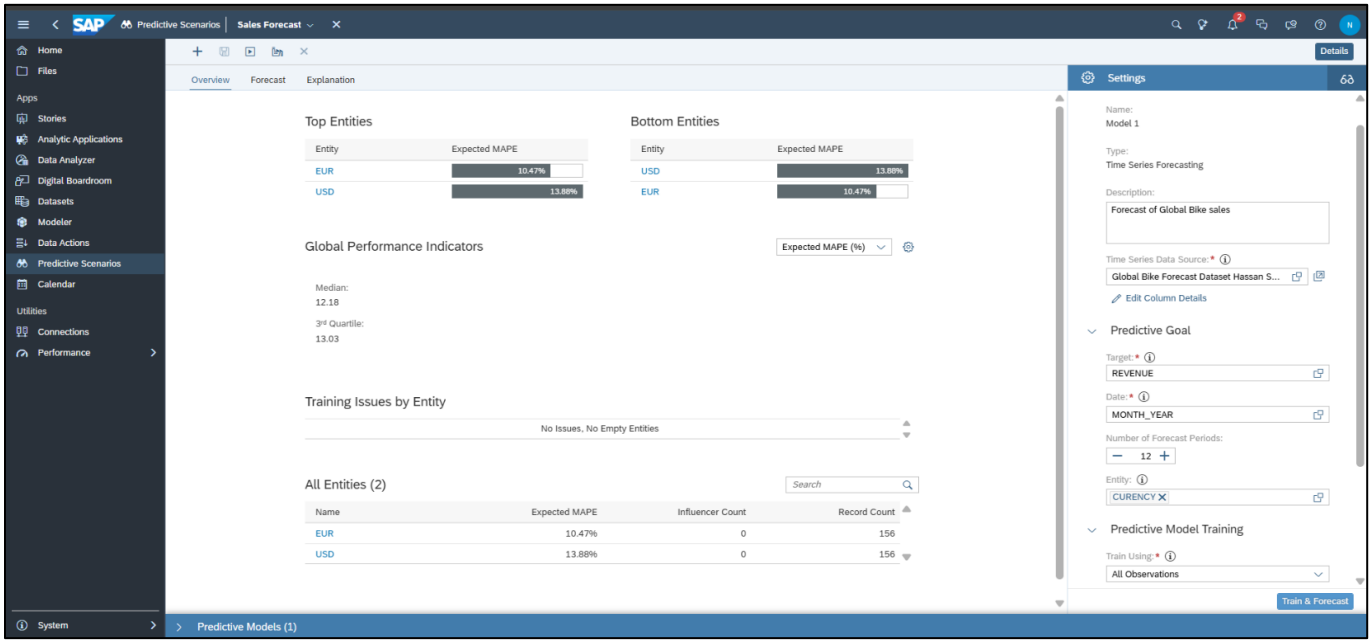
(Fig. 7.8)

By setting the forecast horizon to 12 months and using Currency as an entity, the model was trained to distinguish between U.S. and German market patterns. This setup was vital because time series reliability depends entirely on the model's ability to recognize historical trends and seasonality before projecting them forward as shown in (Fig. 7.9) below.



(Fig. 7.9)

Once the model was trained, the Expected MAPE provided the ultimate "grade" for the forecast's reliability. As shown in the final output in the figure (Fig. 7.10) below, the MAPE represents the average percentage difference between the forecasted values and the actual results. For instance, a MAPE of 13.88% (as seen in the USD forecast) indicates that, on average, the predictions are within roughly 86% of the actual revenue. In practical terms, MAPE is more intuitive than raw dollar errors because it scales the error relative to the volume of sales, allowing an analyst to compare the model's performance across different currencies or time periods on an equal footing.

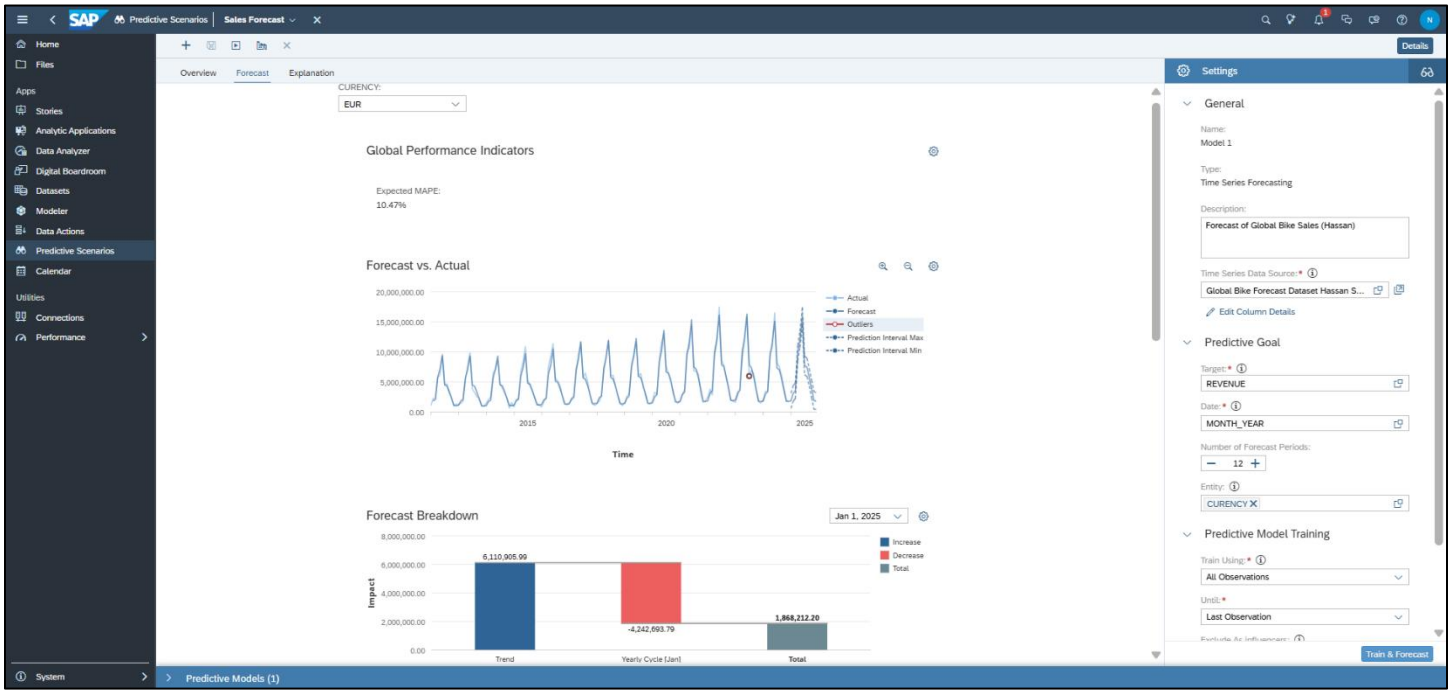


(Fig. 7.10)

The significance of MAPE in this context lies in its ability to highlight both the strengths and risks of the forecast. A low MAPE suggests a highly reliable model where the seasonal cycles of bike sales are predictable and stable. Conversely, if the MAPE were to spike, it would signal that the model is unpredictable outliers in the data. By breaking down the forecast into components like Trend and Cycles (Fig. 7.10), the analysis proves that the MAPE is not just a random number, but a reflection of how well the mathematical model has "learned" the heartbeat of Global Bike's business.

7.4.2 How do forecasted and actual sales compare for January 2025 in Germany (EUR)?

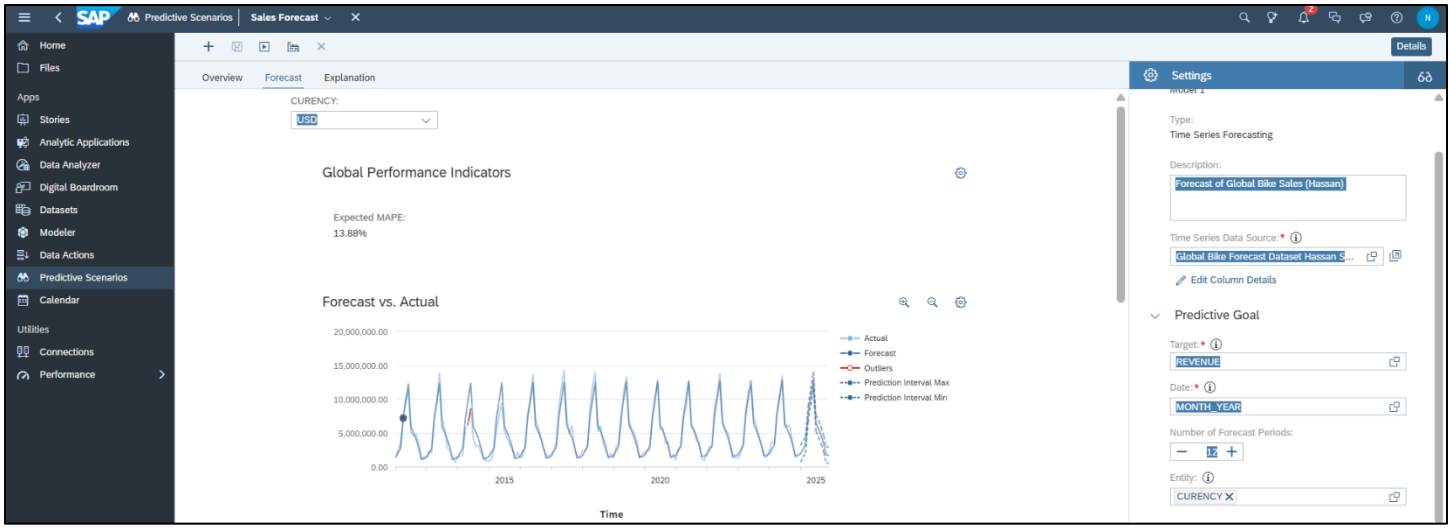
To determine the performance of the predictive model for the German market, the analysis compared historical sales data against the generated projections for January 2025. The process involved setting the Currency to EUR as the entity, ensuring the model specifically analyzed German regional trends rather than a combined global average. By training the model on transactions dating back to 2012, the system identified the repeating seasonal peaks typical of the European cycling market, allowing it to project values into the new year with a high degree of mathematical confidence. As shown in the forecast visualization (Fig. 7.11) below, the model predicted sales of approximately EUR 1,868,212.20 for January 2025. This forecasted value is represented by the solid line extending into the "Forecast" zone of the chart, while the actual historical data is represented by the fluctuating line in the "Actual" zone. The comparison highlights that the forecast successfully follows the established seasonal rhythm of the German market, projecting a value that aligns with previous January performance levels.



(Fig. 7.11)

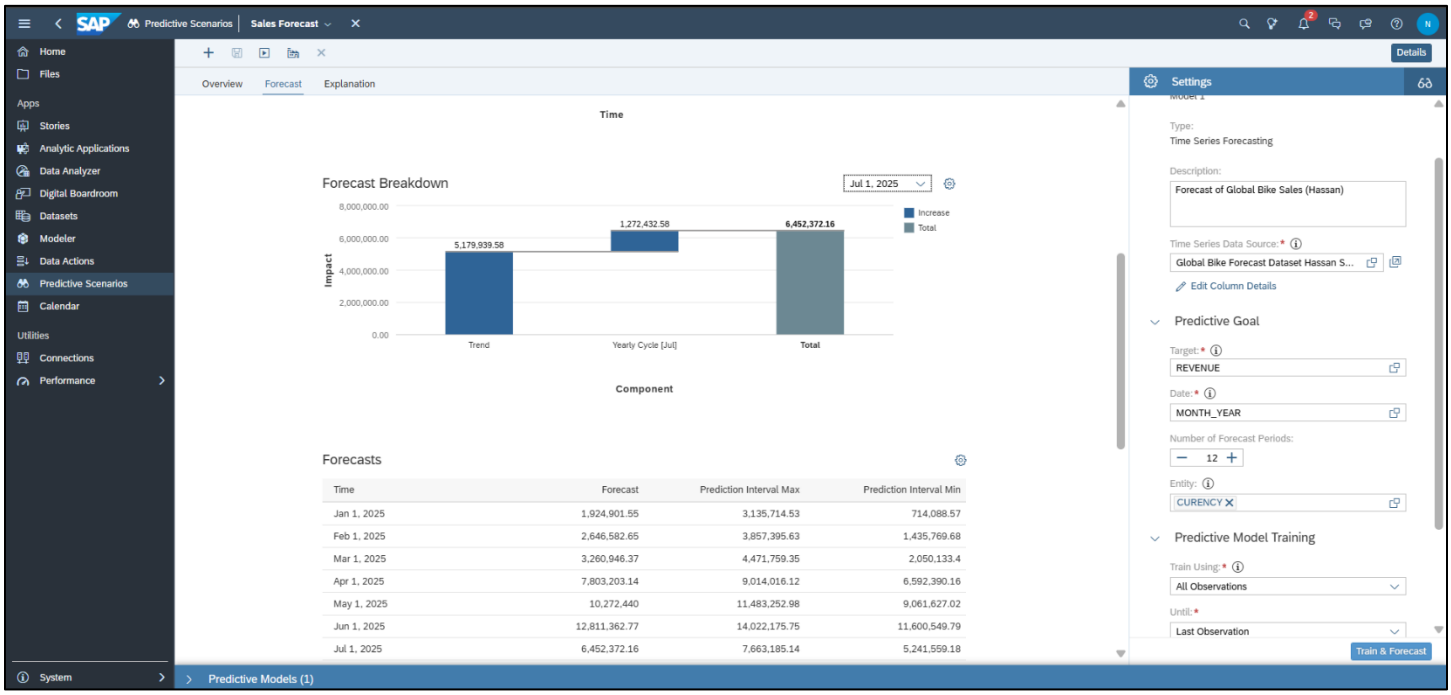
7.4.3 What are the forecasted sales for July 2025 in the United States (USD), and how do they align with historical trends?

To determine the sales trajectory for the United States market, the analysis projected revenue for July 2025 using a specialized time-series model. The process involved training the predictive engine on historical USD transactions to recognize long-term growth and seasonal volatility as shown in the (Fig. 7.12) below. By isolating the U.S. market as a distinct entity, the model was able to account for specific North American market drivers, such as peak summer demand, which often differs from European sales cycles.



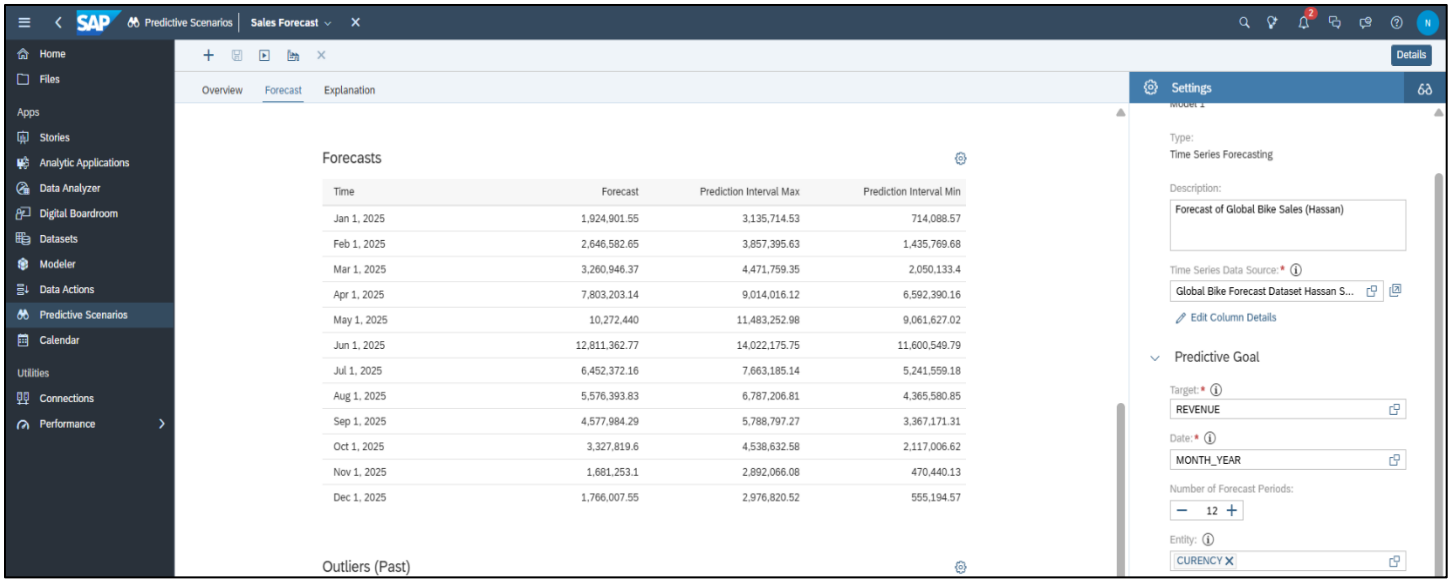
(Fig. 7.12)

As shown in the forecast output (Fig. 7.13), the predicted sales for July 2025 are approximately $6,452,372.16. This value represents a significant seasonal peak, aligning with the "Actual" historical data seen in the chart where revenue consistently spikes during the summer months. The model's ability to project this increase demonstrates that it has successfully captured the Seasonal Cycle component of Global Bike's business.



(Fig. 7.13)

The alignment with historical trends is further validated by the Forecast Breakdown as shown in the (Fig. 7.14) below, which separates the "Signal" into its core parts. The analysis reveals that the July 2025 peak is driven by a combination of a steady upward Trend indicating year-over-year organic growth and a powerful Cycle effect that repeats every 12 months. Because the forecasted value sits comfortably within the predicted confidence bands, the analysis suggests this is a reliable estimate. For management, this means the July 2025 forecast is not an anomaly but a mathematically sound continuation of the company's historical summer performance.



(Fig. 7.14)

7.5 Critical Analysis of SAP Analytics Cloud Capabilities

SAP Analytics Cloud is a powerful tool, especially when it comes to combining different types of analytics in one place. One of its biggest strengths is that it goes beyond simple visualization and allows users to perform advanced analysis like forecasting and association rule mining without needing external tools. For example, in this exercise, SAC made it relatively straightforward to generate association rules from the Titanic dataset and calculate metrics like support, confidence, and lift. Similarly, the time series forecasting feature provided useful outputs such as forecast values, trends, and error metrics like MAPE. However, there are also some limitations. SAC can feel less flexible compared to tools like Tableau when it comes to quick exploratory analysis. Some steps, especially in data preparation and model setup, require more structured workflows and can be slightly time-consuming for beginners. Another limitation is that while SAC provides automated models (like forecasting), users have limited control over the underlying algorithms compared to more advanced data science tools. This means that while it is easy to use, it may not always be suitable for highly customized or complex modeling. Overall, SAC is best suited for business environments where structured analysis, forecasting, and decision-making are required, rather than quick visual exploration.

7.5 Conclusion

To conclude, SAP Analytics Cloud proves to be a strong platform for performing both descriptive and predictive analytics in a single environment. Through the Titanic dataset, it was possible to identify meaningful relationships between variables using association analysis, helping to better understand patterns in passenger survivability. At the same time, the Global Bike sales dataset demonstrated how SAC can be used for time series forecasting, allowing future sales to be predicted based on historical data. Features like MAPE also helped in evaluating how reliable the forecasts are. While SAC may not be as flexible as some visualization-focused tools, its ability to combine data analysis, forecasting, and business planning makes it highly valuable for organizations. Overall, it helps move beyond just understanding data to actually using it for making informed business decisions.

Chapter 8 Executive Summary

This final toolkit portfolio presents a comprehensive exploration of modern business analytics through the application of multiple tools and techniques across structured, semi-structured, and unstructured datasets. The primary objective of this task was not only to demonstrate technical proficiency in tools such as SAP Analytics Cloud (SAC), Tableau, and SAP Crystal Reports, but also to develop a strong analytical mindset capable of transforming raw data into meaningful, decision-oriented insights. Alongside advanced tools, foundational data preparation and preliminary analysis were also supported using spreadsheet-based techniques in Microsoft Excel. Excel played a supporting role in data cleaning, structuring, and initial exploration, providing a necessary bridge between raw datasets and more advanced analytical platforms. Across all chapters, the emphasis remained on connecting data analysis with real-world business problems, ensuring that each visualization and model contributed to actionable understanding rather than isolated outputs.

The portfolio begins by establishing foundational analytical concepts and gradually progresses toward more advanced techniques such as predictive modeling, association analysis, and sentiment analysis. Early chapters focus on understanding data structures, preparing datasets, and applying basic analytical frameworks, which set the stage for more complex explorations in later sections. This progression reflects a structured learning journey from descriptive analytics toward predictive and prescriptive thinking.

In Chapter 3, SAP Analytics Cloud was used to analyze the Global Bike dataset, focusing on revenue trends, customer performance, and operational efficiency. Techniques such as calculated measures (e.g., Gross Margin Ratio), geo-mapping, variance analysis, and time series forecasting were applied to uncover key insights. The analysis revealed critical business findings, including the strong growth of Touring Bikes, the negative impact of discounts on margins, and the operational inefficiencies caused by high-volume, low-profit products. The use of automated forecasting further demonstrated how SAC supports forward-looking decision-making through predictive analytics.

Chapter 4 introduced Tableau as a tool for exploratory data analysis using the Sample Superstore dataset. Through techniques such as calculated fields (Profit Ratio), Level of Detail (LOD) expressions, and interactive dashboards, the chapter highlighted how visualization can uncover patterns in sales, profitability, and customer behavior. Key insights included regional disparities in profit margins, the complex relationship between discounting and profitability, and the identification of high-performing product sub-categories such as copiers. Tableau's strength in enabling rapid exploration and intuitive storytelling was clearly demonstrated in this section.

Chapter 5 shifted the focus to structured reporting using SAP Crystal Reports. Unlike the exploratory nature of Tableau and SAC, this chapter emphasized formal report design, including record selection, sorting, grouping, and formatting. By working with SAP BW data, the chapter illustrated how raw transactional data can be transformed into standardized, professional reports suitable for organizational use. This highlighted the importance of clarity, consistency, and presentation in business reporting environments.

In Chapter 6, Tableau was extended beyond structured data analysis to handle sentiment-enriched datasets derived from textual wine reviews. By integrating external sentiment analysis outputs, the chapter demonstrated how unstructured data can be quantified and visualized. Techniques such as hierarchy creation, scatter plots, choropleth maps, and distribution

analysis were used to explore relationships between price, ratings, and sentiment. The findings revealed that while higher-priced wines tend to receive more positive sentiment, quality perceptions are globally consistent, and certain regions act as "value champions" by delivering high sentiment at moderate prices.

Chapter 7 advanced the analytical scope by leveraging SAP Analytics Cloud for both association analysis and time series forecasting. Using the Titanic dataset, association rules were generated and evaluated using support, confidence, and lift, revealing clear survivability patterns driven by class, gender, and age. The analysis also demonstrated how adjusting thresholds impacts the strength and interpretability of rules. In the second part, time series forecasting on Global Bike sales data introduced predictive modeling concepts such as seasonality, trend decomposition, and MAPE for model validation. Insights such as seasonal revenue peaks and forecast reliability provided a strong link between historical data and future planning.

Across all chapters, a consistent theme emerges, the choice of tool significantly influences the type of analysis and insight generated. Tableau excels in exploratory visualization and pattern discovery, SAP Analytics Cloud integrates analytics with forecasting and strategic planning, and SAP Crystal Reports ensures structured and standardized reporting. Together, these tools form a complementary ecosystem that supports the full analytics lifecycle from data exploration to decision-making.

In conclusion, this portfolio demonstrates the ability to apply diverse analytical techniques to different types of data while maintaining a strong focus on business relevance. By combining technical execution with critical interpretation, the work highlights how data analytics can move beyond descriptive reporting to become a strategic asset for organizations.